

Tetrahedron report number 692

The Reformatsky reaction in organic synthesis. Recent advances

Rogelio Ocampo^a and William R. Dolbier, Jr.^{b,*}

^a*Departamento de Química, Universidad de Caldas, A.A. 275, Manizales, Colombia*

^b*Department of Chemistry, University of Florida, Gainesville, FL 32611-7200, USA*

Received 2 June 2004

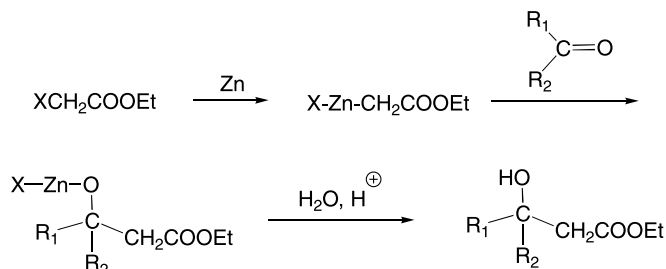
Available online 27 August 2004

Contents

1. Introduction	9326
2. Metals or derivatives and catalysts for Reformatsky reactions	9328
2.1. Zinc metal activation and catalysts	9328
2.2. Metals or derivatives other than zinc	9328
2.2.1. The chromium-Reformatsky reaction	9329
2.2.2. The samarium diiodide-promoted Reformatsky reaction (Kagan reagent)	9329
2.2.3. The indium-mediated Reformatsky reaction	9330
2.2.4. Reformatsky-type reactions involving other metals	9331
3. Non-conventional reaction conditions	9333
3.1. Supported metal-induced Reformatsky reactions	9334
4. Reformatsky reactions involving non-fluorinated substrates	9334
4.1. Precursors of Reformatsky reagents	9334
4.1.1. α -Bromoamides as precursors	9337
4.1.2. Polyhaloprecursors	9338
4.1.3. 'Remote' Reformatsky precursors	9338
4.1.4. α -(Bromomethyl)acrylates as precursors	9339
4.1.5. 4-Bromocrotonates and related precursors	9341
4.1.6. Precursors with leaving groups other than halogen atoms	9341
4.2. Electrophiles	9342
4.2.1. 'Classical' electrophiles	9342
4.2.2. 'Non-classical' electrophiles	9345
4.2.3. Michael or Michael-type acceptors	9348
4.2.4. Carbon electrophiles other than carbonyl or related groups	9349
4.3. Intramolecular Reformatsky reactions	9350
4.4. Advances in stereochemical aspects	9354
4.4.1. Chiral or stereo inductor auxiliaries	9354
4.4.2. Chiral auxiliaries as additives	9359
4.5. Synthetic applications	9359
5. Recent reports of Reformatsky reactions involving fluorinated substrates	9360
5.1. Advances in stereochemical aspects	9363
5.2. Recent applications in synthesis	9364
6. Concluding remarks	9365
References and notes	9366

Keywords: Reformatsky reaction; Organic synthesis; Organozinc reagent.

* Corresponding author. Tel.: +57-352-392-0591; fax: +57-352-846-1962; e-mail: wrd@chem.ufl.edu



Scheme 1. X = halogen; R₁ and R₂ = H, alkyl, aryl.

1. Introduction

The classical form of the well-known Reformatsky reaction, introduced for the first time in 1887,¹ consists of the zinc-induced formation of β-hydroxyalkanoates from ethyl α-haloacetates in a reaction with aldehydes or ketones (Scheme 1).

The reaction was ultimately found to be applicable to alkyl 2-haloalkanoates in general as well as to 'remote' (3-, 4-, 5- or higher) haloalkanoates. The scope of the Reformatsky reaction was also extended beyond the use of aldehydes or ketones as electrophiles. In addition, a number of other metals and catalysts² were found to promote the role analogous to that of zinc. These facts called for a broader, more comprehensive definition, and therefore, as stated in a previous review,³ "Reformatsky reactions are defined as those resulting from metal insertions into carbon–halogen bonds activated by carbonyl-, carbonyl-derived or carbonyl-related groups in vicinal or vinylogous positions with practically all kinds of electrophiles". In a general sense, the Reformatsky reaction can be taken as subsuming all enolate formations by oxidative addition of a metal or a low-valent metal salt or complex into a carbon–halogen bond (or carbon–leaving group bond) activated by a vicinal carbonyl-derived group, followed by a reaction of the enolates thus formed with an appropriate electrophile (Scheme 2).^{4,5} However, an aqueous metal-free electrochemical Reformatsky reaction, presumably proceeding via a radical mechanism, was reported recently.⁶

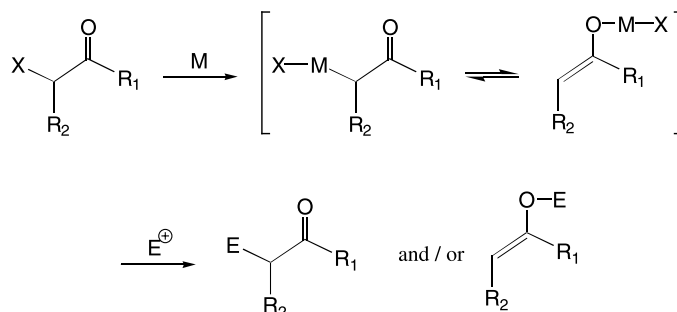
Several recent reviews have appeared, focusing on the structure and nature of the organozinc reagent

(spectroscopically and theoretically),^{3,7} substitution of zinc by other metals,^{2,3} zinc activation,^{3,4b} experimental protocols,^{3–5} different precursors and electrophiles,³ and reactions and applications in organic synthesis.^{3,4,8}

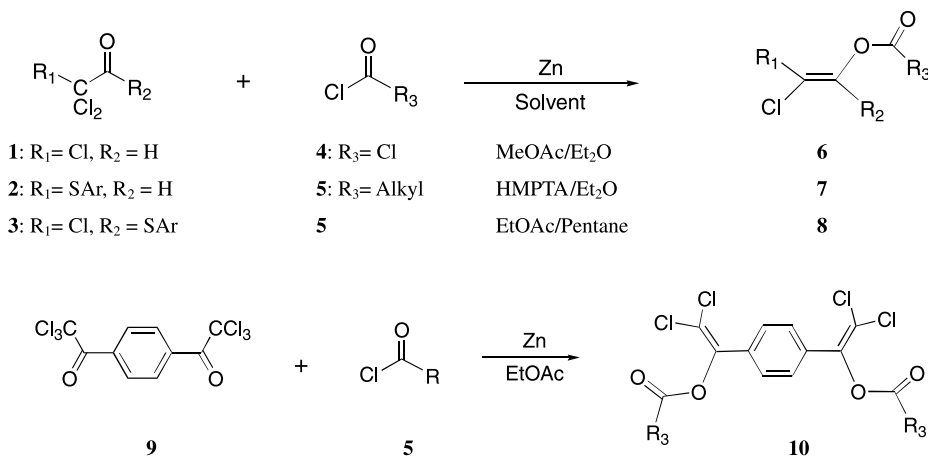
Reformatsky reactions have been recognized as among the most useful methods for the formation of carbon–carbon bonds, becoming a valuable tool in modern organic synthesis with a broad applicability and great versatility in numerous inter- and intramolecular reactions involving a great variety of electrophiles. As such, this methodology is considered a useful alternative to base-induced aldol reactions or, at the least, an important complement to other enolate reactions. A retro-Reformatsky reaction was recently reported and was used as one of the steps in the synthesis of heterocyclic enamines.⁹

One of the advantages of the Reformatsky reaction is that the reaction proceeds under neutral conditions, in contrast to the aldol reaction which, in general, requires a base to generate the enolate or an acid to activate the electrophile. Very important is the fact that in classical Reformatsky reactions no *O*-products are obtained, and even reagents with strong affinities for oxygen such as TMS–Cl afford only C-silylated products in most of the cases. Thus, the site of the reaction is strictly determined by the site of the halogen substituent, although some few exceptions to this rule are known (Scheme 3).^{10–12}

Chloral **1** and other polyhalogenated carbonyl compounds react with phosgene **4** in the presence of zinc dust to afford 2,2-dihalovinyl chloroformate **6** clearly via an *O*-enolate reaction with the phosgene acting as the electrophile.¹⁰



Scheme 2.



Scheme 3.

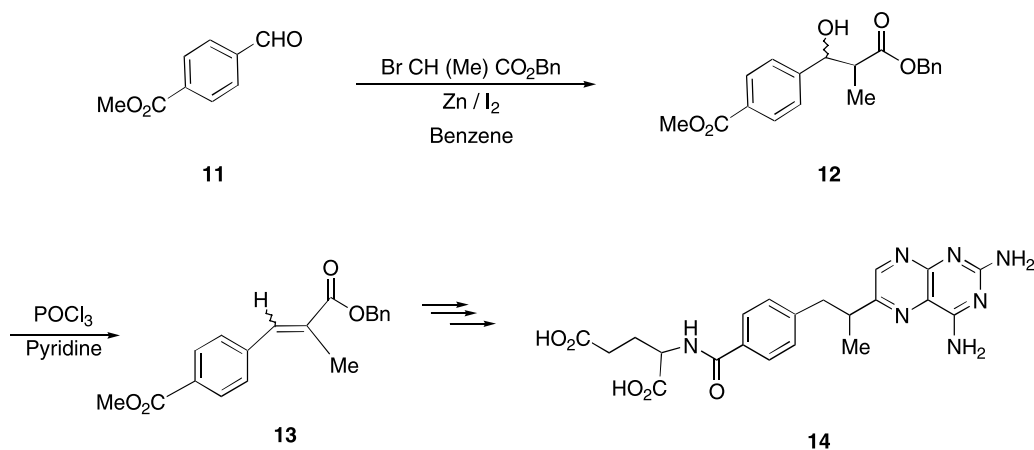
Likewise, *O*-enolate attack of polyhalogenated compounds **2**, **3** and **9** on acyl chlorides **5** in Reformatsky reactions led to vinyl compounds **7**, **8** and **10** (Scheme 3).^{11,12}

However, several disadvantages of Reformatsky reactions have been recognized in the past. A problem encountered in many reactions is the unselective dehydration of the aldols initially formed, but this problem has been overcome either by substantially lowering the reaction temperature or by using silylated starting materials. Perhaps, the most serious limitation attributed to Reformatsky reactions has been the lower yields and stereoselectivities associated with them in comparison to those of aldol reactions.¹³ It is well known that base-induced aldolizations are readily stereocontrolled, and the low stereoselectivities associated with Reformatsky reactions have discouraged synthetic chemists from using them, particularly if asymmetric targets were desired. This perception has often limited the synthetic applications of the Reformatsky reaction, it being used largely for those syntheses that involve subsequent elimination or oxidation steps on the product aldol, as for example in the synthesis of (–)-oudemansins.¹⁴ Also, the Reformatsky condensation of benzyl α -bromopropionate with methyl 4-formylbenzoate **11** is conducted without stereochemical concerns because it

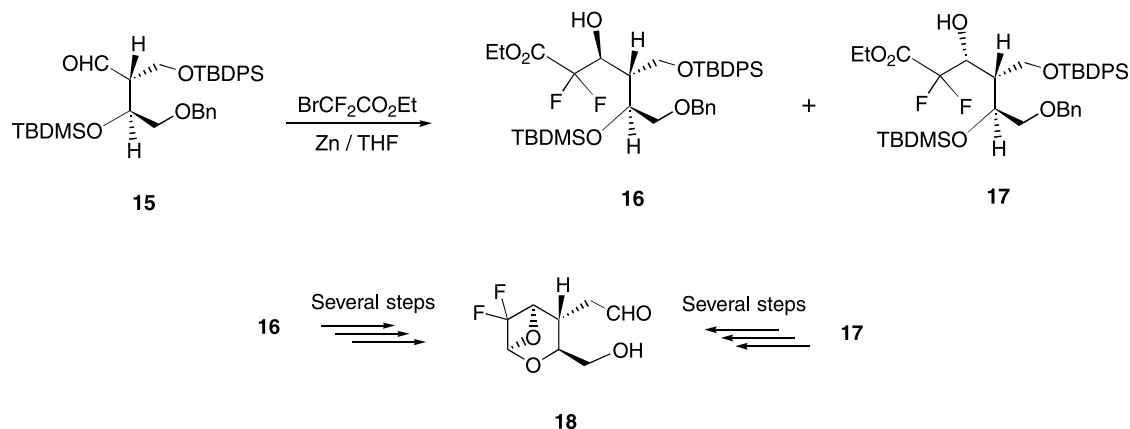
is followed by dehydration of the diastereomeric products **12**, to afford the respective α,β -unsaturated ester **13**, which was used to prepare antifolate 9-alkyl-10-deazaminopterins **14** (Scheme 4).¹⁵ Similarly, β -substituted acrylates $\text{RCH}=\text{CHCO}_2\text{Et}$ ($\text{R}=\text{Ar}$, Cyhex, 3-Py) were synthesized via the reaction of the respective aldehydes with ethyl bromoacetate by way of one-pot Reformatsky/dehydration processes prompted by HMPT and zinc.¹⁶

In the total synthesis of (+)-10,10-difluorothromboxane A₂, synthesis of the key aldehyde intermediate **18** was accomplished in 14 steps. Diastereoselectivity was high in all but the Reformatsky step with aldehyde **15** (Scheme 5).¹⁷ However, both epimers, **16** and **17**, obtained in this reaction could be converted efficiently to key aldehyde **18**. This is a good example of how a non-diastereoselective Reformatsky reaction was able to be useful in a stereo directed synthesis.

Sometimes the diastereomeric mixture resulting from Reformatsky reactions can be resolved by conventional chromatographic techniques, so that the subsequent synthetic steps can be performed on the specific diastereomer.¹⁸ However, such syntheses are obviously not convenient enough to have broad application.



Scheme 4.



Scheme 5.

Stereoselectivity in Reformatsky reactions has become a challenge of interest during the past decade, and a lot of recent progress has been made in this area. As will be illustrated later, there are now a number of examples of highly diastereo- and enantioselective Reformatsky reactions, deriving from the use of chiral ligands, thus circumventing the limitation mentioned above and rescuing the modern synthetic potential of this methodology. Therefore, the Reformatsky reaction is now suited for applications to the synthesis of complex target molecules, including being very applicable to intramolecular aldol processes, even when medium sized rings are desired. It is the intent of this review to summarize the most recent progress in the use of Reformatsky reactions in organic synthesis, including the latest developments in diastereo- and enantioselective reactions.

2. Metals or derivatives and catalysts for Reformatsky reactions

2.1. Zinc metal activation and catalysts

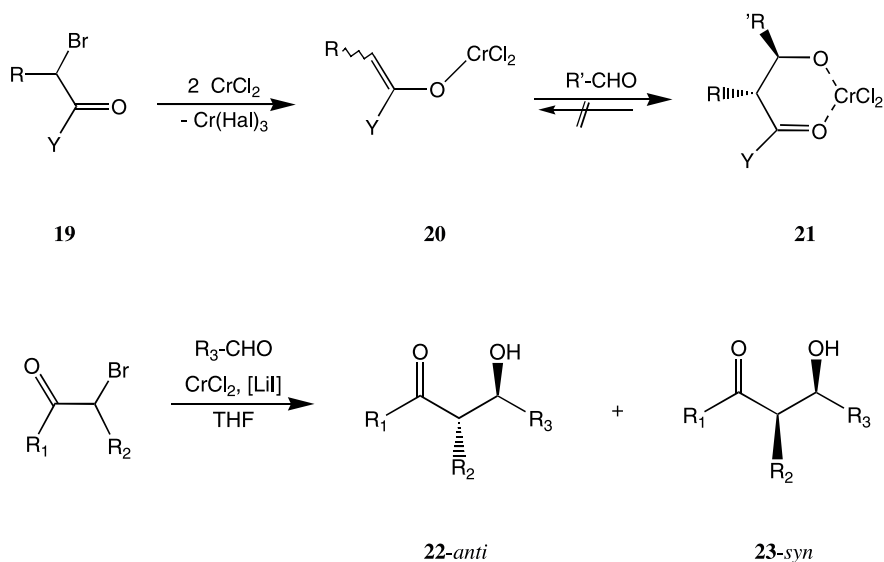
Previous reviews^{3,5,8} have summarized the most common methods of metal activation, which function to either remove the deactivating zinc oxide layer from the metal surface or to achieve a fine distribution of the metal. Such methods generally use procedures that involve simple washing of the zinc (in the form of dust, foil, pellets or turnings),^{3,8} depassivating procedures employing reagents such as iodine,^{8,19} chlorotrimethylsilane,²⁰ iodine-chlorotrimethylsilane,²¹ 1,2-dibromoethane,^{8,22} copper (I) halides,^{3,8,23} mercuric halide,^{3,8,24} or molecular sieves.^{23e,f} Improved classical and non-classic Reformatsky reactions have been accomplished by the use of special types of activated zinc, which include the use of zinc–copper,²⁵ zinc–silver^{23,26} or zinc/silver–graphite couples.²⁷ Good results have also been reported by using reduction-based activation of zinc halides in solution. Examples of this latter category are the so-called Rieke zincTM (zinc chloride reduced by potassium),²⁸ or zinc chloride reduced by lithium with ultrasonic irradiation,²⁹ lithium naphthalenide³⁰ or sodium.³¹ A new protocol of zinc chloride or tin

(II) chloride reduction by sodium in liquid ammonia has been reported³² for the synthesis of β -hydroxyesters, homoallylic and homopropargylic alcohols, with yields ranging from 61 to 96%. The activity of the resulting metal is lower than the corresponding Rieke-zincTM, but remarkably safe and short-time procedures and low cost render this approach particularly attractive for larger-scale applications.

In addition to the different zinc activation protocols available, it is also possible to use diethyl zinc^{33,34} or some additives in combination with zinc,³ for instance tantalum (prepared readily from tantalum(V) chloride and activated zinc)³⁵ or binuclear complexes (for example the vanadium complex $[\text{V}_2\text{Cl}_3(\text{hmpa})_6][\text{ZnCl}_3(\text{hmpa})]$).³⁶ This vanadium binuclear complex proved to be useful for other carbon–carbon formations as well. A combination of $\text{Zn}/\text{Et}_2\text{AlCl}/\text{AgOAc}$ has been used as a modification for Reformatsky reactions of bromodifluoroacetate.³⁷ Cp_2TiCl_2 is reported to be a very special catalyst for zinc-mediated Reformatsky reactions of bromoacetate with 1-cyano-1-phenylsulfonyl alkenes³⁸ and alkyl 4-bromocrotonates with 1,1-dicyanoalkenes.³⁹ Although the catalytic role of Cp_2TiCl_2 is not well understood yet, strong evidence has been presented in favor of an SET mechanism in the course of Reformatsky reagent formation. Also Hg_2Cl_2 ,⁴⁰ triethyl boron,^{41–44} cerium (III) salts,^{3,45–47} disodium telluride,³ titanium(II) chloride/copper,⁴⁸ titanium (IV) chloride,^{42,49} (*o*-tolyl)₃P/TiCl₄,⁵⁰ titanium tetra iodide,⁵¹ titanocene (III) chloride,⁵² trialkylantimony/iodine, tributyl(phenyl)stannyl-lithium and diethyl aluminum chloride⁵³ are recognized examples of additives or inductors for Reformatsky reactions. In the presence of trimethylsilyl triflate, the 1,4-addition of a Reformatsky reagent on carbonyls α,β -unsaturated was observed.⁵⁴ A benzotriazole-mediated Reformatsky reaction has emerged as an interesting alternative for use with fluorinated precursors,^{55,56} as well as with non-fluorinated analogs.⁵⁷

2.2. Metals or derivatives other than zinc

There are a number of well-documented reports of Reformatsky-type reactions that use metals, metal salts, or



Scheme 6. Yields: 44–88%; ratio **22/23**: up to 0:100.

complexes other than zinc. A number of these variants have permitted interesting improvements, particularly in terms of some degree of stereocontrol. The chromium-Reformatsky reaction,⁵⁸ the SmI_2 -promoted Reformatsky reaction (Kagan reagent),⁵⁹ and the indium-mediated Reformatsky reactions⁶⁰ (among several others) deserve special comment.

2.2.1. The chromium-Reformatsky reaction.⁵⁸ α -Halo-ketones, esters, nitriles and other Reformatsky substrates (e.g., **19**) readily react with chromium (II) salts, usually chromium dichloride, in the presence of carbonyl compounds via the intermediates **20** and **21** to give the corresponding aldol compounds **22** and **23** (Scheme 6), whereas crotonates react with high regioselectivity to yield α -products.⁶¹

These reactions usually proceed in THF, DMF or acetonitrile, with 2-haloketones and vinylogous compounds being the most reactive substrates, whereas esters (especially acetates) are much less reactive. Under the usual conditions alkyl, alkenyl and aryl halides are not useful substrates for the chromium-Reformatsky reaction. Unreactive electrophiles in intermolecular reactions include alkyl, alkenyl and aryl halides, esters, amides, imides, imines, immonium salts and α -halocarbonyl compounds; usually unreactive (<5%) are nitriles, Michael acceptors, allyl and benzyl halides, sulfoxides and others. Aldehydes are preferred more than 50–100 times over methyl ketones, and even more so over higher ketones (>200:1) which thus can be used as solvent.

A significant diastereoselectivity is an important feature of the chromium Reformatsky reaction, usually favoring the *syn* products **23** when aldehydes are used as electrophiles, particularly if large α' -residues are used, and this will be explained in Section 4.4.^{61–64} However, by making use of chiral auxiliaries, excellent diastereomeric excess of *anti*

products **22** has been possible.^{65,66} Yields are often higher through LiI addition and by switching from THF to acetonitrile solvent.^{58,67} The nature and bulkiness of the aldehyde seems to have little influence on the reactions and most yields are in 80–98% range.⁶⁷ Even aldehydes prone to self-condensation (i.e., phenylacetaldehyde) react cleanly, while the reaction with ketones yields the kinetic cross aldol products with one or two new quaternary centers. The reaction does not require special activation of the reagent and proceeds at room temperature and is suitable for microscale preparations.⁶¹ The chromium (II) Reformatsky reaction was recently used as a key step in the synthesis of fragments and the complete carbon backbone of the novel diterpenoid tonantzitlone.⁶⁸

2.2.2. The samarium diiodide-promoted Reformatsky reaction (Kagan reagent). The one-electron reducing agent, samarium diiodide, has shown a remarkable versatility in promoting numerous synthetic transformations.⁵⁹ Because of its moderate oxidation potential and high oxophilicity, this divalent lanthanide reagent displays in general functional group selectivity in the reduction step and leads to the formation of products with high diastereoselectivities. Starting with α -bromoesters alone in THF, the reaction conditions can be controlled in such a way that a Reformatsky-type self-condensation takes place affording β -ketoesters,⁶⁹ or by further reduction it leads to β -hydroxyesters. $\text{Sm}(\text{OTf})_2$ has proved to be reactive as well to promote Reformatsky-type reactions using α -haloesters or Peterson-type reactions using halides containing a silyl group.⁷⁰ The samarium diiodide-promoted Reformatsky reaction has been used extensively and efficiently to promote intramolecular Reformatsky reactions leading to medium- and large-sized carbocycles. Also, much of the progress on stereo control in Reformatsky-type reactions has been performed via the use of SmI_2 , particularly for diastereoselective cyclizations. Section 4.3 illustrates several examples of SmI_2 -cyclizations and other

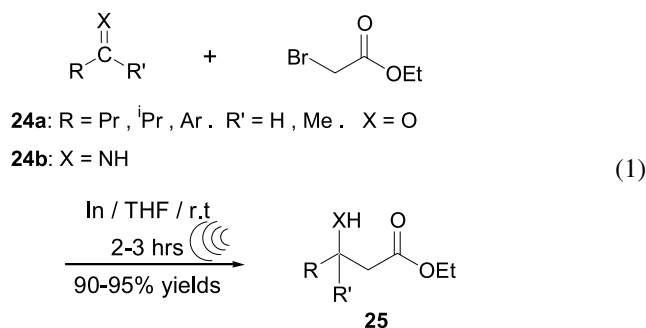
intramolecular Reformatsky reactions, whereas Section 4.4 will discuss the stereochemical aspects of SmI_2 -promoted Reformatsky reactions.

Samarium diiodide or metallic samarium can also be used in combination with other metals. For example, moderate to good yields of β -hydroxy ketones can be obtained from the reaction of α -bromoacetophenone with aldehydes in THF– H_2O under mild and neutral conditions in the presence of cadmium(II) chloride–metallic samarium⁷¹ or bismuth (III) chloride–metallic samarium.⁷²

Lanthanide salts other than samarium derived ones, in combination with lithium benzenetellurolate (PhTeLi) demonstrated good effectiveness for promotion of Reformatsky reactions to afford moderate yields and diastereoselectivity of β -hydroxyesters. Lanthanide chloride or lanthanide trifluoromethanesulfonates,⁷³ or CeCl_3 ⁷⁴ have been used for this purpose, the reaction proceeding with the latter reagents smoothly under mild conditions to give 47–95% yields of β -hydroxy esters. Such reagents were particularly efficient when sterically hindered and enolizable ketones were used. Mischmetall, an alloy of the light lanthanides, has been introduced recently for several kinds of organic procedures including Reformatsky-type reactions.⁷⁵

2.2.3. The indium-mediated Reformatsky reaction.

Indium has emerged as an interesting alternative with respect to Reformatsky reactions. Organoindium reagents have been used in a variety of transformations,^{60,76,77} including allylations, Reformatsky reactions and cyclopropanations, some with a significant degree of diastereoselectivity when using external chiral aminoalcohols such as cinchonine or chinchonidine.⁷⁸ The sonochemical Reformatsky reaction of aldehydes or ketones **24a** with ethyl α -bromoesters in the presence of indium affords β -hydroxyesters **25** in good to excellent yields under mild conditions (Eq. 1).^{79,80}



Interesting advantages of the Reformatsky-type reactions using indium derive from the fact that (i) the organoindium reagent is not reduced by hydroxyl groups (so, the reaction might be conducted in aqueous media),⁸¹ and (ii) the reaction of aldehyde/ketone mixtures proceeds chemoselectively with exclusive addition to the aldehyde.⁷⁹ The analogous indium-mediated reaction of α -haloesters with imines **24b** instead of aldehydes produces 3,4-disubstituted β -lactams **26**.⁸² A highly stereoselective synthesis of 4-octulose derivatives was recently reported via a key indium-mediated Reformatsky reaction using bromoacetone nitrile.⁸³

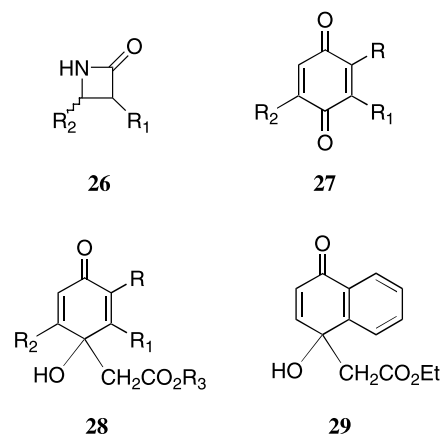


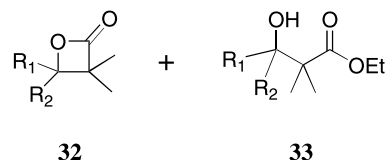
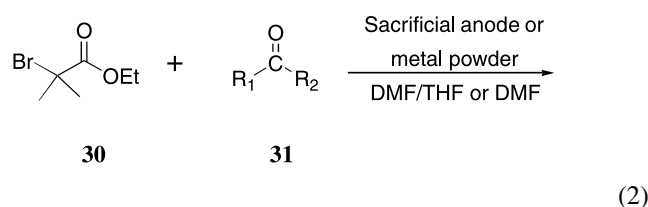
Table 1. Comparison of results from classical and electrochemically supported Reformatsky reactions induced by indium or zinc

Substrate 31 R_1	R_2	Reformatsky reaction conditions and results					
		Indium			Zinc		
		Method	Yield(%) 32 + 33	Ratio 32/33	Method	Yield(%) 32 + 33	Ratio 32/33
Me	Me	Anode	36	7:93	Anode	35	0:100
		Powder	59	19:81	Powder	43	0:100
Et	Et	Anode	61	>99:1	Anode	72	46:54
		Powder	88	100:0	Powder	75	12:88
$-(\text{CH}_2)_5-$		Anode	84	95:5	Anode	76	2:98
		Powder	83	77:23	Powder	96	0:100
Et	Ph	Anode	46	>99:1	Anode	89	100:0
		Powder	72	94:6	Powder	67	36:64
<i>n</i> -Pr	Ph	Anode	56	99:1	Anode	88	100:0
		Powder	76	97:3	Powder	95	62:38
<i>n</i> -Bu	Ph	Anode	38	>99:1	Anode	85	99:1
		Powder	78	99:1	Powder	53	96:4
Et	H	Anode	37	0:100	Anode		
<i>n</i> -Bu	H	Anode	50	0:100	Anode		
Ph	H	Anode	93	0:100	Anode		

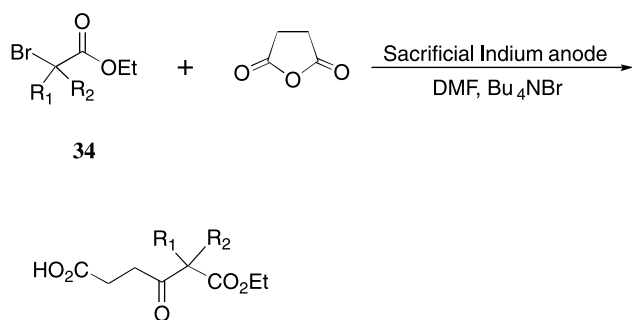
See Ref. 85.

Application of the indium-mediated Reformatsky reaction has been extended to *p*-quinones **27** [$R=R_1=H$, $R_2=H$, Me, MeO] with methyl and ethyl iodoacetate giving good yields of *p*-quinols **28** under mild conditions.⁸⁴ In this manner, naturally occurring quinol esters such as jacaranone **29** are conveniently prepared in a one-pot synthesis.

Indium-promoted, electrochemically-supported Reformatsky reactions of α -bromoalkanoates **30** (typically ethyl α -bromoisobutyrate) with ketones and aldehydes **31** produce di-, tri- and tetra-substituted β -lactones **32**⁸⁵ and negligible amounts of the respective β -hydroxy adducts **33** (Eq. 2 and Table 1). With few exceptions, the yield of β -lactones is significantly lower if zinc is used instead of indium.⁸⁵



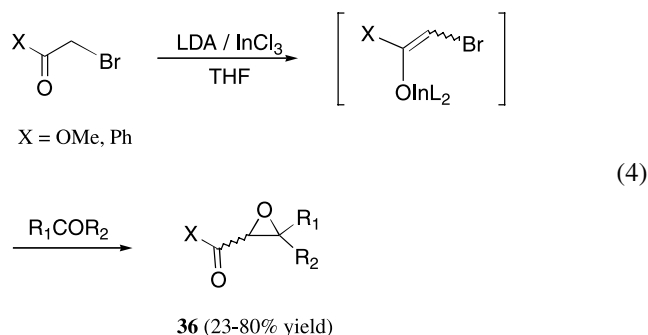
In a similar fashion, the use of zinc, tin, aluminum, indium and iron as sacrificial anodes in an electrochemically assisted Reformatsky reaction of ethyl 2-bromoalkanoates **34** with succinic anhydride produces in all cases the expected 1-ethyl 3-oxohexanedioates **35** in good to moderate yields (Eq. 3).⁸⁶ A recent report establishes that the iron-mediated electrochemical Reformatsky reaction of α -chloroesters with carbonyl compounds efficiently produces β -hydroxyesters, and the reaction is also applicable for the generation of β -hydroxy ketones or nitriles.⁸⁷



35a: $R_1=R_2=\text{Me}$, 80% yield

35b: $R_1=\text{Et}$; $R_2=\text{H}$, 74% yield

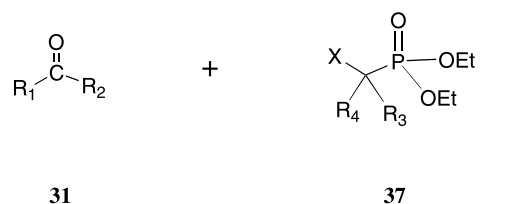
Indium trichloride has been used instead of metallic indium in Reformatsky-type reactions on enamines, and the reaction may be accelerated by addition of acetic acid, although reported yields are only moderate.⁸⁸ Reformatsky- and Darzens-type reactions of indium give rise to epoxy esters **36** in 23–80% yields with selectivities of 52:48 to 82:18 for the *trans*-epoxide stereo isomers (Eq. 4).⁸⁹



2.2.4. Reformatsky-type reactions involving other metals. Reformatsky-type reactions are possible with a lot of metals other than zinc or in combination with it. The electro reductive coupling of α -chloro esters or α -chloro nitriles with carbonyl compounds by means of a sacrificial zinc anode and a nickel catalyst has been reported.⁹⁰ This method has also been applied to fluorinated Reformatsky reagents generated from methyl chlorodifluoroacetate.⁹¹

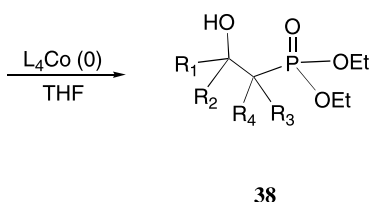
At least one example is known of the use of aluminum metal in combination with bismuth (III) chloride to promote a Reformatsky reaction.⁹² Moreover, use of low-valent metals for Reformatsky-type reactions has provided interesting alternatives. For example, the use of scandium (III) triflate in the presence of triphenylphosphine,⁹³ metal gallium,⁹⁴ or germanium metal⁹⁵ to promote Reformatsky reactions has been reported. In the latter case, germanium metal was prepared in situ by reduction of germanium (II) iodide with potassium metal, these reactions being achieved with high stereocontrol (see Section 4.4).

Cobalt is also being used extensively. Low-valent complexes of cobalt and phosphorus donor ligands (phosphines and phosphinites) proved to be efficient mediators in Reformatsky-type reactions.⁹⁶ This method was used in Reformatsky-type additions of α -halophosphonates **37** to carbonyl compounds **31** to yield β -hydroxyphosphonates **38** (Eq. 5)⁹⁷ and in additions of α -halocarbonyls **39** to (*R*)-2,3-*O*-cyclohexylideneglycerinaldehyde **40** to afford diastereomeric α -hydroxycarbonyls **41** with acceptable *anti* diastereoselectivity (Eq. 6).⁹⁸



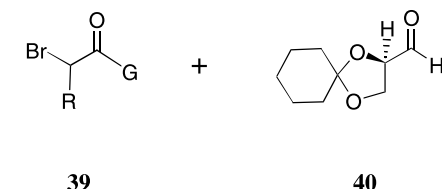
$R_1, R_2 = \text{H, alkyl, aryl}$ $R_3, R_4 = \text{H, F; X = Cl, Br, I}$

(5)



$L = \text{Me}_3\text{P}$

Yields: 22–81%

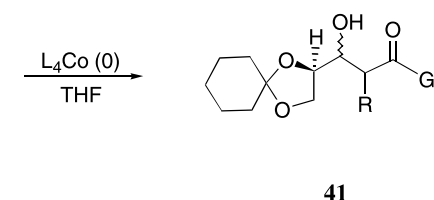


$R = \text{H, CH}_2\text{CH}_2$

$G = \text{alkyl, aryl, OMe}$

$R, G = \text{CH}_2\text{CH}_2\text{OC(O)O}$

(6)



$L = \text{Me}_3\text{P, Ph}_3\text{P}$

Yields= 50–90%

3,4-*anti* / 3,4-*syn* = 67:33 to 98:2

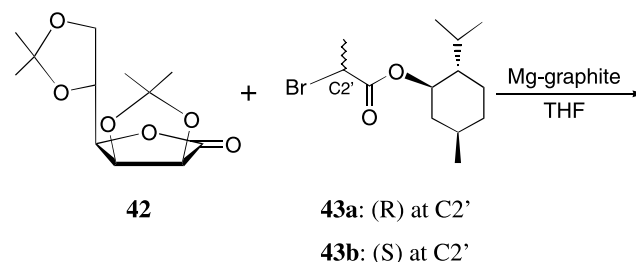
2,3-*anti* / 2,3-*syn* = 57:43 to 78:22

Cobalt–phosphine complex directed Reformatsky methodology can be applied stereospecifically^{99,100} by using chiral auxiliaries, as will be discussed in Section 4.4.

A mild and efficient Reformatsky-type reaction promoted by diethyl zinc with catalytic amounts of $\text{RhCl}(\text{PPh}_3)_3$ has been reported to be useful for both inter- and intramolecular condensations,¹⁰¹ even when using imines as electrophiles.¹⁰²

On the other hand, synthetic highly active metal–graphite surface compounds (for instance magnesium) have been utilized,¹⁰³ with the method being applied to the stereoselective synthesis of α -substituted ulosonic acid derivatives **44** via Reformatsky reactions of chiral precursors **43** with aldonolactones **42** (Eq. 7).¹⁰⁴ This approach to ulosonic

acids represents an alternative method to the SmI_2 -promoted approach.¹⁰⁵

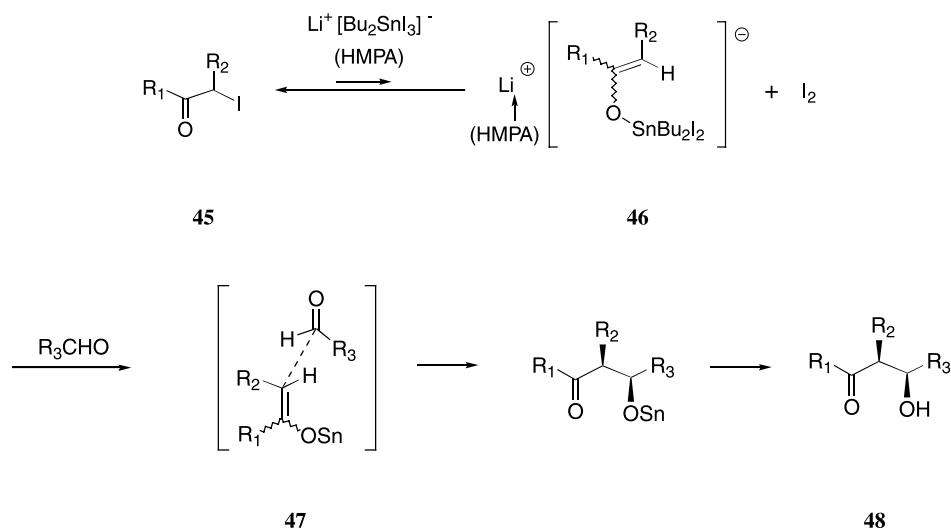


70–86% yields. Ratio (R)/(S) at C2' = 60:40

(7)

Reformatsky-type reactions are also possible with calcium reagents by using calcium atom–THF co-condensates¹⁰⁶ or manganese.^{107,108} Although manganese is a relatively abundant metal and has stronger potential reduction than zinc, it has rarely been utilized in organic synthesis, with a thin but tightly bound oxide layer on its surface being mainly responsible for its low reactivity. The metal-oxide coating is effectively removed by treatment with Me_3SiCl .^{109,110} Manganese metal (activated by Me_3SiCl and PbCl_2 or Et_2AlCl) proved to be a very efficient Reformatsky mediator.¹⁰⁷ The effect of lead on manganese is opposite to its effect on zinc where a catalytic amount of lead deactivates the metal.¹⁰⁹ Another method of activation of manganese was introduced recently,¹¹¹ consisting of the generation of manganese metal in situ, starting from Li_2MnCl_4 (derived from MnCl_2 and LiCl) and magnesium turnings (pretreated with 1,2-dibromoethane).¹¹¹ This method was used for a manganese-mediated reaction of α -haloesters ($X = \text{Cl, Br, I}$) with aldehydes, affording the expected β -hydroxy esters in good yield (78–94%) but with very low stereoselectivity.

Tin metal,^{32,86} tin (II) chloride,¹¹² tin (II) triflate,¹¹³ or tin complexes^{114,115} promote Reformatsky reactions efficiently. For example, the combination of hexabutyldistannane ($\text{Bu}_3\text{SnSnBu}_3$) with dibutyldiiodotin (Bu_2SnI_2) is an effective reagent for Reformatsky-type reactions.¹¹⁴ In its reaction with aldehydes, aldol-type products were obtained in 90% yields with a 54:46 ratio of *erythro*- and *threo* adducts. Even greater diastereoselectivity was obtained in Reformatsky-type reactions promoted by a tin iodide ate complex $\text{Li}^+[n\text{-Bu}_2\text{SnI}_3]$ **46** generated from equimolar amounts of Bu_2SnI_2 and LiI in THF/HMPA (Scheme 7).¹¹⁵ This complex promotes the reaction of α -iodoketones **45** with aldehydes to afford good to excellent yields of the expected products **48** with a diastereoselectivity of up 95:5



$\text{R}_1 = \text{Ph, Me, Et, H; R}_2 = \text{Me, H, n-Bu; R}_3 = \text{alkyl, aryl}$

Yields of **48** = 37–93%

Syn/Anti ratio = 83:17 to 95:5

Scheme 7.

in favor of the *syn* isomers, which is explained in terms of a transition state **47**.

Typical procedure.¹¹⁵ To a THF (1 mL) solution of *n*-Bu₂SnI₂ (0.95 g, 2 mmol) and LiI (0.266 g, 2 mmol) was added HMPA (0.360 g, 2 mmol) at 6 °C (bath temperature), and the solution was stirred for 5 min. To the solution was added α -iodopropiophenone (0.520 g, 2 mmol) and benzaldehyde (0.106 g, 1 mmol). The solution was stirred at 6 °C for 12 h. MeOH (3 mL) was added to quench the reaction and volatiles were removed under reduced pressure. Work-up on the crude product by column chromatography (eluting with hexanes/ethyl acetate 1:1) completed the process.

Other complexes are also able to promote Reformatsky-type reactions. For example, the Fe(CO)₅-promoted reaction of α -bromocarboxylates with aldehydes and ketones has been reported,¹¹⁶ and experimental conditions may be controlled to induce ketone: O addition instead of C:C addition if desired.¹¹⁷ It is also reported that a tetrakis(triphenylphosphine) Pd(0) complex may function as a catalyst for coupling vinyl iodides¹¹⁸ or aryl iodides¹¹⁹ with Reformatsky reagents.

A Reformatsky-type aldol reaction of α -chloroketones with aldehydes has been achieved recently, using reactive barium as a low-valent metal,¹²⁰ and it was considered very effective for obtaining β -hydroxy ketones.

3. Non-conventional reaction conditions

A Reformatsky reaction is run typically at temperatures ranging from room temperature to the boiling point of the solvent in which the reaction is carried out. However, there are numerous reports of Reformatsky reactions (as with a lot of other organic reactions) run under non-conventional

reaction conditions involving ultrasound, high pressure and microwave heating.¹²¹ Solvent-free reactions¹²² or aqueous reactions¹²³ offer obvious advantages which come from avoiding the need for flammable or anhydrous solvents, while mitigating environmental and economic concerns. Under these conditions, reactive functional groups, such as hydroxyl and carboxylic acid can be tolerated in a substrate without the need for time-consuming protection-deprotection schemes. The non-solvent reaction has some additional advantage because solvent removal by distillation is also not necessary. Reformatsky and Luche reactions were found to proceed efficiently in the absence of solvent¹²⁴ by mixing aldehyde or ketone with bromo-compound and Zn–NH₄Cl and by keeping the mixture at room temperature for 1–4 h. Dry medium reaction under microwave irradiation was applied to an alkali metal fluoride-mediated silyl-Reformatsky reaction.¹²⁵ Solid-phase methods have been applied also to Reformatsky reactions and other derivatizations involving C60/C70 fullerenes,^{126,127} and to the synthesis of α,α -difluoro- β -amino acids via the benzotriazole-mediated Reformatsky reaction with resin-bounded amino acids, aldehydes and the α,α -difluoro Reformatsky reagent.⁵⁶ Recently, solid-phase chemical synthesis of phosphonoacetate and thiophosphonoacetate oligodeoxy-nucleotides was accomplished.¹²⁸

Reformatsky reactions of α -bromoesters and carbonyl compounds, via a radical chain mechanism, can be carried out in concentrated aqueous salt solutions using catalytic amounts of benzoyl peroxide or peracids without any co-solvent,^{6,129} with preparative yields being comparable to those of the traditional procedure. Similarly, aqueous medium zinc-mediated Reformatsky reactions of aliphatic aldehydes have been performed successfully by the addition of BF₃·OEt₂ and benzoyl peroxide.⁴³ Aqueous samarium/CdCl₂-,⁷¹ BiCl₃/Sm-,⁷² BiCl₃/Al-,⁹² and indium-mediated⁸¹ Reformatsky reactions have been already mentioned.

The Reformatsky reaction of α -bromoesters with carbonyl compounds, with reproducible yields of β -hydroxyesters (10–92%) are also possible using a mild and effective method of electrochemical zinc activation, based on the cathodic reduction of a catalytic amount of zinc bromide in acetonitrile.¹³⁰ The method is also applicable to the coupling of α -bromoesters with anhydrides. Reformatsky reactions may also be assisted electrochemically,⁸⁶ using zinc, tin, aluminum, indium,⁸⁵ iron⁸⁷ and nickel^{90,131} as sacrificial anodes. The amount of current consumed for the dissolution of 1 equiv of anode metal strongly depends on the kind of metal. Whereas Al and Fe require 1 faraday per equivalent, Sn demands only 0.9 faraday per equivalent, and Zn and In require an even smaller amount of current. It is assumed that in the case of the last three metals, the electrochemical Reformatsky reaction is overlapped by a current-independent normal Reformatsky reaction, in which the anode acts as activated metal.⁸⁶ The electrosynthesis of various β -hydroxy esters, β -hydroxy nitriles, and 2,3-epoxy esters was successfully achieved under extremely mild conditions with a zinc/nickel combination.⁹⁰ The mechanism involved reduction of a Ni(II) complex to a Ni(0) complex, oxidative addition to the α -chloro ester to the Ni(0) complex, and a Zn(II)/Ni(II) exchange, leading to an organozinc Reformatsky reagent.

As already mentioned, ultrasound has become a valuable tool to promote a number of reactions¹²¹ and large-scale Reformatsky reactions may be efficiently conducted in special reactors equipped with an ultrasound generator.^{132,133} The reaction under HIU is found to be concentration dependent, and is particularly useful for substrates with increasing steric demands.¹³⁴ It has been recently used for the synthesis of β -amino esters and β -lactams.¹³⁵ In some cases, it is possible to exert some degree of stereo control, as in the case of the 73:27 *threo*/*erythro* ratio obtained (67% yield) in the ultrasound-promoted Reformatsky-type reaction of trifluoroacetaldehyde with ethyl α -bromopropionate in the presence of Zn.¹³⁶ Ultrasound was also a key tool in the stereoselective Reformatsky reaction (e.e. >98%) of tricarbonyl (η^6 -arene)chromium (0) complexes, which afford β -amino esters and β -lactams¹³⁷ (stereochemistry associated to Reformatsky reactions will be discussed in Section 4.4).

Chiral induction and ultrasound acceleration on Reformatsky reactions have been performed using both non-fluorinated¹⁹ and fluorinated α -bromo precursors.^{138–140} Ultrasound turned out to be an interesting alternative for the generation and storage of the organozinc derived from ethyl bromodifluoroacetate.¹⁴⁰ This is pretty important, specifically for the reaction with NO₂ bearing carbonyl compounds, which require a previous preparation of the organozinc. Several additional reports are known on sonochemically assisted Reformatsky reactions¹⁴¹ and particularly indium-mediated ones.^{79,80} The production of highly reactive Zn powder by simultaneous electrochemical and ultrasound assistance appears to be especially important in medium- or large-scale processes involving Reformatsky reactions, Barbier allylation, alkyne reduction, reductive dehalogenation and 4-nitrophenyl ester cleavage.¹⁴² The results show that Zn powders produced sono-electrochemically are highly reactive.

3.1. Supported metal-induced Reformatsky reactions

'High-surface' alkali metals can be conveniently prepared via deposition of corresponding metals on various supports such as NaCl, polyethylene, polypropylene and cross-linked polystyrene from their solutions in liquid ammonia. Alkali metals deposited on polymeric supports can be stored as stable suspensions in inert solvents, and addition of the suspension of supported alkali metal to a solution of zinc chloride gave an active zinc on polymeric support. This form of zinc is as active as zinc prepared by potassium graphite²⁸ or lithium suspension reduction.²⁹ Polyethylene supported activated zinc was used for Barbier reactions and for the Reformatsky reaction of α -bromoesters and carbonyl compounds, producing β -hydroxyesters in excellent yields.¹⁴³ This method offers considerable benefits: the metal surface is highly developed so its activity is high, the reaction proceeds selectively and the handling of reagents in such form as well as the separation of the reaction products from the support is a facile operation. Under these conditions, the reactions turned out to be equally efficient from α -chloro esters and α -bromo esters.

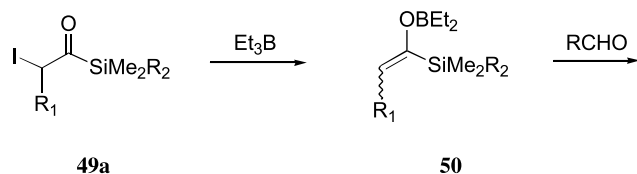
4. Reformatsky reactions involving non-fluorinated substrates

This section deals with application of the classical Reformatsky and Reformatsky-type reactions to a number of synthetic procedures involving non-fluorinated substrates. A lot of progress has been made in this field, particularly with regard to (a) the extension of the reaction to unconventional electrophiles and precursors, (b) the development of alternative methods that provided significant improvements in yields or stereocontrol, shortening of reaction times, simplification of work-up procedures, and (c) the development of multi-step syntheses involving Reformatsky reactions.

4.1. Precursors of Reformatsky reagents

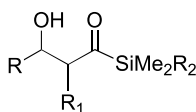
The most familiar version of the Reformatsky reaction consists of the in situ generation of an organozinc intermediate derived from α -halocarbonyl compounds, α -haloesters, α -halothioesters, α -halonitriles, α -haloamides, α -haloimides, α -halo anhydrides,¹⁴⁴ α -halo-lactones or α -halophosphonates.^{3,48,50,93,95,97,98} By treatment with activated zinc or other metals, these substrates generate carbon nucleophiles that add to a large number of electrophiles. Typically these reactions take place in polar solvents, particularly THF, DME, acetonitrile, dioxane, DMF, DMSO, HMPT or mixtures of such solvents.³ Examples are also known of Reformatsky reactions being carried out in halogenated solvents, such as CH₂Cl₂.^{42,48,50,93,95,145–148}

α -Haloacylsilanes **49** (X = Cl, Br, I) react with aldehydes in the presence of triethylboron with moderate to good diastereoselectivity, usually favoring the *erythro* isomer **51** (Eq. 8).⁴¹



$R^1 = \text{H, Me, n-C}_6\text{H}_{13}$; $R^2 = \text{Me, } t\text{-Bu, Ph}$

$R = \text{Me, Ph, } t\text{-Bu, PhCH:CH}$



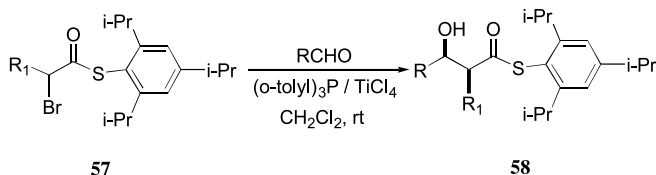
Yields: 32–79%

erythro/threo ratio: 6:94 to 96:4

(8)

The so-called Tischenko reaction¹⁴⁹ sometimes accompanies the Reformatsky reaction of benzaldehyde with acylsilanes **49** affording 1,3-diol monoester **56** via the Cannizzaro-like intermediates **52** through **55** (Scheme 8).⁴¹

Aryl α -hydroxythioesters **58** have been prepared from aldehydes and α -bromothioesters **57** in the presence of a (*o*-tolyl)₃P/TiCl₄ combination (Eq. 9),⁵⁰ with high yields and good *syn/anti* stereoselectivity (up to 95:5).

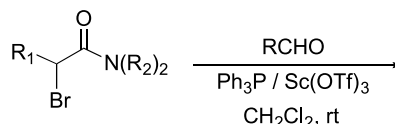


Yields: up to 93%. *Syn/anti* ratios up to 98:2

(9)

As expected, the larger the sulfenyl group in the thioester, the greater the diastereoselectivity obtained. The same

observation holds for the Reformatsky reaction of α -bromoamides **59** with aldehydes in the presence of triphenylphosphine/scandium (III) trifluoromethanesulfonate (Eq. 10)⁹³ or triphenylphosphine/germanium tetrachloride combinations,⁹⁵ which gives rise to α -hydroxyamides **60** with *syn/anti* ratios of up to 98:2⁹³ or up to 99:1.⁹⁵

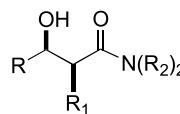


59a, $R^2 = \text{Ph}$

59b, $R^2R^2 = (\text{CH}_2)_5$

59c, $R^2R^2 = (\text{COOCH}_2\text{CH}_2)$

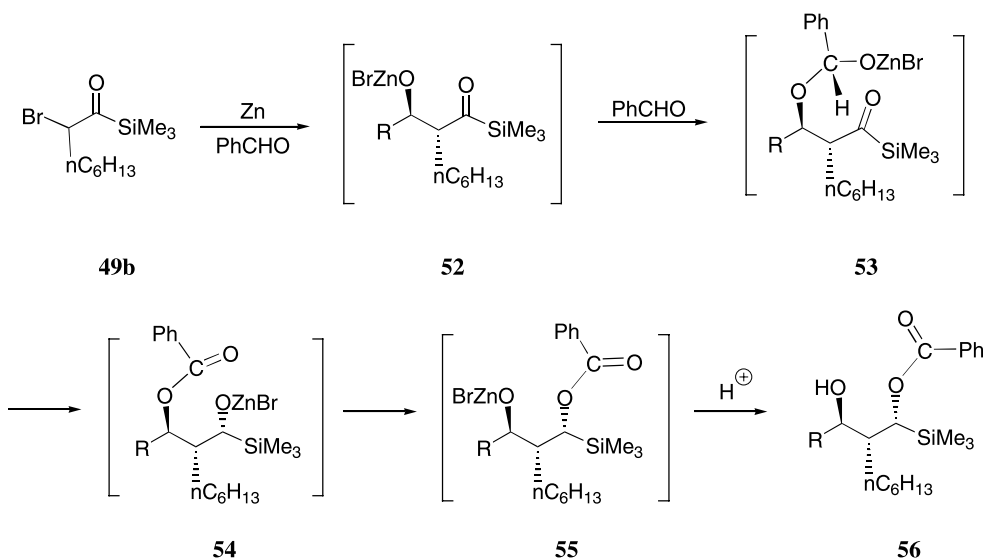
(10)



Yields: up to 93%. *Syn/anti* ratios up to 98:2

In addition to the above-mentioned examples, Table 2 contains other selected examples of classical intermolecular Reformatsky reactions affording β -hydroxy-derivatives or the respective dehydrated products.

On the other hand, in the presence of zinc, α -bromo-malonates **62** react with *N*-arylpiperidine-2-thiones **61** to give the respective α,β -unsaturated derivatives **63** (Eq. 11). This unprecedented Reformatsky reaction has been used for the synthesis of tricyclic analogues **64** of quinolone antibacterial agents.¹⁵⁹



Scheme 8.

Table 2. Selected examples of classical Reformatsky reactions

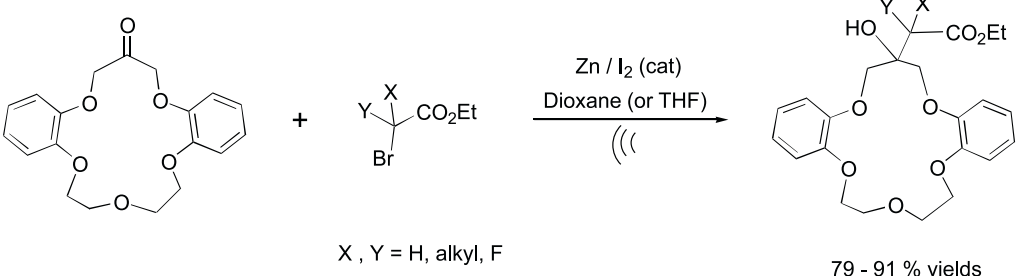
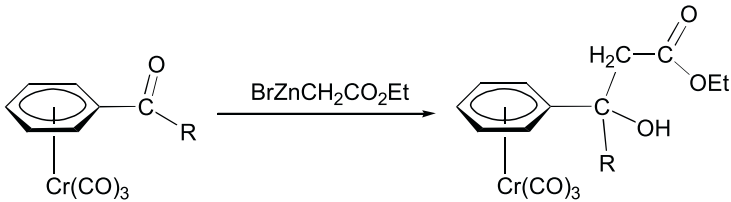
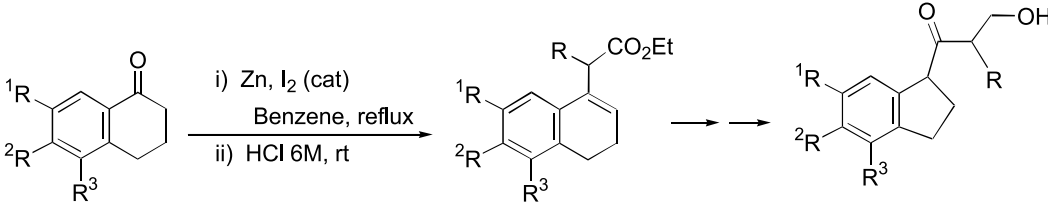
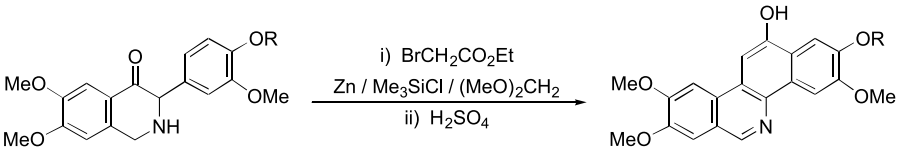
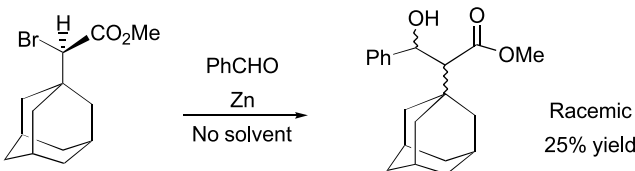
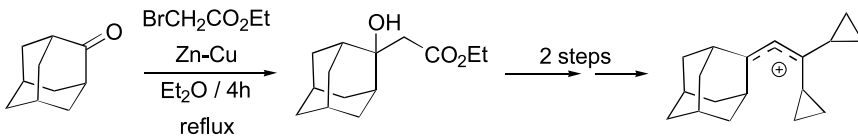
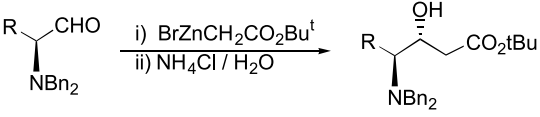
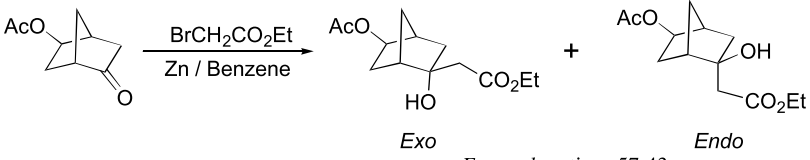
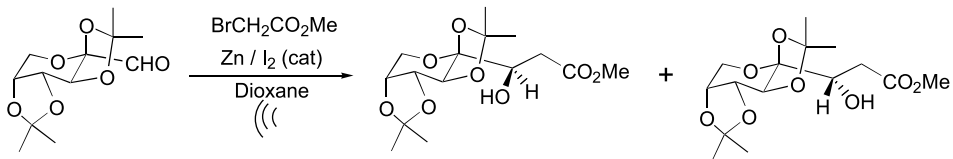
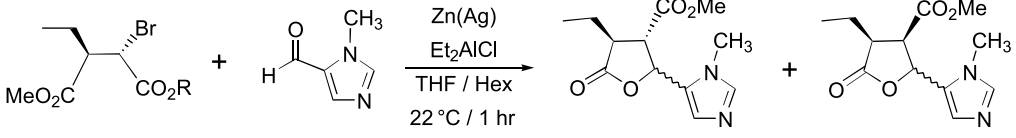
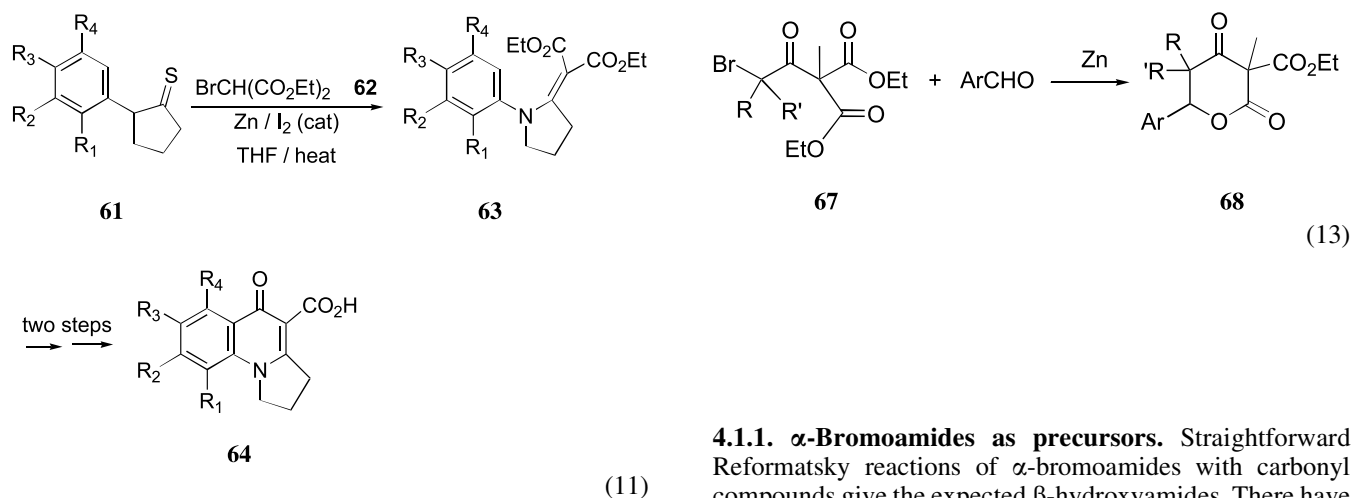
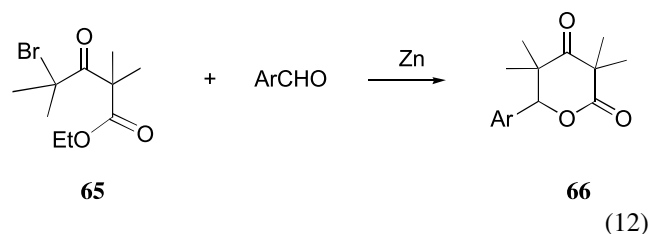
Entry	Reactions	Refs.
1	 X, Y = H, alkyl, F 79–91 % yields	141
2	 Analog reactions performed on formyl and acetyl derivatives of ($\eta^5\text{-C}_5\text{H}_5$)Mn(CO)₃ and ($\eta^5\text{-C}_5\text{H}_5$)Re(CO)₃	150
3		151
4		152
5	 Racemic 25% yield	153
6		154
7	 Yields= 30–87%; <i>anti:syn</i> ratio= 62:38 to 80:20	155
8	 Exo Endo Exo:endo ratio = 57:43	156

Table 2 (continued)

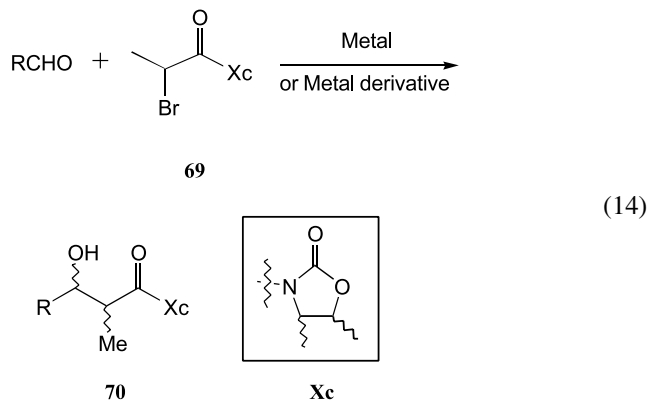
Entry	Reactions	Refs.
9	 Product ratio = 10 : 1	157
10	 Combined yield = 94% ; Product ratio = 91:9	158

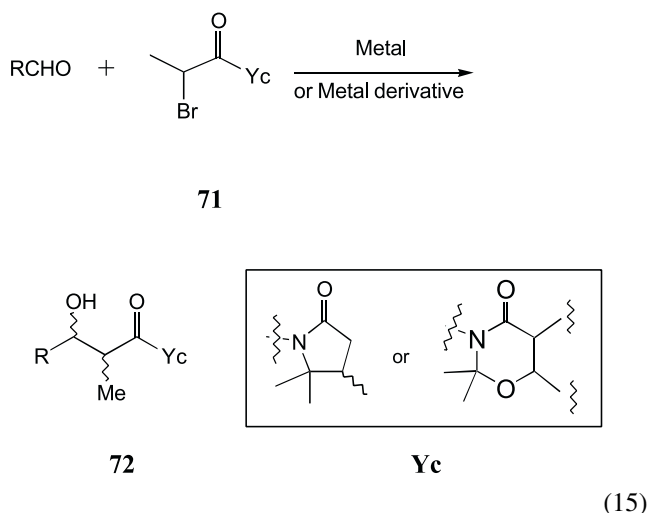


γ -Bromoderivatives of β -ketoesters or malonates have also been used as substrates for Reformatsky reactions with aldehydes, producing δ -lactones that result from ring closure of the respective adducts. In this manner, 6-aryl substituted tetrahydro-3,3,5,5-tetramethyl-2,4-pyranones **66** or 6-substituted 3-alkoxycarbonyl-3,5,5-trimethyl-2,3,5,6-tetrahydropyran-2,4-diones **68** have been synthesized, starting from aromatic aldehydes and alkyl 4-bromo-2,2,4-trimethyl-3-oxopentanoates **65** (Eq. 12)¹⁶⁰ or dialkyl 2-methyl-2-(2-bromoacyl)malonates **67**,¹⁶¹ respectively (Eq. 13).



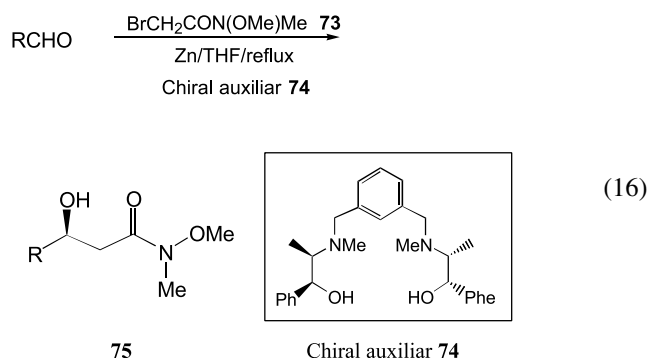
4.1.1. α -Bromoamides as precursors. Straightforward Reformatsky reactions of α -bromoamides with carbonyl compounds give the expected β -hydroxyamides. There have been some interesting developments on Reformatsky reactions of α -bromoacyl-derived oxazolidones **69** (Eq. 14)^{65,66,73,74,93,95,99,162–166} or α -bromoacyl-derived oxazinones **71** (Eq. 15)^{167–170} as special classes of α -bromoamides to be added to carbonyl compounds.



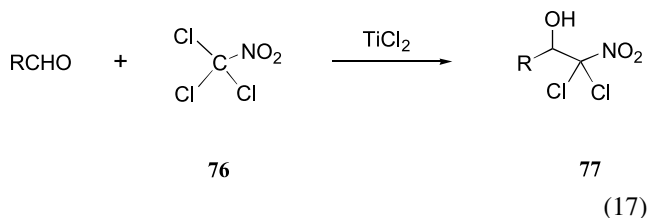


Introduction of chiral auxiliaries based on these amide moieties has been the key to significant improvements in terms of stereocontrol of the Reformatsky reactions induced by zinc,^{162–164,167,168} chromium,^{65,66} SmI₂ or other lanthanides,^{73,74,166} germanium-⁹⁵ or cobalt.⁹⁹ Stereochemical aspects of the Reformatsky reactions of fluorinated precursors, making use of chiral auxiliaries, will be discussed in Section 5.1.

Chiral β-hydroxy Weinreb amides **75** are obtained in a similar fashion by the Reformatsky reaction of α-bromo Weinreb amides **73** with carbonyls in the presence of chiral β-aminoalcohols **74** as auxiliaries (Eq. 16)¹⁷¹ (more details on asymmetric approaches are included in Section 4.4).

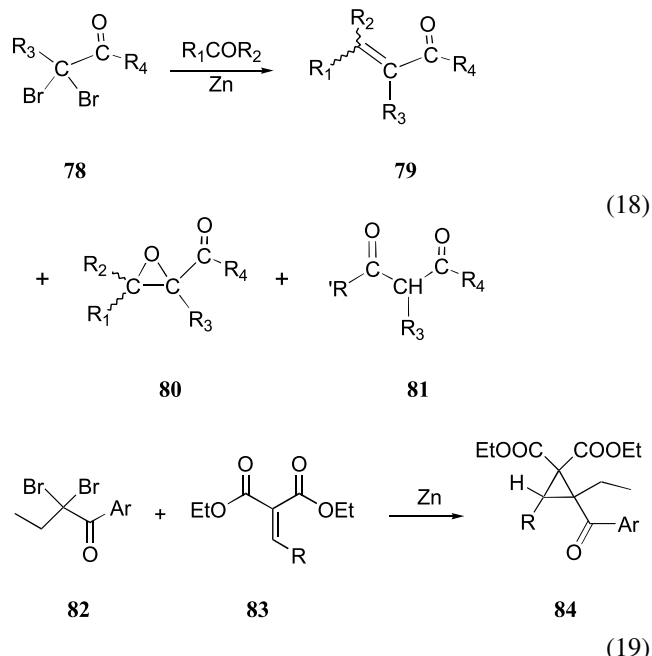


4.1.2. Polyhaloprecursors. In a Reformatsky-like reaction, trichloronitromethane **76** adds to aldehydes in the presence of tin(II) chloride to yield α,α-dichloro-α-nitroalcohols **77** (Eq. 17), with aliphatic aldehydes affording higher yields than aromatic aldehydes.¹¹²



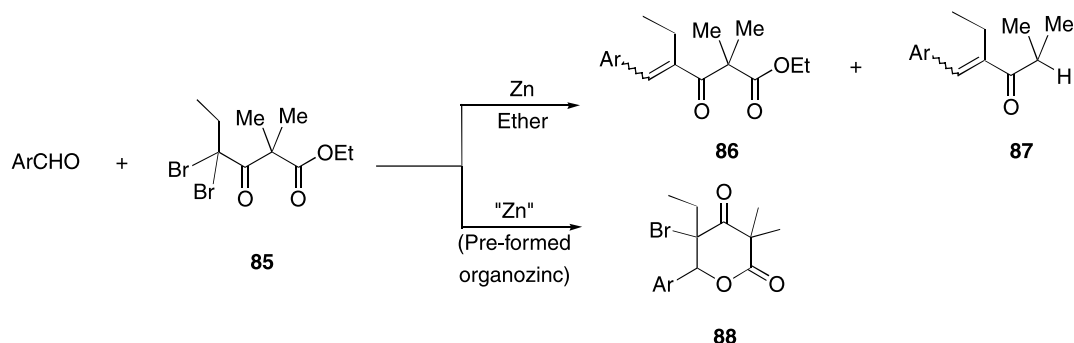
Zn-mediated reactions of α,α-dibromo ketones **78** with carbonyl compounds, depending on the starting reagents, the nature of the solvent, and the reaction conditions, give

rise to α,β-unsaturated ketones **79**, α,β-epoxyketones **80** or α-alkyl-β-diketones **81** (or mixtures of such products) (Eq. 18).¹⁷² However, 1-aryl-2,2-dibromobutane-1-ones **82** react with zinc and α-arylmethylidene-, or α-alkylidene-malonates **83** via a Reformatsky reaction followed by an internal malonic alkylation to give the respective cyclopropane-1,1-dicarboxylates **84**, which are formed mainly as Z isomers (Eq. 19).¹⁷³

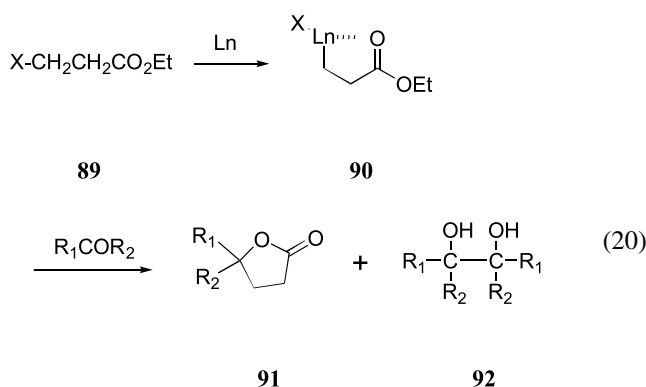


Reaction of γ,γ-dibromo-β-ketoesters (e.g., ethyl 4,4-dibromo-2,2-dimethyl-3-oxohexanoate) **85** with Zn and aromatic aldehydes, depending on the specific reaction conditions, affords the reduced non-halogenated α,β-unsaturated ester **86**, the respective decarboxylated ketone **87** or a mono-halogenated δ-lactone **88** (i.e., a pyrandione derivative) (Scheme 9).¹⁷⁴ This kind of reaction sometimes is prone to giving rearrangement products.

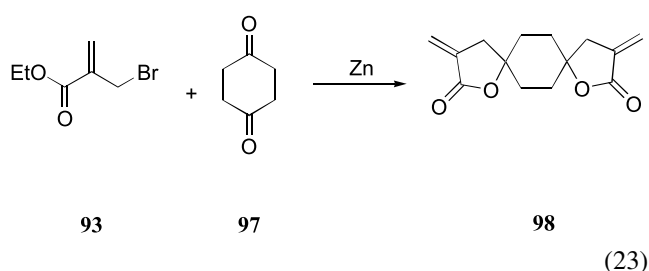
4.1.3. 'Remote' Reformatsky precursors. β-Metal enolates, commonly known as homoenolates, may be involved in Reformatsky like reactions as nucleophiles. Generation of homoenolates has been accomplished by several methods,¹⁷⁵ those being preferable that use a direct reaction from 3-halo carbonyl compounds and a metal. Zinc ester homoenolates, readily prepared in benzene–DMF or dimethylacetamide from ethyl 3-iodopropanoate by means of a zinc–copper couple, can be added to carbonyl compounds.¹⁷⁵ An improved procedure achieves the direct preparation of lanthanoid ester homoenolates **90** from 3-haloesters **89** and lanthanoid metals (La, Ce, Nd, Sm), and in turn **90** is added to carbonyl compounds without the need of additives or catalysts. In this manner, γ-lactones **91** have been prepared in good yields under mild conditions, even at room temperature (Eq. 20).¹⁷⁶ Pinacol derivatives **92** are often side products, which are generated by reaction with the carbonyl substrate.¹⁷⁶ As high as 63% yield of a pinacol **92** derived from acetophenone may be obtained as the only product if acetophenone is added to the homoenolate ester **90** previously prepared from the 3-iodoester **89** with Sm.¹⁷⁶



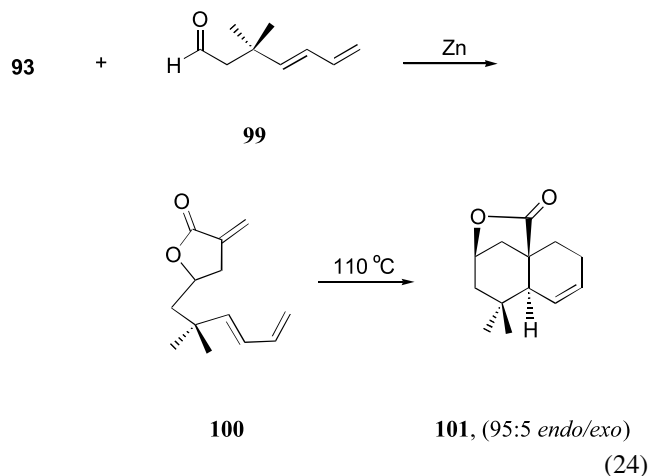
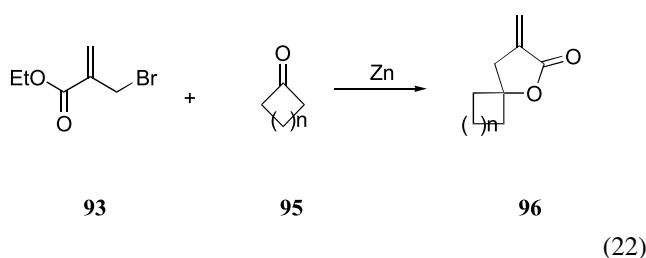
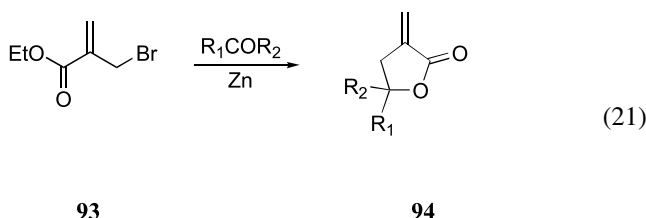
Scheme 9.



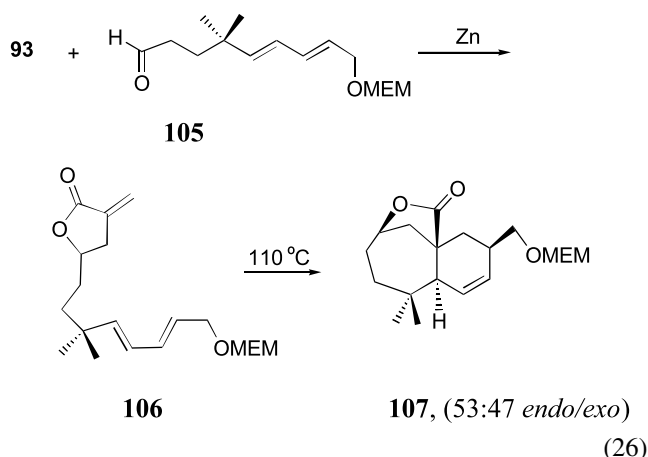
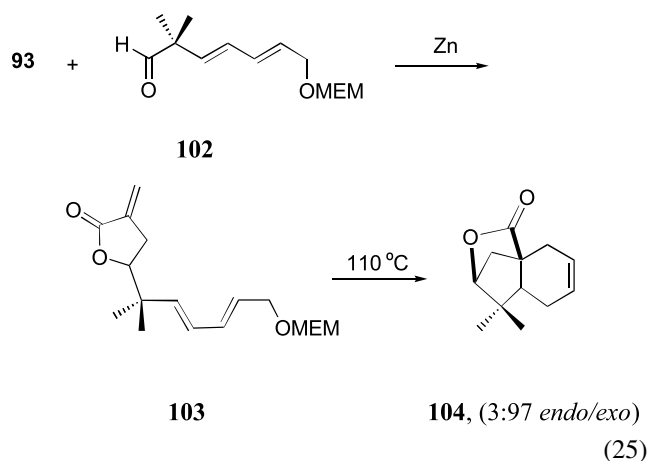
Combined % yields of (**91** + **92**) = 63–93%;
product ratio **91/92** = 66:34 to 93:7



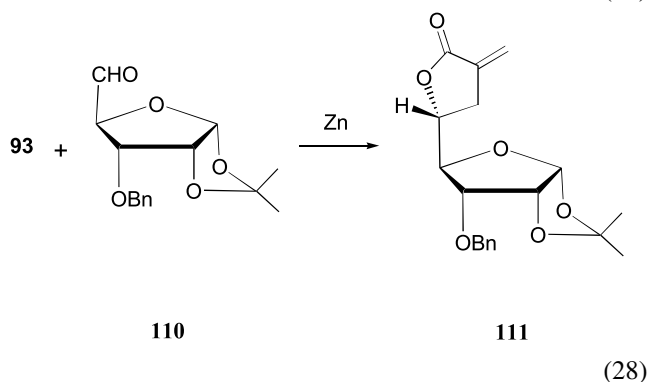
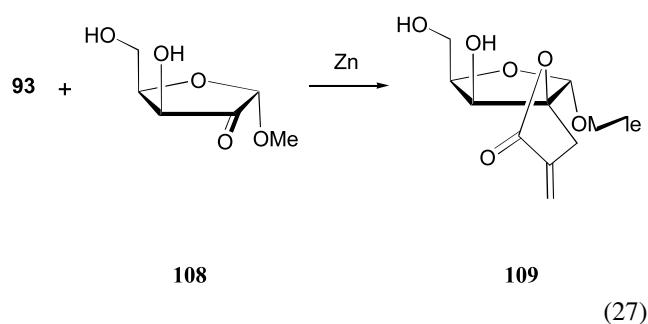
4.1.4. α -(Bromomethyl)acrylates as precursors. The Dreiding–Schmidt reaction¹⁷⁷ consists of the Zn-induced Reformatsky-like additions of α -(bromomethyl)acrylates **93** to carbonyl compounds followed by subsequent ring closure to form α -methylidene- γ -butyrolactones **94** (Eq. 21).¹⁷⁷ Although cyclic ketones **95** straightforwardly give rise to spiro derivatives such as **96** (Eq. 22),¹⁷⁷ it has been reported that piperidone does not generate the anticipated spiro α -methylene- γ -butyrolactones.¹⁷⁸ When starting from dicarbonyl compounds (for instance **97**), several spiro moieties like **98** may be formed in the same molecule (Eq. 23).¹⁷⁷



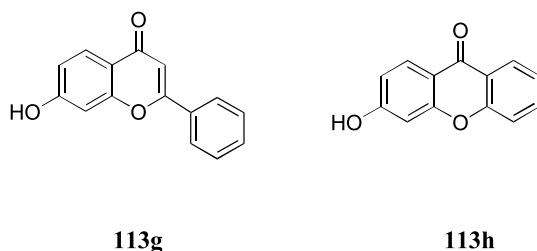
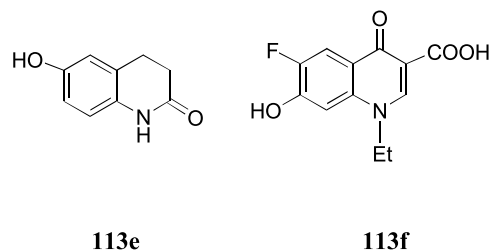
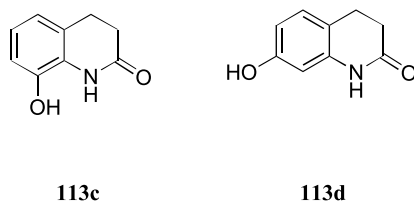
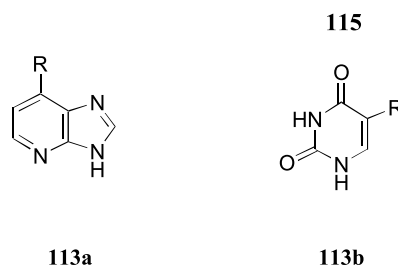
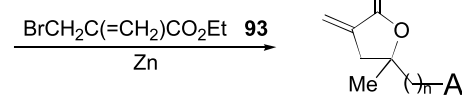
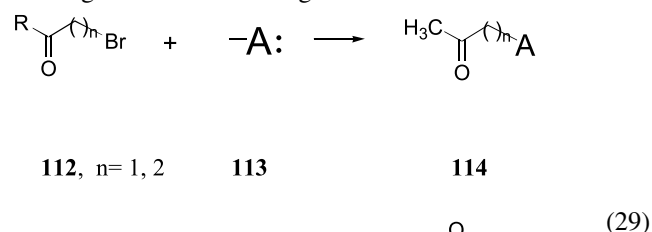
The preparation of α -methylene- γ -butyrolactones has been reviewed separately,¹⁷⁹ and additional comments and selected examples have been included in another recent review on Reformatsky reactions.³ Recent developments in applying this methodology have used dienals **99** to generate α -methylene- γ -butyrolactones **100** bearing a dienyl side chain,¹⁸⁰ useful substrates for preparing tricyclic cyclo-adducts **101** via a Diels–Alder reaction (Eq. 24),¹⁸¹ or the analogous tricyclic adducts **104** and **107** (Eqs. 25 and 26).¹⁸⁰ It has been observed that the size of the tether connecting the α -methylene- γ -butyrolactone moiety and the diene system defines the size of the resulting ring (Eqs. 24–26). It is also important to mention that highly functionalized tethers or use of a furan ring as diene system prevent the intramolecular Diels–Alder pathway. Also, sometimes the product is not the α -methylene- γ -butyrolactone moiety itself but the respective β -hydroxy adduct, which can be cyclized very easily with NaH .¹⁸⁰

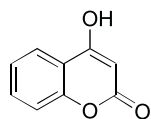


This use of α -(bromomethyl)acrylates for the generation of α -methylene- γ -butyrolactones has been extended to the preparation of sugar derivatives^{182,183} such as the antifungal active compounds **109** and **111**, starting from their respective carbonyl precursors **108** and **110** (Eqs. 27 and 28).¹⁸²

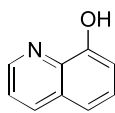


Moreover, this methodology has been exploited by attaching addends **A**, **113**, for example pyrimidine-¹⁸⁴ or purine-derived nucleic acid bases,¹⁸⁵ γ -aryloxy- (with aryl groups such as phenyl-derivatives, biphenyl, coumarin, flavone, xanthone, carbazole, dibenzofuran or quinolinone moieties),^{186,187} or steroidal residues¹⁸⁸ to the tail of ketones before the Drieding–Schmidt reaction is performed with the respective carbonyl derivatives **114** (Eq. 29). The resulting products **115** are useful for different kinds of biological activities, such as antifungal, antitumor, antiplatelet, anti-viral agents or vasorelaxing molecules.

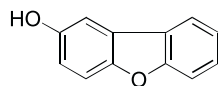




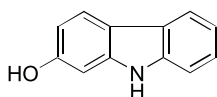
113i



113j

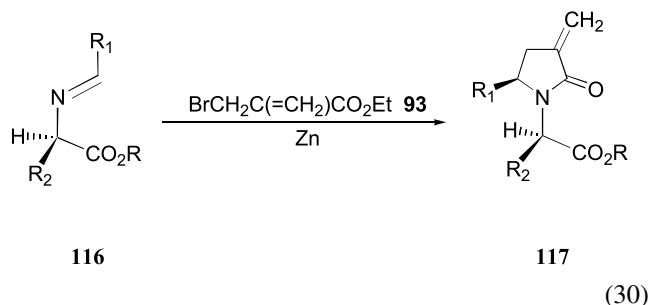


113k



113l

The Dreiding–Schmidt reaction, when applied to α -iminoiminoesters **116**, furnishes analogous α -methylene- γ -butyrolactams **117** with stereoselectivities as high as 100% e.e. (Eq. 30).¹⁸⁹

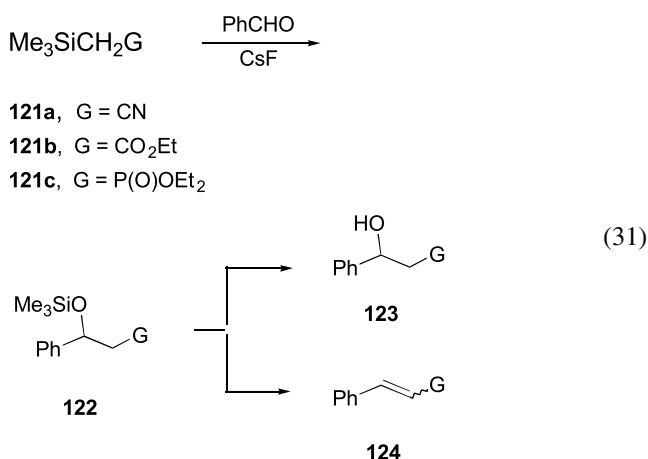


4.1.5. 4-Bromocrotonates and related precursors. It can be considered that precursors bearing halogen atoms in γ -position of α,β -unsaturated esters, with Zn, give rise to ambidentate organozinc intermediates which are able to be added to carbonyls (or to other electrophiles) either by γ - or α - attack (Scheme 10).

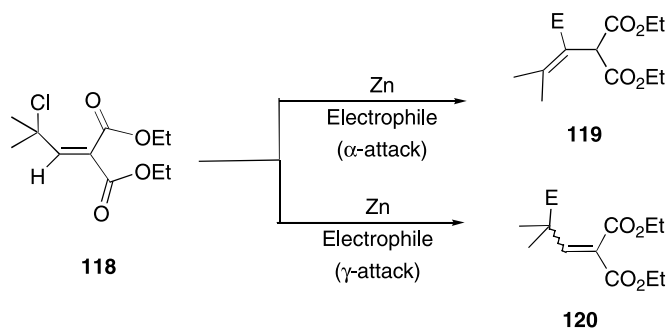
However, in practice several examples demonstrate that Reformatsky reagents derived from 4-bromocrotonates and 3-bromomethylcrotonates add to electrophiles regioselectively through the γ -carbon (see later),^{98,190–193} and some bromomethyl-substituted heteroaromatic compounds behave like 4-bromocrotonates.¹⁹⁴ The analogous α,β -unsaturated- γ -haloesters [for example 2-(2-chloroalkylidene)-malonates **118**], on quenching with electrophiles, give

products **119** or **120** of either α - or γ -attack, respectively (Scheme 10).¹⁹⁵

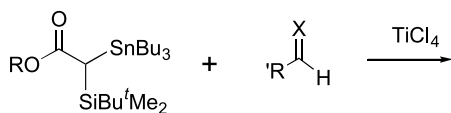
4.1.6. Precursors with leaving groups other than halogen atoms. Several substrates bearing leaving groups other than halogens may undergo reaction in a Reformatsky-like fashion, giving rise to β -hydroxy functionalized compounds. In these cases zinc or other metals needs to be replaced by salts or analogous agents to promote the reaction. For example α -trimethylsilyl-acetic or phosphonic compounds of the form $\text{Me}_3\text{SiCH}_2\text{G}$ ($\text{G}=\text{CN}$, CO_2Et , $\text{P}(\text{O})(\text{OEt})_2$) **121** condense with aromatic aldehydes in the presence of fluorides (KF , CsF , Bu_4NF) unsupported or supported,^{125,196} under microwave irradiation in heterogeneous dry media. The silyl-Reformatsky reaction leads to the O-silylated adduct **122**, which can either be subsequently hydrolyzed to give desilylated derivatives **123** or dehydrated to give the respective alkenes **124** (Eq. 31).^{125,197} Syntheses of α -methylene- γ -lactones are available by this method,¹⁹⁸ which represents an alternative approach to the Dreiding–Schmidt reaction.^{177–180}



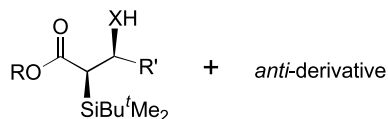
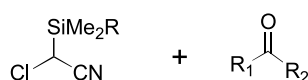
Similar reactions utilize α -silyl- α -stannylacetic derivatives **125** and aldehydes or aldimines **126** in the presence of TiCl_4 ¹⁹⁹ to diastereoselectively produce *syn* products **127** (Eq. 32), or zinc-induced Reformatsky–Peterson reactions of $\text{Me}_3\text{SiCHClCN}$ **128** with carbonyls to produce alkene-nitriles **129** (Eq. 33).²⁰⁰



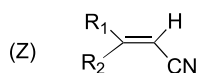
Scheme 10.

**125****126**, $X = O$ or $N\text{-Bn}$

(32)

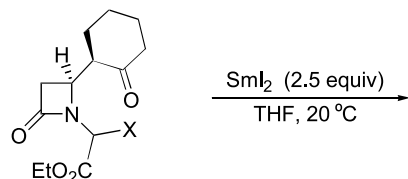
**127****128**

(33)

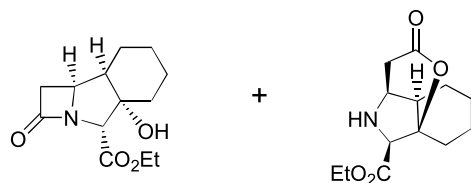
**129**, 65–99% yields

Z/E ratio: 58:42 to 100:0

In addition, precursors **130** bearing benzoyloxy,²⁰¹ or S-Pyr²⁰² groups may act as leaving groups in intramolecular SmI_2 -mediated Reformatsky-type reactions to afford the respective products **131** (Eq. 34), although reaction conditions may be controlled to favor rather the rearranged products **132**. SmI_2 -promoted intramolecular Reformatsky reactions will be mentioned in Section 4.3.

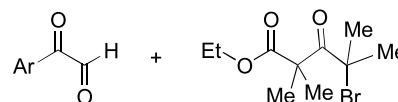
**130**, $X = \text{OBz}$ or S-Pyr

(34)

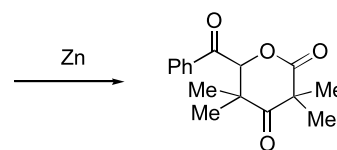
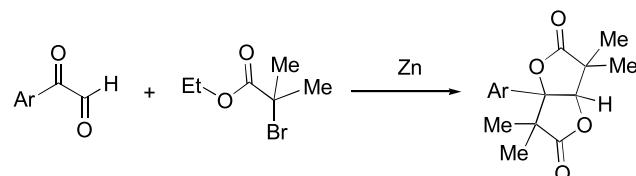
**131****132**

4.2. Electrophiles

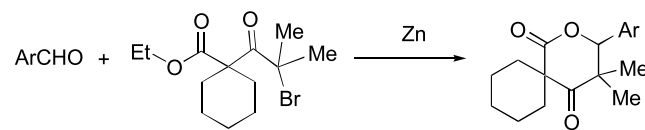
4.2.1. 'Classical' electrophiles. It is well recognized^{3,4,8} that Reformatsky reagents act as nucleophiles with a number of electrophiles including aldehydes, ketones, nitriles, phosphonates, amides and imides, affording the expected β -hydroxy- or α,β -unsaturated products. If key functional groups are appropriately located, adducts may spontaneously close up to γ -lactones,¹⁵⁸ or δ -lactones.²⁰³ Also, specific precursors give rise to monocyclic pyran-2,4-diones, for example, **135** (Eq. 35),^{160,174,204} dilactones, for example, **137** (Eq. 36),²⁰⁵ and pyran-2,4-diones bearing spiro moieties, for example, **139**, **141** and **143** (Eqs. 37–39).^{161,206–208}

**133****134**

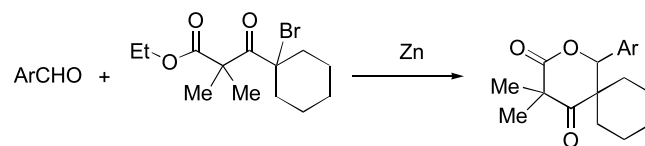
(35)

**135****133****136****137**

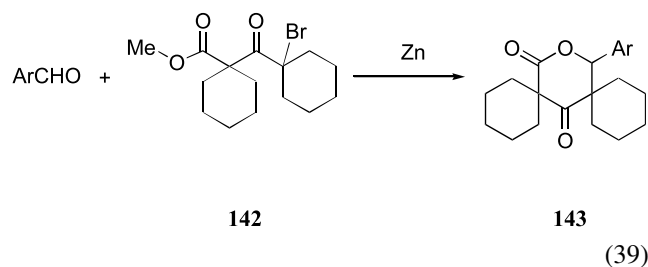
(36)

**138****139**

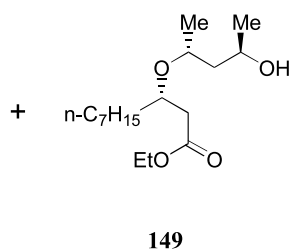
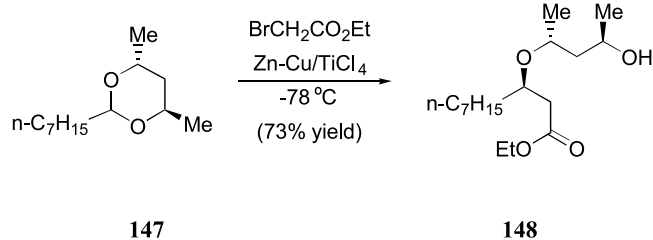
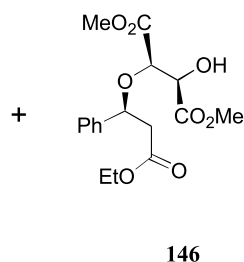
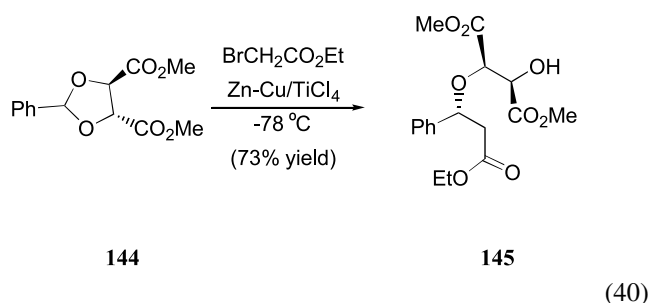
(37)

**140****141**

(38)

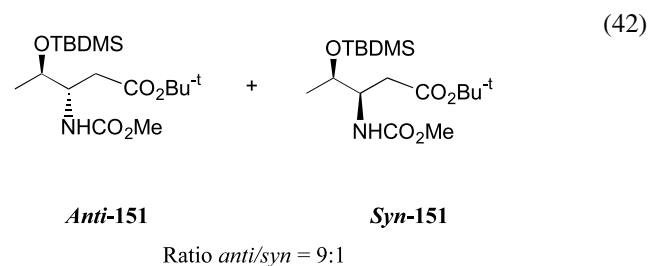
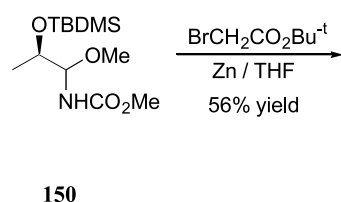


In the presence of Lewis acids such as TiCl_4 or $\text{BF}_3 \cdot \text{Et}_2\text{O}$, acetals may release an alkoxy group upon reaction with organozinc Reformatsky reagents providing 50–90% yields of β -alkoxyesters with variable levels of *erythro/threo* diastereoselectivity. Up to 84% e.e. is reported for reactions starting from chiral acetals.⁴² Thus, chiral six-membered- and five-membered-cyclic acetals (e.g., **144** and **147**) are opened by Reformatsky reagents (Eqs. 40 and 41).

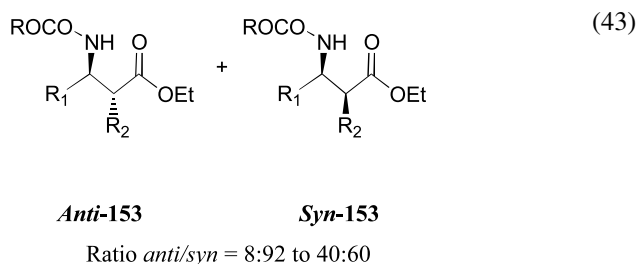
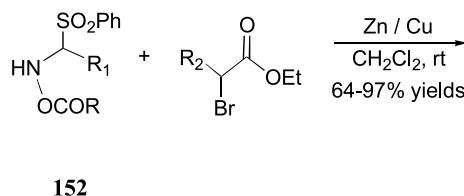


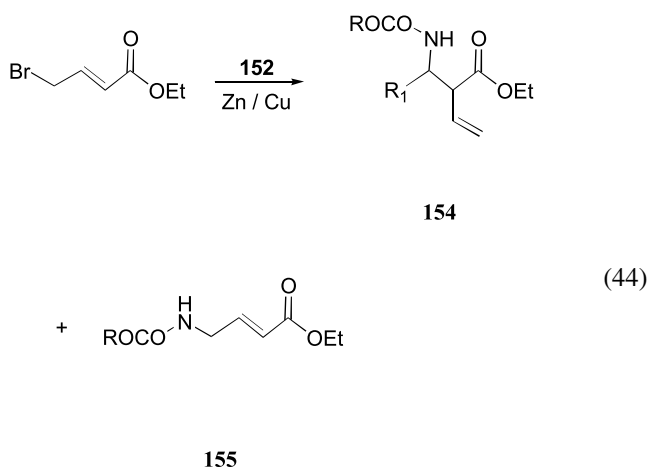
(41)

Likewise, a methoxy group present at the α -position in *N*-methoxycarbonyls (e.g., **150**) may be displaced by Reformatsky reagents or other organozinc compounds through a probable imine-mediated mechanism to furnish β -(*N*-methoxycarbonyl)aminoesters **151** with high *anti* diastereoselectivity, which can then be converted to β -amino esters (Eq. 42).



Displacement of a leaving group from the structurally related *N*-protected amidoalkylphenyl sulfones **152** can be effected by zinc-based Reformatsky reagents. The reaction has become an alternative entry to β -amino esters (e.g., **153**) with variable stereoselectivity (Eq. 43).¹⁹³ As such, α -amidoalkylphenyl sulfones are considered to be imine equivalents. The analogous reaction of **152** with 4-methylcrotonates gives rise to a mixture of **154** and **155** resulting from α - and γ -attack, respectively, with a modest regioselectivity (Eq. 44).





Ratio **155/154** = 49:51 to 30:70

1),⁴⁹ 4-acetoxy-2-azetidinones (entry 2),^{163,164,167–170,191} cyclic tertiary 2,2-bis(alkoxy)amines and 2-alkoxyimmonium ions bearing an acetal moiety (entries 3 and 4),¹⁹⁴ oxazolidinones (entry 5),^{145,210} and oxazinanes (entry 6).²¹¹ The reaction shown in entry 2 is reported to be a crucial step in the synthesis of key intermediates (e.g., **156**) to 1 β -carbapenems (e.g., **157**),^{163,164,167–170,191} while the reactions of entries 3 and 4 have been used for the synthesis of antimalarial quinoline-4-methanols such as **158**.¹⁹⁴

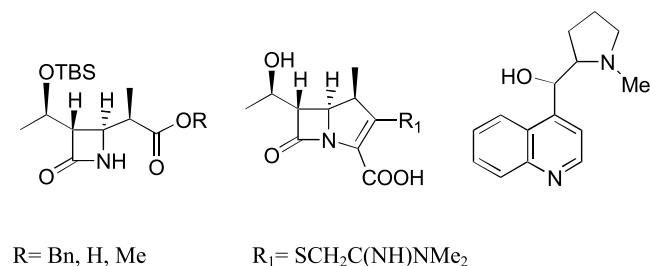
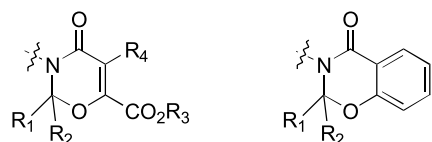
**156****157****158**

Table 3 summarizes some additional examples of Reformatsky reactions that make use of acetal-, *N*-acetal-derived, or structurally related substrates, such as 2-acetoxytetrahydrofuran (or 2-acetoxytetrahydropyran) derivatives (entry

Table 3. Selected examples of Reformatsky reactions involving acetal-related substrates

Entry	Reactions	Refs.
1	n = 1, 2	49
2 ^a	20–97% yield 89–98% diastereoselectivity	163, 164, 167–169, 191
3	58% yield	194
4	56–63% yield	194
5	60–92% yields [Redn]	145, 210
6	72–96% yields	211

^a In entry 2, the following groups hold for L.



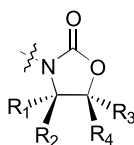
$R_1 = \text{Me, Bu, Bn, 4-furan}$

$R_2 = \text{Me, Bn}$

$R_3 = \text{Me, Et}$

$R_4 = \text{Ar}$

$R_1R_2 = (\text{CH}_2)_5$



$R_1 = \text{H, Me, } ^i\text{Pr, Bu, Bn}$

$R_2 = \text{H, Me, Bu, Bn}$

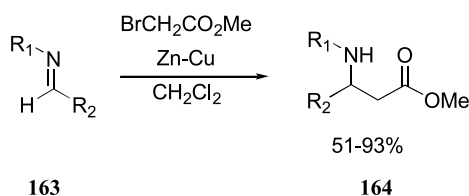
$R_3 = \text{H, Me}$

$R_4 = \text{H, Me}$

$R_3R_4 = (\text{CH}_2)_5$

On the other hand, α -oxoketene dithioacetals **159** leads to phenol derivatives **160** under classical Reformatsky conditions by using an excess of the organozinc precursor (Scheme 11).²¹² The reaction can be used for the synthesis of substituted ethyl 3-hydroxy-5-(methylthio)stilbene-carboxylates **161** and for the synthesis of 4-, and 4,5-disubstituted 6-(methylthio)pyran-2-ones **162** in moderate to good yields.

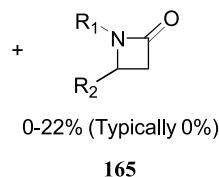
4.2.2. ‘Non-classical’ electrophiles. A previous review³ briefly mentioned some Reformatsky reactions involving ‘uncommon electrophiles’, such as azomethines, nitriles, acyl chlorides, anhydrides, lactones, esters, orthoformates, oxiranes, azirines, amins, nitrones, alkynes and inorganic acid chlorides. Also, as illustrated in Eqs. 45–48, Reformatsky reagents add to imines and related compounds, for example, **163** and **166** (Eqs. 45–48), in the same fashion as they do with aldehydes, affording β -amino carbonyl derivatives such as β -amino esters (like **164**) or β -lactams **165**, with β -amino esters being preferred especially when an *ortho*-substituent is present in an arylamine-derived imine (Eqs. 45 and 46).^{102,137,148,193,199,213–215}



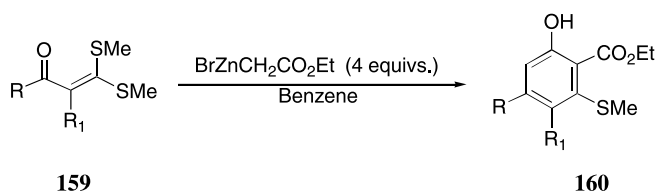
163

164

(45)

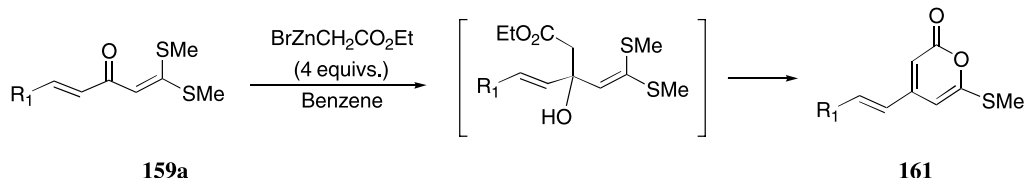


165



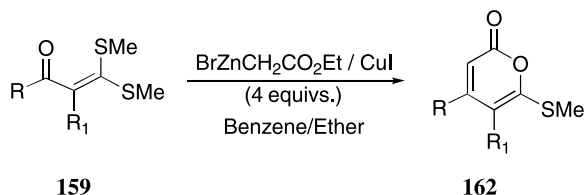
159

160



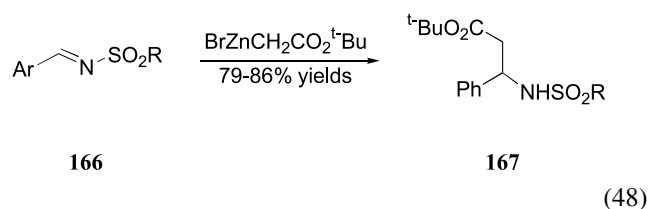
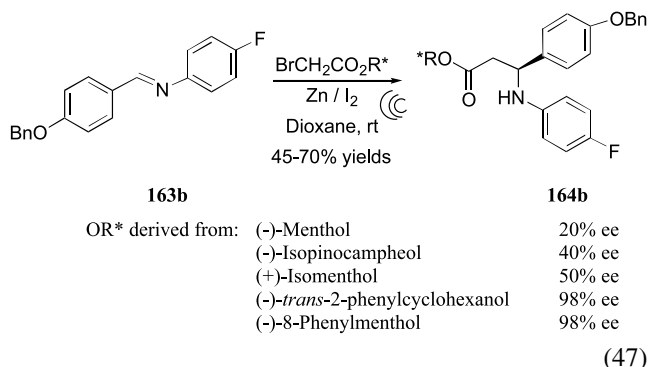
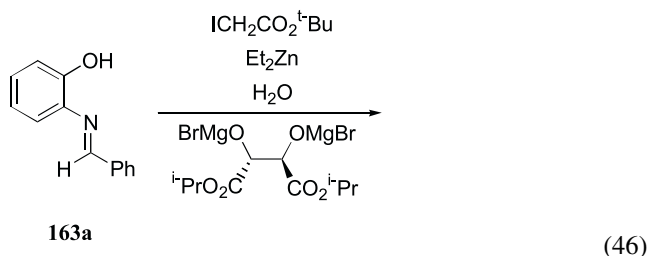
159a

161



159

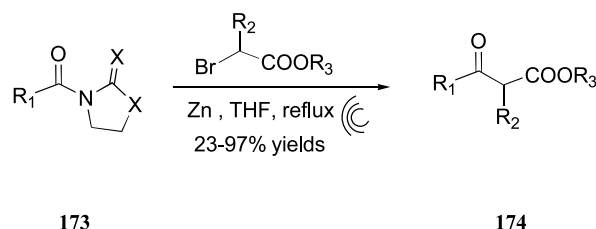
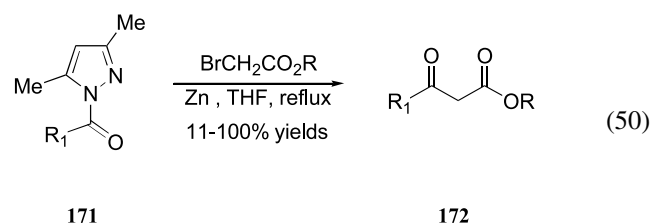
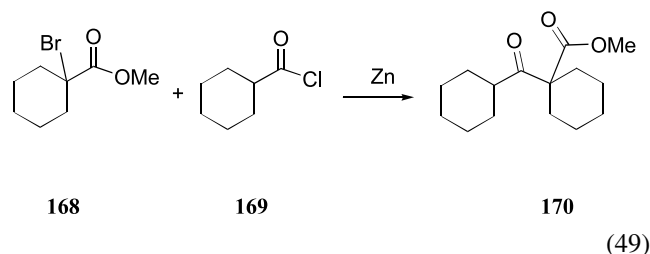
162



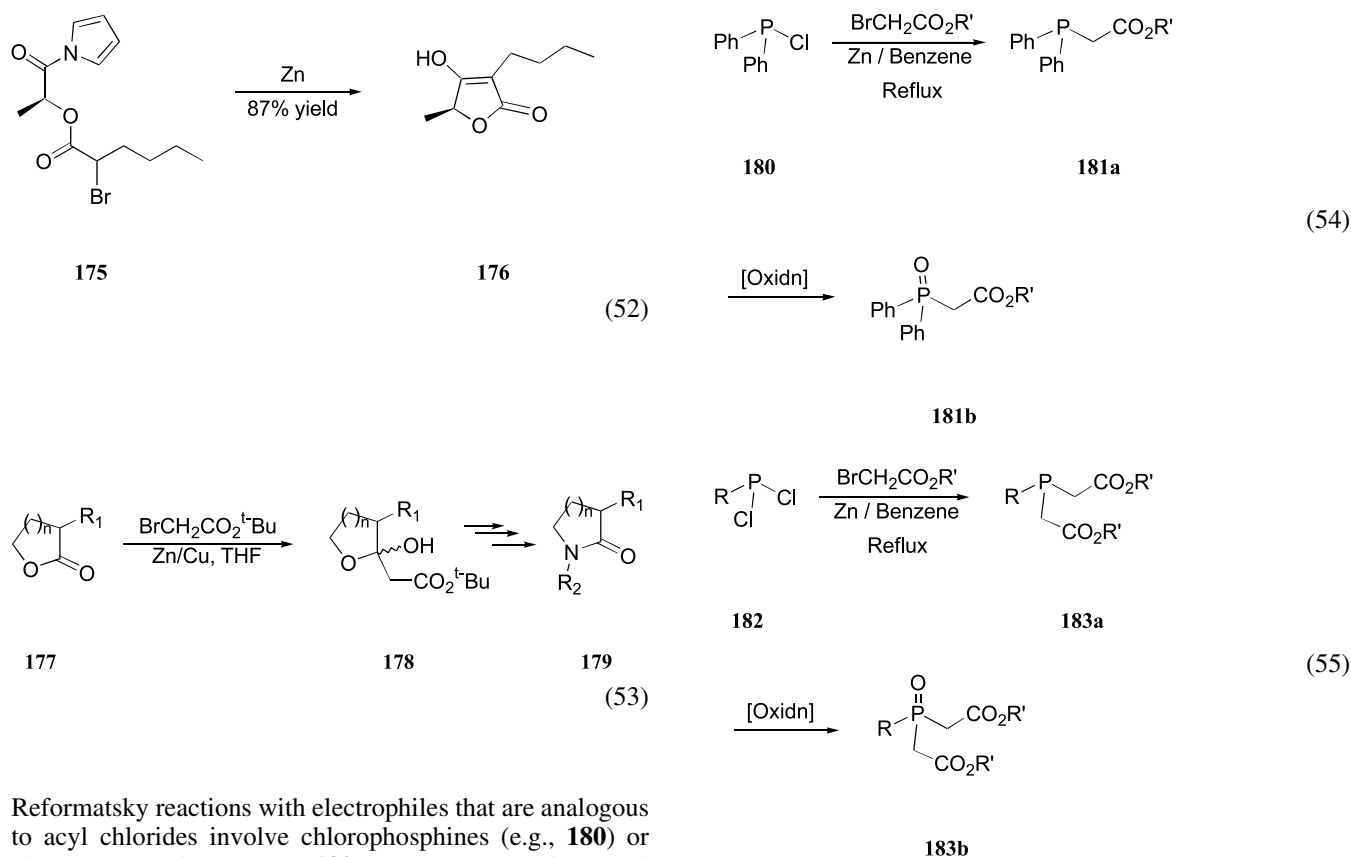
Sulfenamides **163a** (where $R_1 = \text{SPh}$) exhibit quite similar behavior and they also afford β -amino esters upon reaction with Reformatsky reagents.²¹⁶ Ordinary imines also condense with Reformatsky reagents in the presence of (1-trimethylsilyl)-benzotriazole, affording good yields of the corresponding β -amino esters.⁵⁷ The Reformatsky reaction with imines is commonly known as the Gilman-Speeter reaction,²¹⁷ and by further deprotection methods, the products may be readily converted to β -aminoacids. It was already mentioned (Eq. 30) that imines may be converted to α -methylene- γ -lactams by condensation with Reformatsky reagents derived from 2-(bromomethyl)acrylates via a Dreiding–Schmidt-type reaction.¹⁸⁹ Reformatsky reactions of imines involving fluorinated precursors and electrophiles will be mentioned in Section 5.

It is well recognized that leaving groups from acyl derivatives and related compounds are prone to be replaced also by Reformatsky reagents affording β -ketoesters,

β -keto-related compounds or their respective enols. A number of examples of straightforward Reformatsky reactions involving acyl chlorides,^{208,218,219} anhydrides,^{86,220} lactones,²²¹ and amides bearing potential leaving groups^{222–225} are reported, and some selected examples are shown in Eq. 49–53. Eq. 49 represents the first step for the preparation of some pyran-2,4-diones targets containing spiro systems, (e.g., **143** in Eq. 39),²⁰⁸ and the reaction depicted in Eq. 53 was recently used for the synthesis of heterocyclic enamines via the ring transformation reaction of γ -lactones **177**.²²¹ Some intramolecular Reformatsky reactions produce a Dieckman-like product or its enol, for example, **176** (Eq. 52), and this kind of Reformatsky–Claisen path which results from the good leaving ability of the pyrrole group is reported to be more efficient than the analogous Dieckman reaction.²²² As expected, oxazolidine- or oxazinane-based amides (e.g., **173**) react with Reformatsky reagents to afford β -ketoesters **174** (Eq. 51),²²⁵ and the same observation is true for amides (e.g., **171**) bearing good leaving groups (Eq. 50).^{225b}



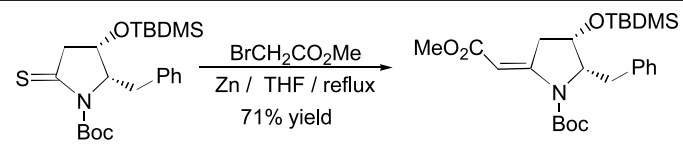
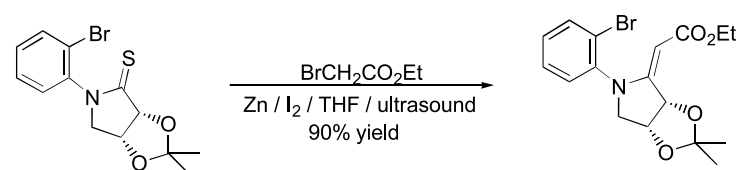
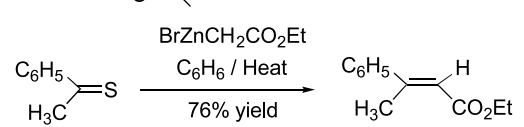
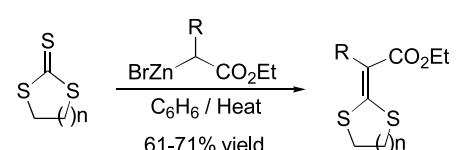
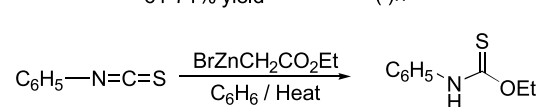
With X being: O, S



Reformatsky reactions with electrophiles that are analogous to acyl chlorides involve chlorophosphines (e.g., **180**) or dichlorophosphines (e.g., **182**). They react with alkyl bromoacetates in the presence of Zn to form (alkoxycarbonylmethyl)phosphines **181a** (Eq. 54) or bis-(alkoxycarbonylmethyl)phosphines **183a** (Eq. 55), which can be converted to the respective oxides **181b** or **183b** by oxidation.²²⁶

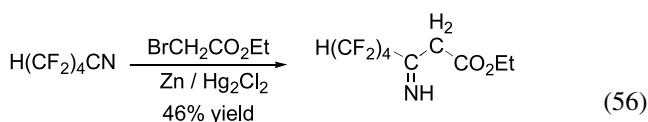
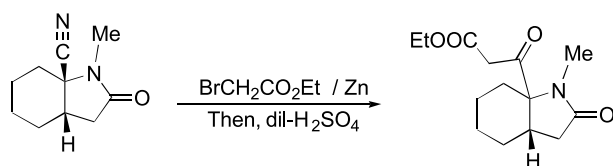
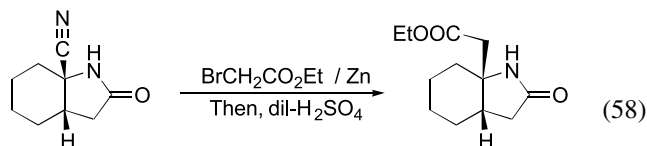
Reformatsky reactions involving thiocarbonyl compounds, particularly γ -thioalactams^{159,225,227,228} usually give rise to the respective elimination products. Organozinc reagents are shown to undergo facile C–C bond formation on trithiocarbonates, xanthates, thione and dithioesters through

Table 4. Selected examples of Reformatsky reactions involving thiocarbonyl derived substrates

Entry	Reactions	Refs.
1		227
2		228
3		229
4		229
5		229

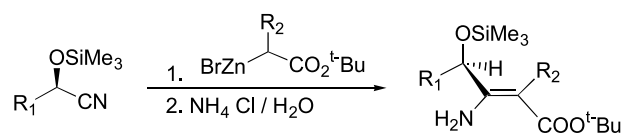
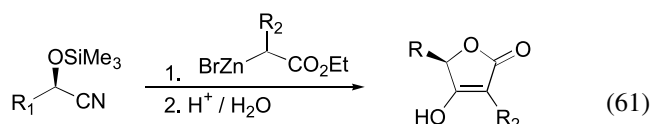
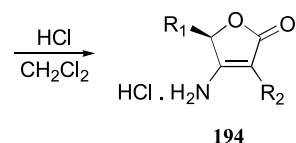
carbophilic addition yielding products formed by elimination of either sulfur or an alkylthio group.²²⁹ A list of selected examples appear in Table 4. As seen in entry 5, a rare Reformatsky reaction involving phenylisothiocyanate produces ethyl *N*-phenylthiocarbamate, probably through a ketene elimination on the intermediate Reformatsky adduct.²²⁹

Reformatsky reactions with nitriles as electrophiles, commonly known as the Blaise reaction,²³⁰ give rise to β -imino ester intermediates (e.g., **185**), which afford β -ketoesters (e.g., **187**) or the respective enols after hydrolytic work-up (Eqs. 56 and 57). If desired, the cyano group may be removed during the course of the Reformatsky reaction (Eqs. 58 and 59).^{231,232} Removal of the nitrile group starting from tertiary β -amino nitriles represents a very facile access to functionalized amines when a β -amino ester synthon is required.²³²

**184****185****186****187****188****189****190****191**

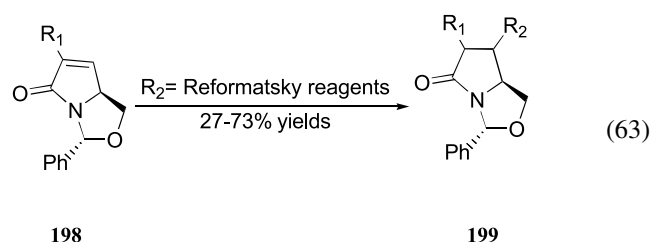
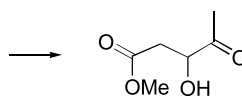
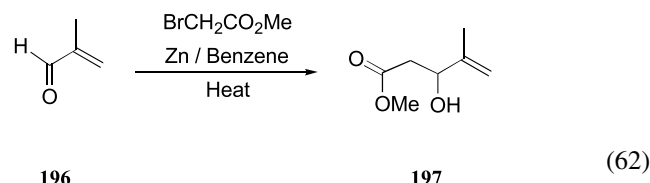
The Blaise reaction on chiral (*R*)-*O*-silyl-protected cyanohydrins **192** starting from *tert*-butyl bromoesters gave good yields (54–83%) of the corresponding *tert*-butyl (*R*)-3-amino-4-trimethylsilyloxy-2-alkenoates **193** without racemization (e.e. up to 95%), which were then cyclized to the corresponding α,β -unsaturated β -amino- γ -hydroxybutyrolactones **194** in the presence of HCl (Eq. 60).¹⁴⁷ When starting from ethyl bromoesters the Blaise reaction gave rise to the analogous α,β -unsaturated β -hydroxy- γ -butyrolactones **195** (Eq. 61).^{233,234} Tetrone acids prepared this way

have interest as pharmaceuticals and agrochemicals, and they may be synthesized on a large scale.

**192****193****192****195**

N,N-Disubstituted α -aminonitriles are reactive with activated halides under Reformatsky conditions in the presence of zinc and 10 mol% of acetic acid, and the reaction is used as a key step for the synthesis of tertiary homoallyl amines and α -amino esters.²³⁵

4.2.3. Michael or Michael-type acceptors. There is not a reliable way to predict the regiochemistry (1,2- or 1,4-) during the addition of Reformatsky reagents to α,β -unsaturated carbonyls. In fact, there are some examples of classical carbonyl attack (1,2-addition),^{212,236–238} particularly on α,β -unsaturated aldehydes (for instance see Eq. 62),^{236,239,240} whereas some other examples illustrate conjugate additions (such as that of Eq. 63).^{54,173,241}

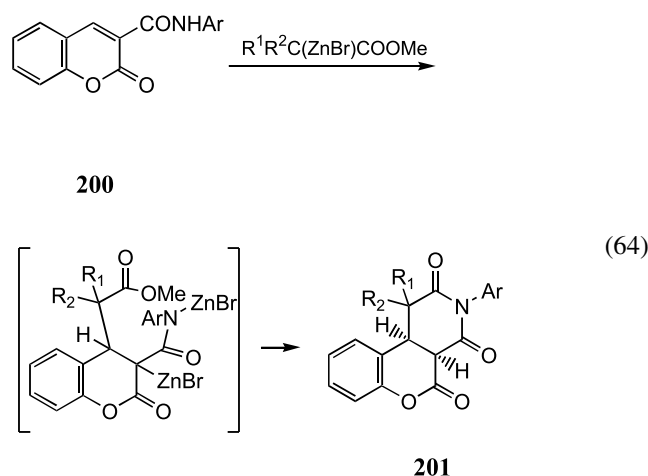
**198****199**

A clear 1,2-addition pathway with methacrolein **196**

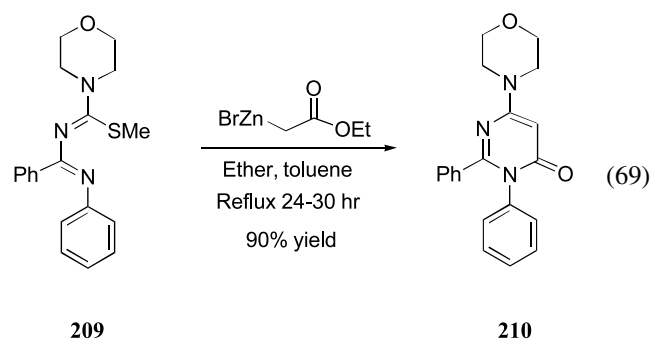
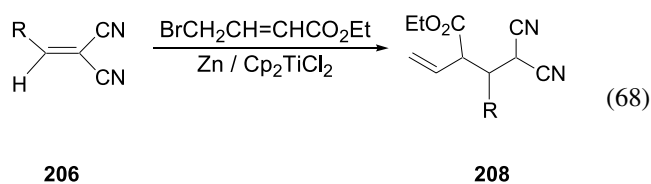
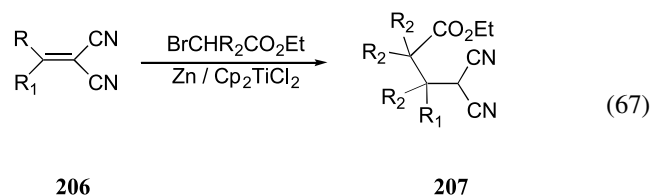
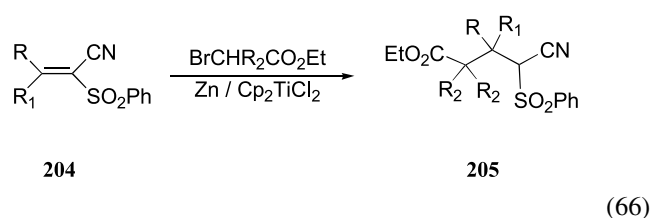
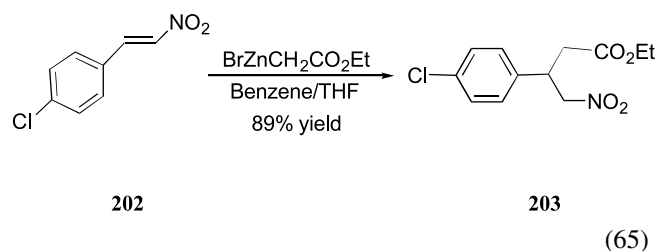
(Eq. 62) can be exploited as a ‘masked ketone’ in such a way that further cleavage of the double bond generates the carbonyl group, instead of using methyl glyoxal as the starting material. This method was efficiently used as the key step for the synthesis of 4-acetoxy-1-oxazetidin-2-ones.²³⁶

Sometimes a minor structural modification on the substrate induces the regiochemistry to switch. For example, the 1,2-Reformatsky addition with 3-arylmethylene-4,5-dihydrofuran-2(3*H*)-ones is reported versus 1,4-addition to 4-arylmethylene-3-methyl-isoxazol-5(4*H*)-ones.²⁴² However, there is some suggestion that bulky Reformatsky reagents tend to undergo conjugate additions, especially if the reaction is carried out on α,β -unsaturated ketones in the presence of trimethylsilyl triflate in THF at -78°C .⁵⁴ This trend was observed in reactions with 2-cyclopenten-1-one, 2-cyclohexan-1-one and carvone.

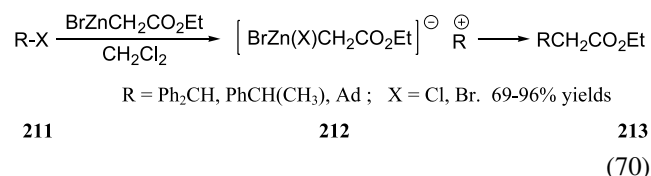
Heterocyclization of *N*-arylamides **200** of 2-oxochromen-3-carboxylic acids via a conjugate Reformatsky reaction with conventional reagents has led to 4 α -10 β -dihydro-1*H*-chromeno[3,4-*c*]pyridine-2,4,5-triones **201** after hydrolysis (Eq. 64).²⁴³



Michael-type addition of Reformatsky reagents has been observed on electron-deficient alkenes, for example α -nitrostyrenes **202** (Eq. 65),²⁴⁴ 1-cyano-1-phenylsulfonyl alkenes **204** (Eq. 66),³⁸ and 1,1-dicyanoalkenes **206** (Eqs. 67 and 68).³⁹ The reactions with 1-cyano-1-phenylsulfonyl alkenes or 1,1-dicyanoalkenes require catalytic amounts of Cp_2TiCl_2 , and the reaction gives rise to γ -functionalized esters in each case. Conjugate addition of Reformatsky reagents to 1,3-diaza-1,3-butadienes **209** is also possible, and such reaction results in the formation of dihydropyrimidinones **210** via a spontaneous ring closure of the respective adduct (Eq. 69).²⁴⁵



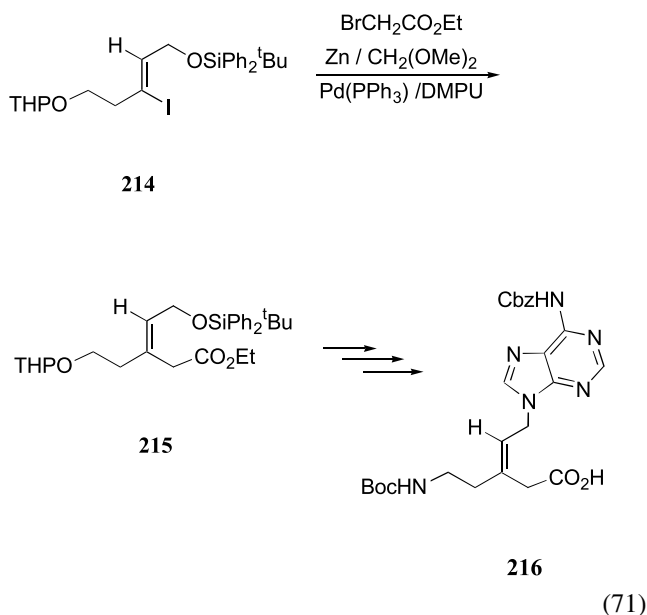
4.2.4. Carbon electrophiles other than carbonyl or related groups. In addition to the electrophiles already mentioned as active substrates for Reformatsky reactions, very reactive alkyl halides may substitute halogen for Reformatsky reagents. Thus, treatment of diphenylchloromethane **211** with $\text{BrZnCH}_2\text{CO}_2\text{Et}$ in CH_2Cl_2 gives rise to the respective substitution product ethyl 3,3-diphenylpropionate **213** in 96% yield, probably mediated by a ionic pair **212** (Eq. 70).¹⁴⁶ Similar behavior is exhibited by 1-bromoadamantane and 1-phenylethyl chlorides, so that this is a useful method to produce α -substituted ethyl acetates in good yields (69–96%).



C60 and C70 fullerenes are also prone to add organozinc reagents including not only the classic Reformatsky reagents, but also those which derive from α -bromoketones, allyl bromide, benzyl bromide, α -bromoacetonitrile and 1-iodobutane, to form the respective mono-alkylated derivatives along with minor by-products including the

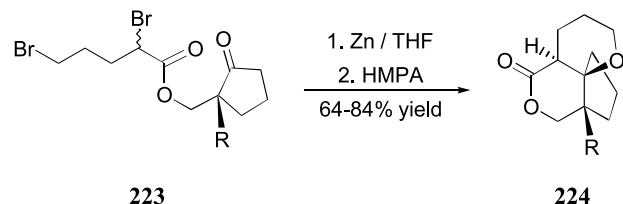
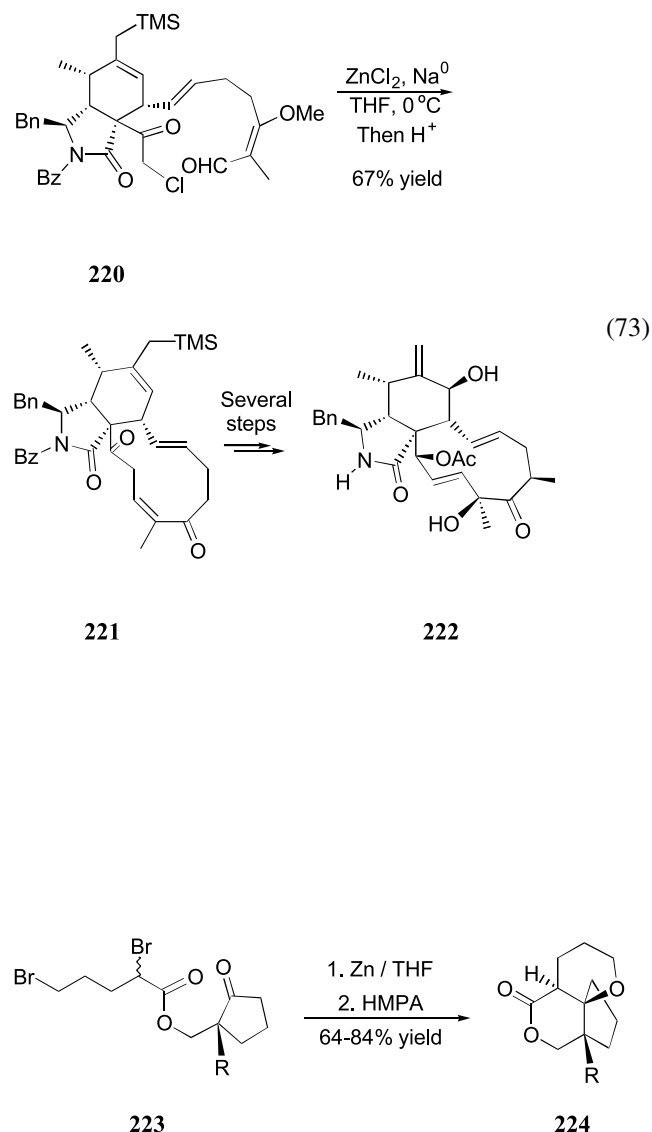
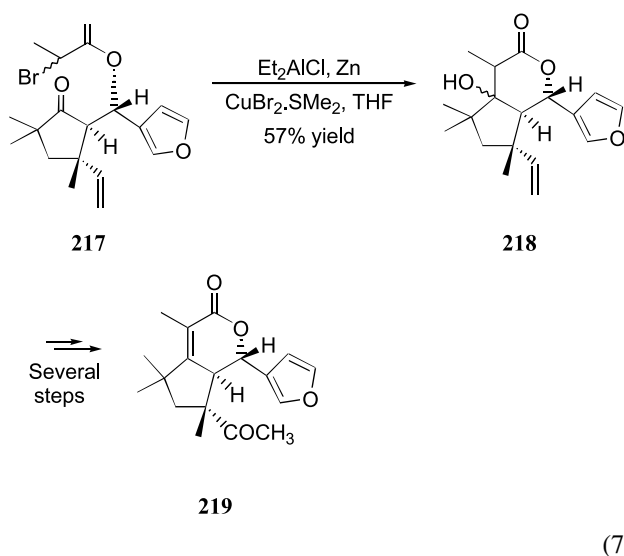
1,4-disubstituted ones.^{126,127} Typically the reactions are carried out in the absence of any solvent.

On the other hand, vinyl and aryl iodides (e.g., **214**) couple with Reformatsky reagents in the presence of palladium (0) catalysts (Eq. 71),^{118,119} and the reaction has been used as the key step for the synthesis of the Boc-protected *Z*- and *E*-olefinic peptide nucleic acid analogues (*Z*-OPA and *E*-OPA) monomer **216** containing the base thymine.¹¹⁸

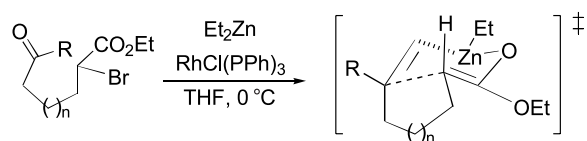
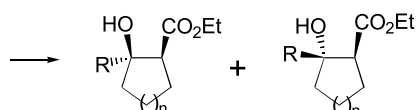


4.3. Intramolecular Reformatsky reactions

Zn/Et₂AlCl-mediated or Rieke-zinc-promoted intramolecular Reformatsky reactions are well known procedures, and several examples have been presented in a previous review.³ Key steps for the synthesis of the BCD framework of Richardianidins **219** (Eq. 72),⁵³ C(16),C(18)-*epi*-cytochalasin D **222** (Eq. 73),²⁴⁶ and the daphniphyllum alkaloids framework **225**²⁴⁷ make use of this methodology.

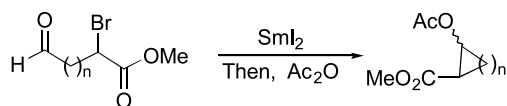


RhCl(PPh₃)₃/Et₂Zn couple has proved to be an efficient agent for both inter- and intramolecular Reformatsky reactions with some degree of stereocontrol favoring *cis* isomers (e.g., **228**). Such stereoselectivity is enhanced when the intramolecular reaction leads to five-membered rings (Eq. 75),¹⁰¹ and this effect has been attributed to a sterically and electronically more favored rigid conformation of transition state **227** when *n* = 1.

**226a** : R = H , n = 1**227****226b** : R = Me , n = 1**226c** : R = H , n = 2**226d** : R = Me , n = 2**228a** (63% yield)**229a** (5% yield)**228b** (59% yield)**229b** (5% yield)**228c** (41% yield)**229c** (33% yield)**228d** (65% yield)**229d** (26% yield)

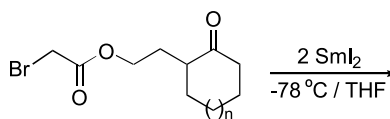
(75)

As briefly mentioned on a previous review³ and also in Section 2.2.2, samarium diiodide⁵⁹ is among the most exploited agents for intramolecular Reformatsky-type reactions, especially because of the observed high yields and stereocontrol. For instance, acetoxy carbonyl precursors **230** (*n* being equal to 6, 7, 9, 12, or 13) were converted to the respective eight-membered to 15-membered cyclic β -acetoxyesters **231**, with yields ranging from 63 to 82% (Eq. 76).²⁴⁸

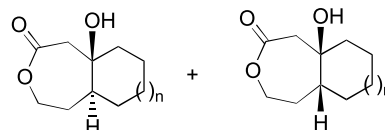
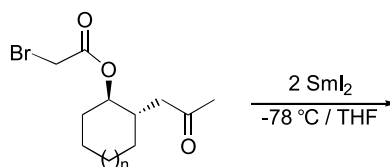
**230****231**

(76)

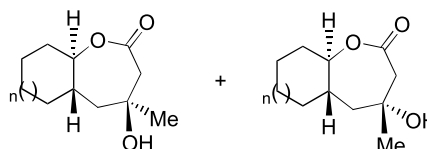
Identical chemistry occurs for α -bromoesters with compounds such as **232** which bear a properly placed carbonyl group for the generation of five-, six- and larger-membered rings, a particularly useful reaction for the generation of stereo-defined lactones with fused rings. Diastereomeric ratios **233/234** of up 200:1 have been observed (Eqs. 77 and 78).²⁴⁹ This kind of closure was used for the synthesis of (2*R*,4*R*)-supellapyrone, the sex pheromone of the brown-banded cockroach *supella longipalpa*.²⁵⁰

**232a**

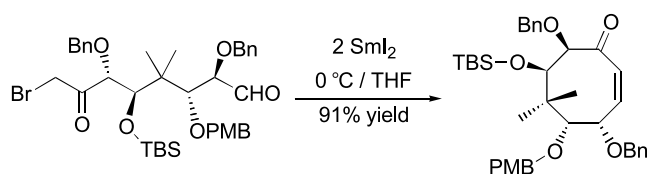
(77)

**233a****234a****232b**

(78)

**233b****234b**

This methodology has been applied also to the asymmetric synthesis of **235**, which is a fully functionalized B ring system of taxol (Eq. 79).²⁵¹

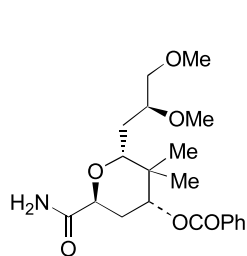
**232c****235**

(79)

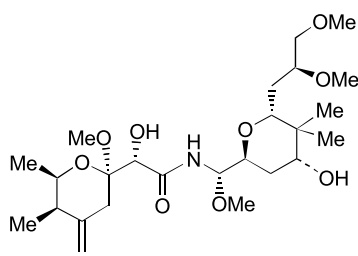
Selected examples of practical applications of intramolecular SmI_2 -promoted Reformatsky reactions to synthesis are listed in Table 5. The reaction shown in entry 1 constitutes the key step to the synthesis of (+)-benzoylpedarin **236**, which is a synthetic intermediate in the preparation of the potent insect toxin pederin **237**,²⁵² while the synthesis of the bioactive (–)-octalactin **A** **238** is mediated by reaction illustrated in entry 2.²⁵³ The reaction in entry 3 is useful for the synthesis of (+)-compactin lactone, a potent competitive inhibitor of the HMG–CoA,²⁵⁴ and the reaction of entry 4 involves the crucial step in the approach to the F–M ring framework of ciguatoxine (CTX1B) **239** or model compounds,²⁵⁵ while entry 5 illustrates a carbon–carbon bond forming reaction in nucleoside chemistry.²⁵⁶

Table 5. Examples of synthetic applications of SmI₂-promoted intramolecular Reformatsky reactions

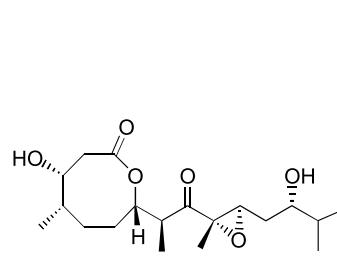
Entry	Reactions	Ref.
1		252
2		253
3		254
4		255
5		256



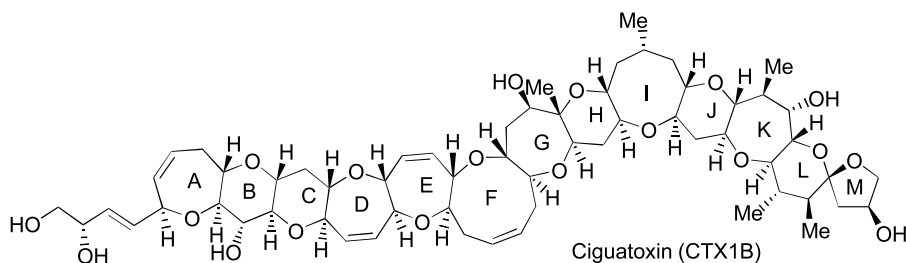
236



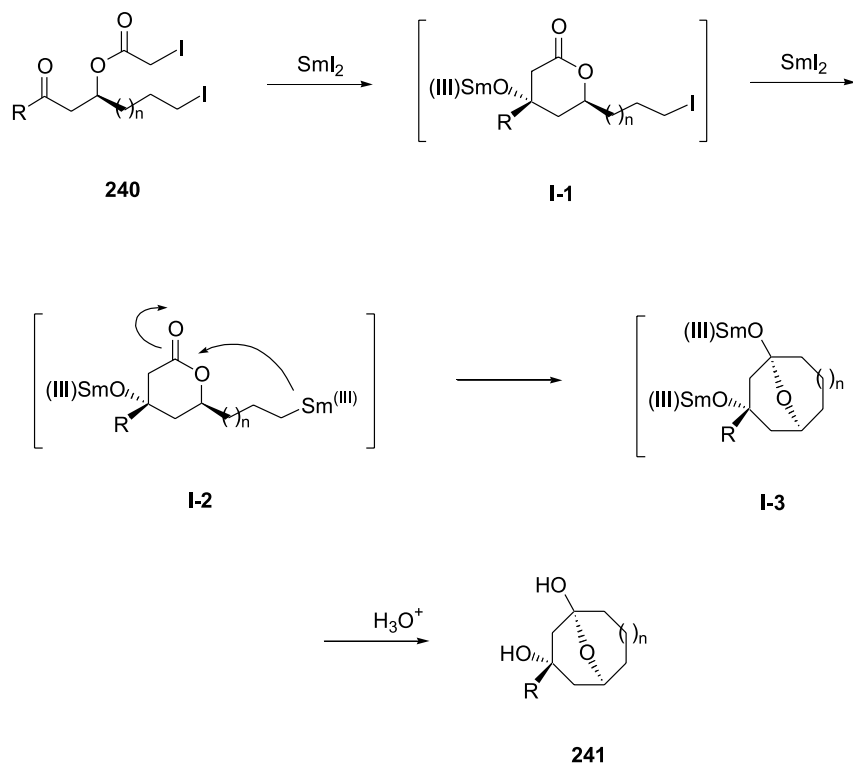
237



238



239



Scheme 12.

A similar SmI_2 -promoted intramolecular Reformatsky reaction constitutes the key step in the synthesis of the novel inhibitor of cholesterol biosynthesis, (–)-decaestreictine,²⁵⁷ and of (–)-borrelidin.²⁵⁸

Samarium(II) iodide is also used to access eight- and

nine-membered carbocycles, for example, **241**, via a domino reaction, comprised of a Reformatsky reaction from **240** to form intermediate **I-1**, which is followed by an internal nucleophilic acyl substitution reaction that is undergone by intermediate **I-2** (Scheme 12).²⁵⁹ This method represents a general and efficient approach to

Table 6. Examples of stereoselectivity accomplished by using chiral auxiliaries attached to precursors

Entry	Chiral precursor	E ⁺	Reformatsky agent	Reaction conditions	Products	Yield (%)	Stereo selectivity ^a	Refs.
1	242a	245	Zn	THF, 0 °C 30 min	249a + 250a	99	91:9 ^b	163
2	242b	245	Zn	cf. entry 1	249b + 250b	91	90:10 ^b	163
3	242c	245	Zn	cf. entry 1	249c + 250c	99	90:10 ^b	163, 164
4	242c	PhCHO	Zn	cf. entry 1	251 + 252	98	98:2 ^b	162
5	242d	246	CrCl ₂ /LiI (cat)	THF, rt, 5 h	253 + 254	63	8:92	65
6	242d	<i>i</i> -PrCHO	CrCl ₂ /LiI (cat)	cf. entry 5	255 + 256	91	4:96	65, 66
7	242b	<i>i</i> -PrCHO	CrCl ₂ /LiI (cat)	THF, 20 °C	257 + 258	88	5:95 ^b /(98:2) ^c	65, 66
8	242e	PhCHO	GeI ₂ /K	THF, rt, 18 h	259 + 260	94	99:1 ^b /(99:1) ^c	95
9	242e	247	Co[P(Ph) ₃] ₄	THF, –0 °C 2 h	261	70	> 96% d.e.	99
10	242f	<i>t</i> -BuCHO	SmI ₂	THF, –78 °C, 0.5 h	262	87	> 99% d.e.	166
11	242g	248	SmI ₂	THF, –78 °C, 1 h	263 + 264	94	96:4	260
12	243a	245	Zn	THF, reflux, 20 min	265a + 266a	87	98:2 ^b	168
13	243b	245	Zn	cf. entry 12	265b + 266b	76	99.6:0.4 ^b	168
14	243c	245	Zn	cf. entry 12	265c + 266c	85	98:2 ^b	169a
15	244a	163b	Zn/I ₂	Sonication, rt 48 h	267a	45	98% d.e.	214
16	244b	163b	Zn/I ₂	cf. entry 15	267b	55	98% d.e.	214

^a Ratios of products in the respective order.

^b Ratios (2,3-*syn/anti*) in entries 1–4, 7–8 and 12. Ratios of isomers α/β in entries 12, 13 and 14.

^c Diastereomeric ratio (i.e., *R/S* in respect of carbon α) in parenthesis.

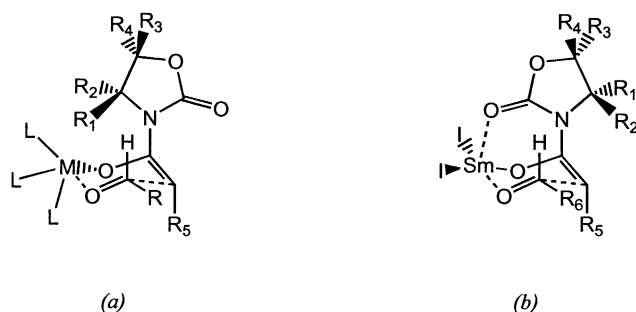


Figure 1. (a) Transition state according to the Zimmerman–Traxler model.^{61,64,95,99} (b) Transition state according to Nerz–Stormes–Thornton model.^{166,262}

a variety of highly functionalized, stereo-defined carbocycles.

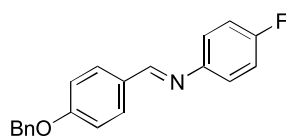
SmI₂-promoted cyclization reactions involving substrates containing benzyloxy or S-pyridyl moieties as leaving groups, which lead to fused bicyclic or tricyclic (β -lactams, have already been mentioned (Eq. 34).²⁰¹

4.4. Advances in stereochemical aspects

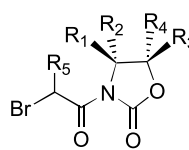
The chromium-,^{58,61–67} SmI₂-,^{59,166,201,202,249–259,260–263} scandium-,⁹³ germanium-,⁹⁵ indium-,⁷⁸ cobalt-,^{98–100} rhodium-,¹⁰¹ and tin-^{48–52,115,199,200} versions of the Reformatsky reaction, combined with introduction of chiral or stereo inductor auxiliaries into the precursors have rescued the Reformatsky methodology as a synthetic tool even for asymmetric targets. Some brief mention was made on this matter in Section 2.2, Eqs. 6, 8–10, 14–16, 32, 34, 40–43, 46, 47, 72, 73 and 77–79, and Schemes 6–8 and 12.

4.4.1. Chiral or stereo inductor auxiliaries. Some degree of diastereoselectivity and enantiofacial differentiation has been made possible through the use of halo precursors bearing chiral or stereo inductor auxiliaries attached in the form of esters or amides (e.g., **242** through **244**). These chiral moieties derive mainly from alcohols such as menthol and related compounds,^{214,264} oxazolidines,^{163,164,166} and oxazinanes.^{167–169} A relevant equation appear in Table 3 (entry 2). In addition, Table 6 lists some selected examples of the best stereoselectivities attained by these methods. As seen in Table 6, *syn* products are regularly favored, and this *syn* preference is explained in terms of an enantiofacial *re* differentiation which is probably derived from a chair-like, six-membered transition state, according to the Zimmerman–Traxler,^{61,64,95,99} or Nerz–Stormes–Thornton models (Fig. 1).^{166,262} The chromium Reformatsky reactions favor *anti* products when starting from α -bromo amides or α -bromo esters (entries 5–7), but the opposite trend is observed when the reaction is carried out using α -halo ketones.^{65,66}

Further functional group conversions or removal of the chiral auxiliary makes possible the synthesis of asymmetric targets. For instance, products **249** and **265**, prepared according to conditions specified in entries 1–3 and 12–14, have been appropriately converted to 1 β -methyl-carbapenems.^{163–169,191} Also, products **256**, **258**, **262** or **267**, which were readily prepared as illustrated in entries 6, 7, 10, 15 and 16, may be hydrolyzed to afford the respective β -hydroxyacids with high enantiomeric excesses,^{166,264} and products **267** may be converted to the respective β -lactams with up to 99% e.e., a key procedure in preparing 3-substituted azetidin-2-ones, which exhibit cholesterol

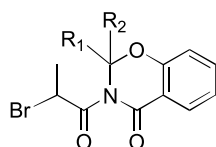


163b



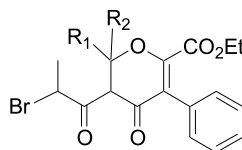
242

- a: R₁ = ⁱPr, R₂ = R₃ = R₄ = H, R₅ = Me
 b: R₁ = Bn, R₂ = R₃ = R₄ = H, R₅ = Me
 c: R₁ = R₂ = ⁿBu, R₃R₄ = (CH₂)₅, R₅ = Me
 d: R₁ = Bn, R₂ = R₃ = R₄ = R₅ = H
 e: R₁ = H, R₂ = Ph, R₃ = H, R₄ = Ph, R₅ = Me
 f: R₁ = ⁱPr, R₂ = R₃ = R₄ = R₅ = H
 g: R₁ = H, R₂ = Bn, R₃ = R₄ = R₅ = H

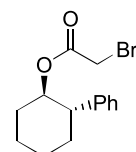


243a: R₁ = R₂ = ⁿBu

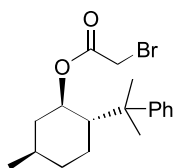
243b: R₁ = R₂ = Bn



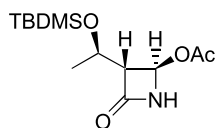
243c: R₁R₂ = (CH₂)₅



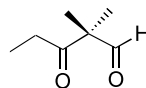
244a



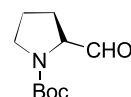
244b



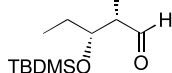
245



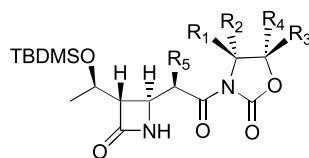
246



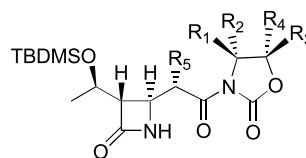
247



248

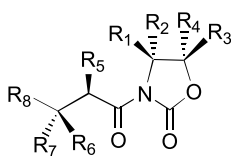
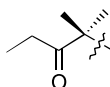
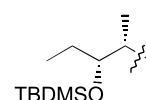


249



250

- a: $R_1 = {}^i\text{Pr}$, $R_2 = R_3 = R_4 = \text{H}$, $R_5 = \text{Me}$
b: $R_1 = \text{Bn}$, $R_2 = R_3 = R_4 = \text{H}$, $R_5 = \text{Me}$
c: $R_1 = R_2 = {}^n\text{Bu}$, $R_3R_4 = (\text{CH}_2)_5$, $R_5 = \text{Me}$
d: $R_1 = \text{Bn}$, $R_2 = R_3 = R_4 = R_5 = \text{H}$
e: $R_1 = \text{H}$, $R_2 = \text{Ph}$, $R_3 = \text{H}$, $R_4 = \text{Ph}$, $R_5 = \text{Me}$
f: $R_1 = {}^i\text{Pr}$, $R_2 = R_3 = R_4 = R_5 = \text{H}$
g: $R_1 = \text{H}$, $R_2 = \text{Bn}$, $R_3 = R_4 = R_5 = \text{H}$

R₈ moiety for **253** and **254**R₈ moiety for **261**R₈ moiety for **263** and **264**

251 (2,3-*syn*), $R_1=R_2 = {}^n\text{Bu}$, $R_3R_4 = (\text{CH}_2)_5$, $R_5 = \text{Me}$, $R_6 = \text{OH}$, $R_7 = \text{H}$, $R_8 = \text{Ph}$

252 (2,3-*anti*), $R_1=R_2 = {}^n\text{Bu}$, $R_3R_4 = (\text{CH}_2)_5$, $R_6 = \text{H}$, $R_7 = \text{OH}$, $R_8 = \text{Ph}$

253 $R_1 = \text{Bn}$, $R_2=R_3=R_4=R_5 = \text{H}$, $R_6 = \text{OH}$, $R_7 = \text{H}$

254 $R_1 = \text{Bn}$, $R_2=R_3=R_4=R_5 = \text{H}$, $R_6 = \text{H}$, $R_7 = \text{OH}$

255 $R_1 = \text{Bn}$, $R_2=R_3=R_4 = \text{H}$, $R_5 = \text{Me}$, $R_6 = \text{OH}$, $R_7 = \text{H}$, $R_8 = {}^i\text{Pr}$

256 $R_1 = \text{Bn}$, $R_2=R_3=R_4 = \text{H}$, $R_5 = \text{Me}$, $R_6 = \text{H}$, $R_7 = \text{OH}$, $R_8 = {}^i\text{Pr}$

257 (2,3-*syn*), $R_1 = \text{Bn}$, $R_2=R_3=R_4 = \text{H}$, $R_5 = \text{Me}$, $R_6 = \text{OH}$, $R_7 = \text{H}$, $R_8 = {}^i\text{Pr}$

258 (2,3-*anti*), $R_1 = \text{Bn}$, $R_2=R_3=R_4 = \text{H}$, $R_5 = \text{Me}$, $R_6 = \text{H}$, $R_7 = \text{OH}$, $R_8 = {}^i\text{Pr}$

259 (2,3-*syn*), $R_1 = \text{H}$, $R_2 = \text{Ph}$, $R_3 = \text{H}$, $R_4 = \text{Ph}$, $R_5 = \text{Me}$, $R_6 = \text{OH}$, $R_7 = \text{H}$, $R_8 = \text{Ph}$

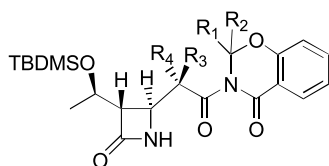
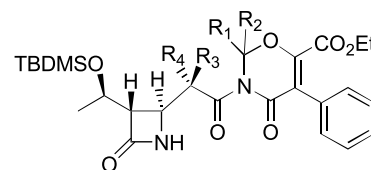
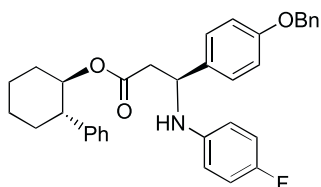
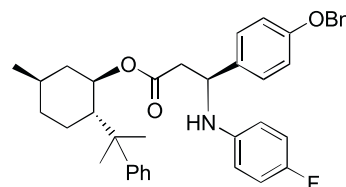
260 (2,3-*anti*), $R_1 = \text{H}$, $R_2 = \text{Ph}$, $R_3 = \text{H}$, $R_4 = \text{Ph}$, $R_5 = \text{Me}$, $R_6 = \text{H}$, $R_7 = \text{OH}$, $R_8 = \text{Ph}$

261 (2,3-*syn*), $R_1 = \text{H}$, $R_2 = \text{Ph}$, $R_3 = \text{H}$, $R_4 = \text{Ph}$, $R_5 = \text{Me}$, $R_6 = \text{OH}$, $R_7 = \text{H}$

262 $R_1 = {}^i\text{Pr}$, $R_2=R_3=R_4=R_5 = \text{H}$, R_6 and $R_7 = \text{OH}$ and H ; configuration at C3 not determined, $R_8 = {}^t\text{Bu}$

263 $R_1 = \text{H}$, $R_2 = \text{Bn}$, $R_3=R_4=R_5 = \text{H}$, $R_6 = \text{H}$, $R_7 = \text{OH}$

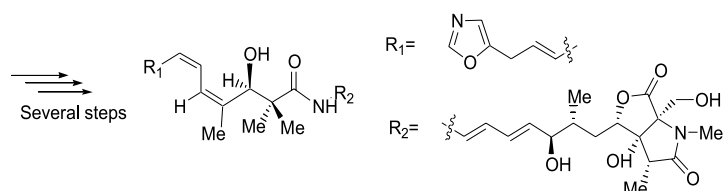
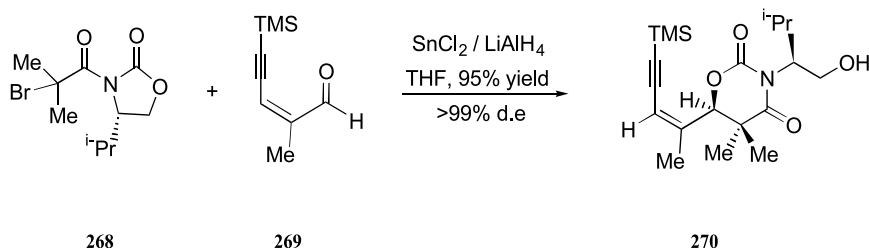
264 $R_1 = \text{H}$, $R_2 = \text{Bn}$, $R_3=R_4=R_5 = \text{H}$, $R_6 = \text{OH}$, $R_7 = \text{H}$

**265a** $R_1=R_2=^n\text{Bu}$, $R_3=\text{H}$, $R_4=\text{Me}$ (α -isomer)**265b** $R_1=R_2=\text{Bn}$, $R_3=\text{H}$, $R_4=\text{Me}$ (α -isomer)**266a** $R_1=R_2=^n\text{Bu}$, $R_3=\text{Me}$, $R_4=\text{H}$ (β -isomer)**266b** $R_1=R_2=\text{Bn}$, $R_3=\text{Me}$, $R_4=\text{H}$ (β -isomer)**265c** $R_1R_2=(\text{CH}_2)_5$, $R_3=\text{H}$, $R_4=\text{Me}$ (α -isomer)**266c** $R_1R_2=(\text{CH}_2)_5$, $R_3=\text{Me}$, $R_4=\text{H}$ (β -isomer)**267a****267b**

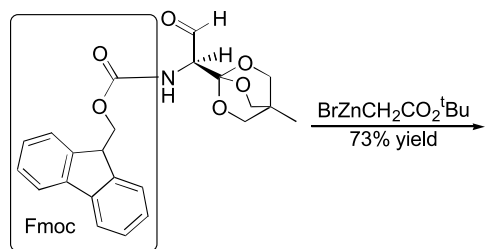
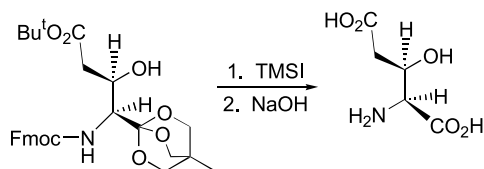
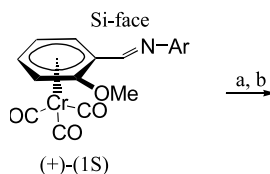
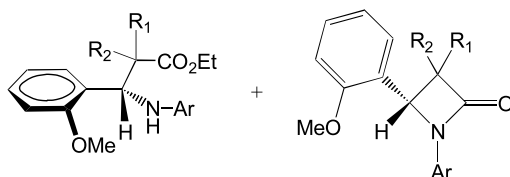
absorption inhibitory activity.²¹⁴ Products **259**, which resulted from the germanium Reformatsky reactions (entry 8), have been converted to the respective methyl esters with optical purities $>99\%$,⁹⁵ whereas the adduct **261**, which resulted from the cobalt-phosphine Reformatsky approach (entry 9), was stereospecifically converted to the dolastatin 10 unit dolaproine (Dap) useful as a human anticancer agent.⁹⁹ Similar conversions from products **254** of chromium Reformatsky reactions (entry 5) were used for the synthesis of fragment A of epothilons, macrocyclic lactones with taxol-like mitose inhibition profile.⁶⁵ An efficient, related synthetic route was developed to prepare substituted δ -lactones in enantiopure forms which are potentially useful for biosynthetic studies with genetically engineered modular polyketide synthase (PKS). The method involves hydrolysis of products **263** that were prepared by SmI_2 -promoted Reformatsky reactions starting from the enantiomeric (*S*)-4-benzyloxazolidinone (entry 11 in Table 6).²⁶⁰

On the other hand, a Sn(II) -enolate asymmetric Reformatsky-type condensation of an acyloxazolidinone **268** to $\text{Me}_3\text{Si}-(\text{acetylene})-\text{CH}=\text{C}(\text{Me})\text{CHO}$ **269**, producing adduct **270**, is involved as the key step in the convergent enantioselective total synthesis of the antitumor antibiotic neooxazolomycin **271** (Eq. 80).²⁶⁵

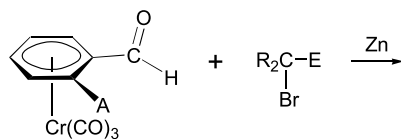
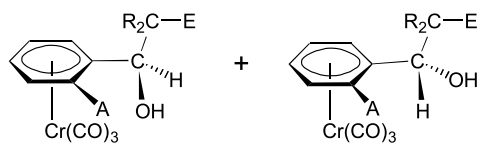
Further deprotection of the products **273** of highly stereocontrolled Reformatsky reactions, makes possible the synthesis of all four diastereomers of β -hydroxy- α -amino acids **274**, starting from a chiral serine aldehyde **272** via a variety of carbonyl addition reactions including the Reformatsky reaction (Eq. 81).²⁶⁶ The method readily allows for stereospecific incorporation of both C and H isotopes in amino acid side chains, and a similar procedure has been reported with the analogous fluorinated Reformatsky reaction in the synthesis of L-4,4-difluoroglutamic acid involving an analogous N-protected derivative of L-serine (see later).²⁶⁷

**271**

(80)

**272****273**, (2*S*,3*R*), ratio *threo*/*erythro* 92:8**274**, (2*S*,3*R*), 98% ee
(81)**278****279****280**70–94% combined yield; ee of **279** and **280** up to 98%With R₁ = H, Me. R₂ = H, Me. Ar = Ph, p-(MeO)Ph

Cp-based complexes exert some diastereo control. For example, it was reported recently that titanocene (III)-promoted Reformatsky reactions occur with *anti* diastereoselectivity.⁵² Tricarbonyl (η^6 -arene)chromium (0) (e.g., **275**) and ferrocene-based related complexes may function as chiral auxiliaries, (Eqs. 82 and 83),^{33,137,268} giving products with high enantiomeric excesses, in such a way that it is possible to correlate the absolute configuration of the obtained products with the configuration of the starting tricarbonylchromium complexed chiral substrates. Also, a highly stereoselective synthesis (e.e. > 98%) of a mixture of β -amino esters **279** and β -lactams **280** was accomplished using enantiomerically pure tricarbonyl (η^6 -benzaldehyde)-chromium complexes **278** in a Reformatsky-type reaction promoted by ultrasound (Eq. 83).¹³⁷

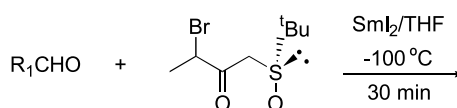
**275****276** (major)**277** (minor)

62–88% combined yield. %d.e ranging from = 2–100%

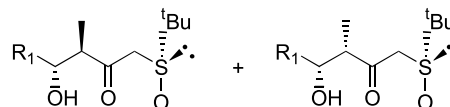
With E = CO₂Me, CO₂^tBu, CO₂Et, CN. R = H, Me. A = Me, OMe
(82)

Conditions: (a) R₁R₂C(Br)CO₂Et/Zn/dioxane, 20–25 °C, ultrasound. (b) CH₂Cl₂, *h* ν , air (for decomplexation).

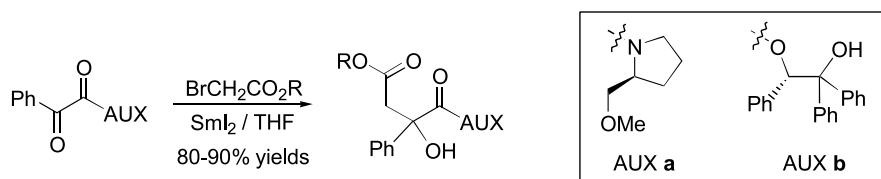
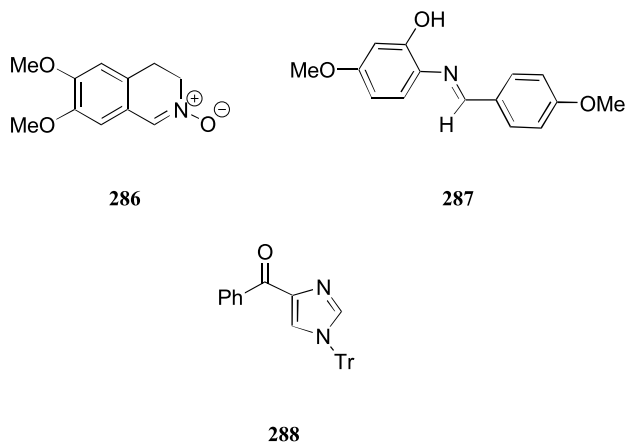
A recent development on diastereoselective Reformatsky reactions involving chiral α -bromo- α' -sulfinyl ketones **281** and linear aldehydes in the presence of SmI₂ gives rise to the respective adducts (**282** + **283**) with of up to 98% *syn* diastereomeric excess (Eq. 84).²⁶³ Further reduction of the Reformatsky adducts furnishes *anti* and *syn*-2-methyl-1,3-diols moieties in excellent yields and diastereoselectivities. Related chiral sulfinylimines have been used in Reformatsky reactions of fluorinated precursors (see Section 5).^{269,270}

**281**

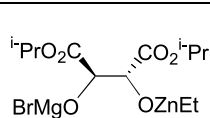
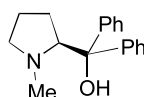
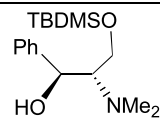
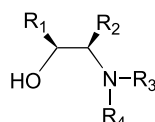
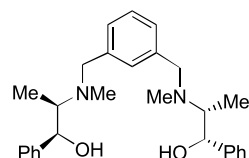
(84)

**282****283**Yields: 55–87%. *Syn/anti* ratios: 45:55 to 98:2. Ratios **282**/**283**: 80:20 to 96:4.

SmI₂-induced Reformatsky reactions of chiral diols- or chiral (2*S*)-methoxymethylpyrrolidine-based α -ketoesters/amides **284** afforded the corresponding adducts **285** with 80–90% yields and 60–89% diastereomeric excess (Eq. 85).²⁶¹

**284a****285a** (R= Me, 72% d.e.; R= ^tBu, 89% d.e)**284b****285b** (R= Me, 60% d.e)**Table 7.** Selected examples of the best enantioselectivities accomplished by adding chiral auxiliaries

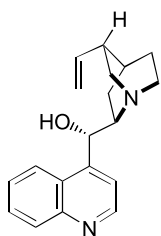
Entry	Precursor	E ⁺	Reformatsky agent	Reaction conditions	Chiral auxiliary	Product	Yield (%)	e.e. (%)	Ref.
1	ICH ₂ CO ₂ CEt ₃	286	Et ₂ Zn	CH ₂ Cl ₂ , 0 °C, 2 h	289	297	99	86	34
2	ICH ₂ CO ₂ Bu	287	Et ₂ Zn	CH ₂ Cl ₂ , 0 °C, 12 h	289	298	58	96	213
3	BrCH ₂ CO ₂ Bu	PhCHO	Zn–Cu	THF, 0 °C	290	299a	91	75	274
4	BrCH ₂ CO ₂ Bu	PhCHO	Zn	THF, 0 °C, 15–20 h	291	299a	100	65	276
5	BrCH ₂ CO ₂ Bu	PhCHO	Zn–Cu	THF, reflux, 5 h	292	299b	57	64	272b
6	BrCH ₂ CO ₂ Bu	PhAc	Zn–Cu	THF, Tol, –13 °C, 25 h	293	300	65	74	277
7	BrCH ₂ CO ₂ Bu	PhCHO	Zn	THF, 0 °C, 24 h	294	299a	90	62	273
8	BrCH ₂ CO ₂ Bu	288	Zn	THF, Pyr, –40 °C, 4 h	295	301	> 99	97	279
9	BrCH ₂ CO ₂ Et	PhCHO	Zn/TMSCl	THF, 50 °C, 20 min	296	299a	49	34	280

**289****290****291****292**, R₁=R₂= Ph, R₃= Bn, R₄= Me**293**, R₁= Ph, R₂= Me, R₃=R₄= allyl**294**

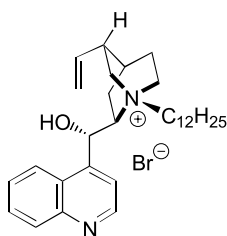
4.4.2. Chiral auxiliaries as additives. From catalytic to stoichiometric amounts of chiral auxiliaries such as carbohydrates,²⁷¹ amino alcohols (ephedrine and *N*-alkyl-ephedrine),^{171,272–278} cinchonine and cinchona alkaloids,^{78,279} chiral micelles,²⁸⁰ chiral α -imino esters,¹⁸⁹ α -amino acid esters and peptide derivatives²⁸¹ may be added as ligands to a reaction mixture, in order to exert some extent of stereocontrol. In some specific cases, good enantiomeric excesses have been obtained, as listed in Table 7. Entries 1 and 2 illustrate the use of chiral tartrate derivatives **289** as chiral auxiliaries, particularly for the asymmetric addition of Reformatsky-type reagents to isoquinoline *N*-oxide derivatives **286** and imines **287**,^{34,213} whereas entries 3–7 demonstrate the use of chiral aminoalcohols or derivatives **290–294**.^{273–276} However, the best results are observed when cinchona alkaloids **295** are used as chiral ligands (entry 8), although in this particular case the high enantioselectivity is attributed to the enantiofacial discrimination which results from chelation with the sp^2 -nitrogen adjacent to the reactive carbonyl center.²⁷⁹ Quaternary salts **296** deriving from cinchonines bearing long chains, as chiral micelles, create a chiral environment for asymmetric Reformatsky reactions, affording moderate enantiomeric excesses (entry 9).²⁸⁰ Similarly, micelles deriving from chiral amino alcohols of the form **290–294** exert a related behavior.²⁸⁰ Chiral β -(*N*-arylamino)esters **298** from entry 2 may be easily converted to the respective chiral β -amino esters by simple deprotection procedures.²¹³

4.5. Synthetic applications

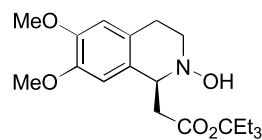
A number of examples have been mentioned throughout this review dealing with important synthetic targets which can be synthesized via multistep procedures involving the Reformatsky methodology as at least one of the key steps. Several of those targets are molecules with some biomedical potential, or synthetic intermediates for their preparation, which include for instance 9-alkyl-10-deazaminopterin,¹⁵ richardianidins,⁵³ epothylons,⁶⁵ tonantzitlolone,⁶⁸ dolaproine,⁹⁹ nucleoside- and nucleotide derivatives,^{128,256} a monomer for a polymer analog of a nucleic acid,¹¹⁸ (+)-pilocarpine,¹⁵⁸ 1 β -carbapenems,^{163–169,191} peptides (with projection to peptide libraries),²⁰² cadinanolide,¹⁹⁸ a tetracyclic model for the aziridinomitosenes,²²⁸ (+)-pederin,²⁵² α - and β -amino acids and their respective esters and β -lactams,⁵⁷ (–)-octalactin,²⁵³ (+)-compactin lactone,²⁵⁴ ciguatoxine (CTX1B) or model compounds,²⁵⁵ (–)-decarestrictine,²⁵⁷ (–)-borrelidin,²⁵⁸ neooxazolomycin,²⁶⁵ trienomycin,²⁸² bremazocine,²⁸³ the cytotoxic macrolide 15-*epi*-haterumalide NA,²⁸⁴ 4-octulose derivatives,⁸³ 8-azapsoralens and other coumarin-derived products,^{203,285} quinol esters such as jacaranone,⁸⁴ quinoline 4-methanols,¹⁹⁴ tetronic acids,^{233,234} ulosonic acids,^{104,105} analogues of quinolone antibacterial agents,¹⁵⁹ C-glycosyl derivatives,²⁸⁶ phenolic derivatives (with aryl groups such as phenyl-derivatives, biphenyl, coumarin, flavone, xanthone, carbazole, dibenzofuran or quinolinone



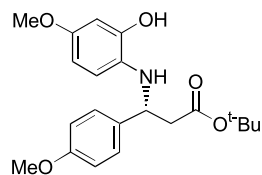
295



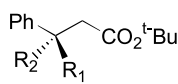
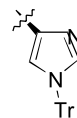
296



297



298

299a, R₁=H, R₂=OH299b, R₁=OH, R₂=H300, R₁=Me, R₂=OH301, R₂=OHR₁ for 301

As may be anticipated, the quality of the results in terms of the stereocontrol is affected by the ratio of the chiral ligand with respect to the starting materials.

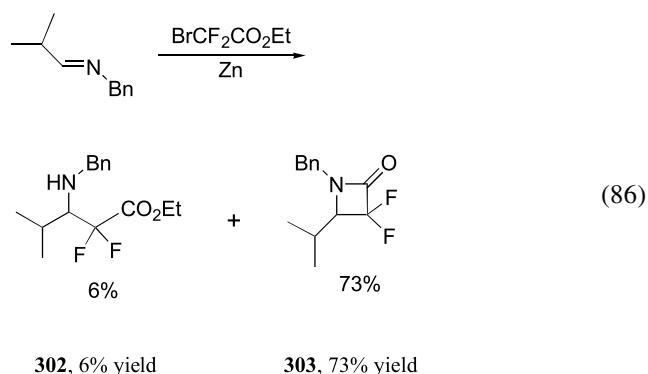
moieties),^{186,187} steroidal residues,¹⁸⁸ taxoid rings,²⁵¹ *epi*-cytochalasin D,²⁴⁶ daphniphyllum alkaloids,²⁴⁷ supellapyrone (a pheromone),²⁵⁰ agrochemicals,^{233,234} 3-hydroxycitronellic

acid,²⁸⁷ (±)-heritol as insecticide,²⁸⁸ (±)-nimbidiol and (±)-nimbiol,²⁸⁹ 3,4-diphenylbutanoic acids with antifungal activity,²⁹⁰ (Z)- and (E)-4-amino-2-(trifluoromethyl)-2-butenic acids as potential mechanism-based inactivators of γ -aminobutyric acid aminotransferase (GABA-AT),²⁹¹ liposidomycin-B (nucleoside antibiotic),²⁹² (+)-castanospermine,²⁹³ equine metabolites of anabolic steroids,²⁹⁴ fuscilol (a marine diterpene with potential as anti-inflammatory),²⁹⁵ a vitamin A analogue²⁹⁶ of pyrimidine-,¹⁸⁴ purine-derived nucleic acid bases,¹⁸⁵ and other targets with medicinal or broad biological interest.^{112,297–302} Some such targets are associated as agents that are presumably or certainly bioactive as antitumoral, antiviral, antifungal, antibiotic, antimalarial, vasorelaxing molecules, metabolic inhibitors of cholesterol biosynthesis or antifolates, pesticides, pheromones, etc.

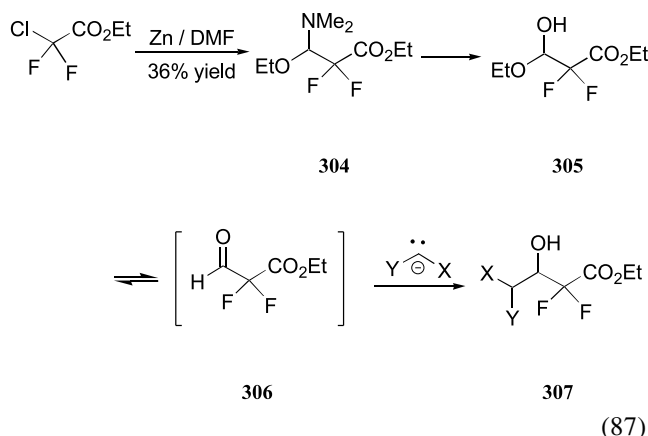
Additional miscellaneous examples which can be mentioned include: enantioselective synthesis of indane acetic derivatives,³⁰³ γ -amino- β -hydroxyacids,¹⁵⁵ formation of highly oxidized terpenes derived from abietic acid,³⁰⁴ and other terpenoids,³⁰⁵ butenolides such as (±)-mint-lactone,³⁰⁶ synthetic approaches to homochiral bicyclo-[5.2.1]decanes based on D-camphor,³⁰⁷ synthesis of a stable neutral hydrocarbon radical 2,5,8-tri-*tert*-butylphenalenyl,³⁰⁸ adamantyl derivatives for generation of carbocations,¹⁵⁴ cyclopentenones,³⁰⁹ quinoline derivatives,³¹⁰ tetrahydrofuran units and five-membered ring lactones fused to hexopyranosides,³¹¹ 3-arylidene and 3-cyclohexylidene benzofuran-2(3*H*)-ones,³¹² superferrocenophane,³¹³ and targets for applications in cosmetics.³¹⁴

5. Recent reports of Reformatsky reactions involving fluorinated substrates

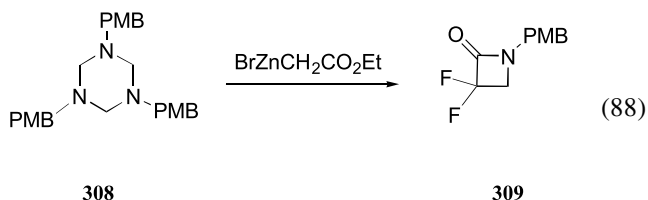
The subject of Reformatsky reactions of fluorine-containing precursors and substrates has been recently reviewed.³¹⁵ Classical reactions analogous to those of non-fluorinated systems lead to the synthesis of α,α -difluoro- β -hydroxyesters and α,α -difluoro- β -hydroxyketones, and the replacement of carbonyl compounds for imines^{269,270,316,317} produces the anticipated α,α -difluoro- β -amino esters, for example, **302**, or their respective α,α -difluoro- β -lactams, for example, **303** (Eq. 86).³¹⁷ The Reformatsky reaction of fluorinated precursors with imines was carried out recently using rhodium catalysis, and it was used for the synthesis of functionalized (Z)-fluoroalkene-type dipeptide isosteres.³¹⁸ *N*-(α -aminoalkyl)benzotriazoles derived from aldehydes behave as imine equivalents, and as such they condense with monofluorinated Reformatsky reagents to afford the expected α -fluoro- β -amino esters.⁵⁵ The benzotriazole methodology with $\text{BrZnCF}_2\text{CO}_2\text{Et}$ has been also applied under solid-phase conditions for the preparation of α,α -difluoro- β -amino acids on a solid support (MBHA linked amines).⁵⁶ The carboxyl function gives access to a large range of possible transformations making these derivatives a useful tool for the generation of large libraries of fluorinated compounds including peptides, β -peptides and peptidomimetics.



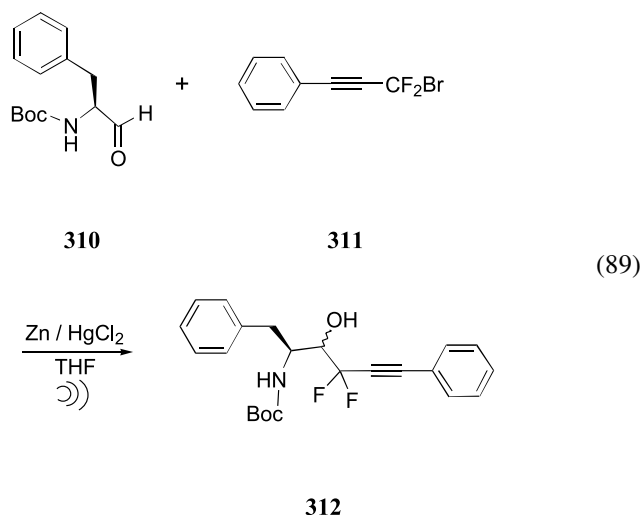
DMF can also work as electrophile with the zinc-based Reformatsky agent derived from ethyl chlorodifluoroacetate, thus giving a Vilsmeier-type N–O acetal product **304**, convertible into the respective ethyl hemiacetal **305**, which is chemically equivalent to a difluoromalonaldehydic derivative **306**. This latter product condenses with active methylene compounds via aldol-like reactions (Eq. 87),³¹⁹ and the procedure is useful for the synthesis of diverse functionalized α,α -difluorinated esters and amides.



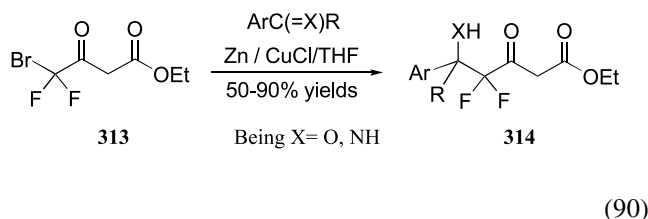
It is expected that N–O acetals or related substrates condense with fluorinated Reformatsky reagents, and specifically with 1,3-oxazolidines, which upon reaction generate laterally functionalized fluorine-containing β -lactams (see Section 5.1).³²⁰ Similarly, ethyl bromodifluoroacetate condenses with *N,N',N''*-trisubstituted hexahydro-1,3,5-triazines, for example, **308** (Schiff base trimer) under Reformatsky conditions, which is a useful route toward *N*-protected 3,3-difluoroazetidin-2-ones **309** (Eq. 88).³²¹



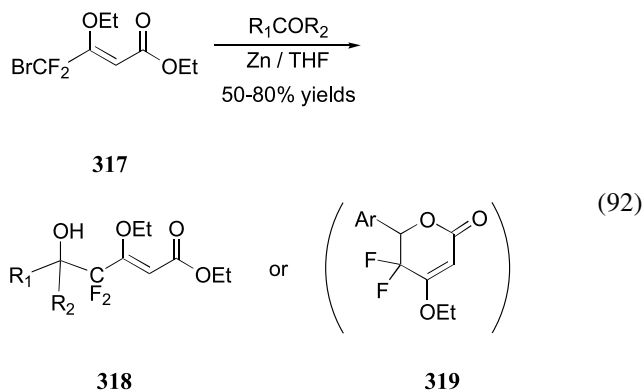
Chlorodifluoro- and bromodifluoroacetylene derivatives (e.g., **311**) have been used in Reformatsky-like reactions as well (Eq. 89).³²² The product **312** of this Reformatsky-like reaction has been used for the synthesis of a key intermediate of potent proteinase inhibitors.³²²



In the presence of zinc and cuprous chloride, straightforward reaction of 4-halo-4,4-difluoroacetoacetate **313** with aromatic aldehydes or aryl ketones³²³ (or their respective imines³²⁴) under mild conditions gives good to excellent yields of the corresponding δ -hydroxyl- γ,γ -difluoro- β -keto esters or their related δ -amino- γ,γ -difluoro- β -ketoesters **314** (Eq. 90).^{323,324}

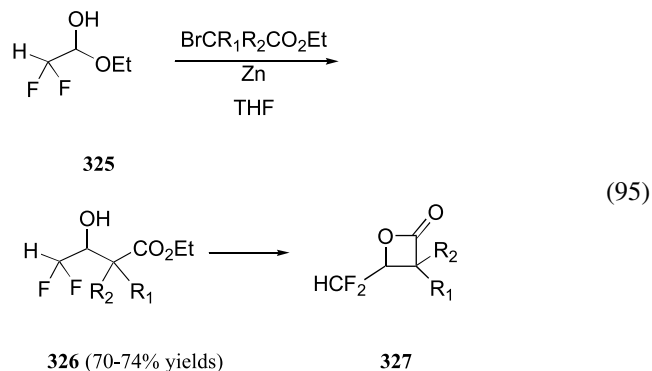
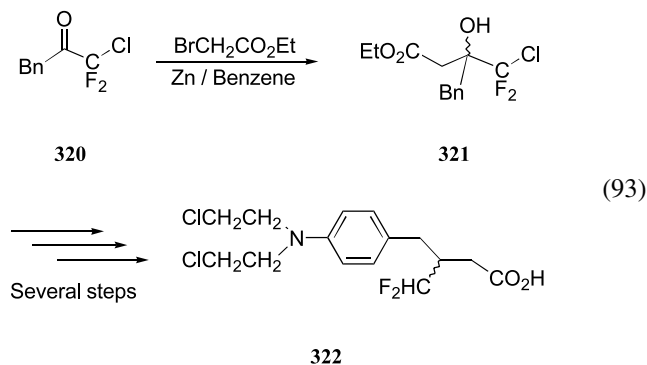


Analogous to how 4-bromocrotonates behave in Reformatsky reactions, 4-bromo- or 4-chloro-4,4-difluorocrotonates **315** react either in the α -regiochemical mode or by γ -attack, giving rise to α -substituted 4,4-difluoro-3-butenates (e.g., **316**, Eq. 91)³²⁵ or γ substituted 4,4-difluoro-2-butenates (e.g., **318**, Eq. 92), respectively.³²⁶ In the latter case, the expected α,β -unsaturated γ,γ -difluoropyran-2-one **319** was formed starting from aromatic aldehydes with prolonged reaction times.

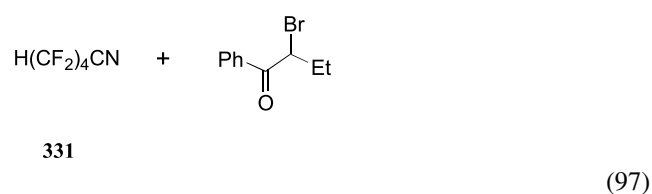
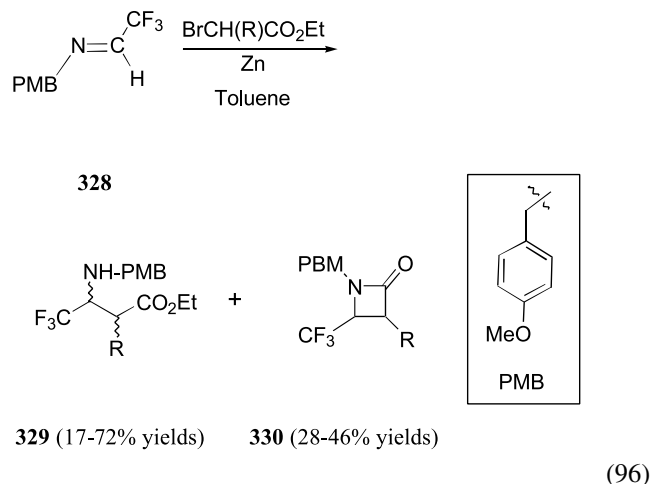
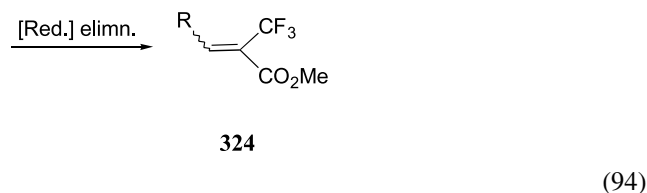
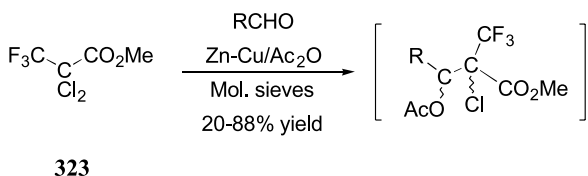


The reaction of ethyl bromodifluoroacetate with Michael acceptors in the presence of copper powder has been studied recently, and the method has been used for the synthesis of 4,4-difluoro- α -tocopherol.³²⁷

In general, it is recognized that chlorodifluorocompounds releases chlorine in Reformatsky reactions very much slower than do the analogous bromo- or iodo-compounds, but the chloro-derived precursors are cheaper. Some reports encourage usage of chlorodifluoro- precursors, introducing little experimental variants that allow the reactions to proceed favorably to products. Representative examples of slight experimental modifications include the catalysis of nickel under electrochemical conditions,⁹¹ the use of couples such as Zn/CuCl or Zn/AgOAc,³²⁸ or the combination Zn/AgOAc/Et₂AlCl,³⁷ and the indium-mediated Reformatsky reactions starting from 2,2-difluoro-2-halo-1-furan-2-ylethanones,⁸¹ the latter case being useful for the synthesis of α,α -difluoro- β -hydroxy ketones in binary aqueous media. As already mentioned, ultrasound is an additional important tool in Reformatsky reactions of both non-fluorinated^{19,79,80,141} and fluorinated^{138–140} precursors, and it is indispensable with NO₂ bearing aldehydes or ketones which require prior generation of the organozinc BrZnCF₂CO₂Et, which is only stable when generated under ultrasonication conditions.¹⁴⁰ Lanthanide salts have been used successfully, and such is the case for the SmI₂-induced reaction of chloro- and bromodifluoroacetates with aldehydes, giving rise to the expected classical α,α -difluoro- β -hydroxyesters in good yields.³²⁹ SmI₂ has also been used with perfluorinated Reformatsky and related nucleophilic intermediates.³³⁰ Alternatively, by using catalytic amounts of lanthanide salts (e.g., cerium trichloride) Reformatsky methodology starting from chloro- and bromodifluoroacetates⁴⁵ and from ethyl chlorodifluoroacetate,⁴⁶ or ethyl bromodifluoroacetate⁴⁷ has become recognized as an efficient and simple procedure. However, the relative inertness of the chlorodifluoro moiety becomes an advantage if additions of Reformatsky reagents to chlorodifluoroketones are desired. For instance, the key step for the preparation of chlorambucil derivatives **322**, involves a Reformatsky reaction with the chlorodifluoroketone **320**, providing the adduct **321** with the CF₂Cl moiety remaining unchanged (Eq. 93).³³¹

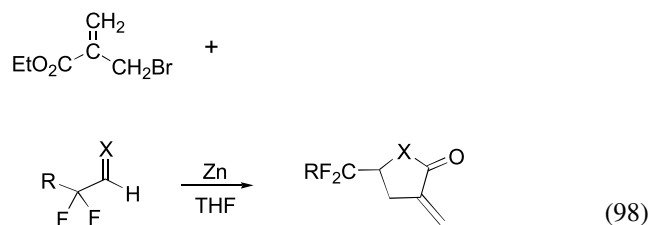


One-pot Reformatsky reaction/acylation/reductive elimination starting from the readily available $\text{CF}_3\text{CCl}_2\text{CO}_2\text{Me}$ **323** with aldehydes constitutes a general synthetic scheme for the preparation of mixtures (*E/Z*) of α -(trifluoromethyl)- α,β -unsaturated carboxylates **324**, mostly in favor of the *Z* isomers with selectivities ranging from 53:47 to 66:34, useful as fluorine-containing building blocks (Eq. 94).³³²



On the other hand, Eqs. 95–97 illustrate common reactions of non-fluorinated Reformatsky reagents with fluorinated electrophiles, giving rise to the anticipated products. Eq. 95 deals with the reaction of a classical Reformatsky reagent with a fluorinated hemiacetal **325** to afford γ,γ -difluoro- β -hydroxyesters **326**, with low *threolerythro* selectivity,³³³ and Eq. 96 shows a typical Reformatsky reaction with a fluorinated imine **328**, which affords a mixture of the β -amino esters **329** and the respective β -lactams **330**, with the β -lactams being preferred over the β -amino esters.³³⁴ β -Lactones **327** bearing a lateral difluoro moiety can be obtained from **326** by standard procedures. The Reformatsky reaction of a fluorinated nitrile **331** gives a 46% yield of the rare tautomeric form **332**, stable at temperatures of up 145 °C (Eq. 97).⁴⁰

The version of the Dreiding–Schmidt reaction^{177–180} of methyl 2-(bromomethyl)acrylate applied to difluoro- or trifluoroacetaldehyde **333a–b** (or their acetals) gives rise, respectively, to the expected γ -difluoromethyl- α -methylene- γ -butyrolactones **334a**³³⁵ or γ -trifluoromethyl- α -methylene- γ -butyrolactones **334b** (Eq. 98),³³⁶ potent in vitro antitumor compounds, even stronger than 5-fluorouracil. Analogous reaction of methyl 2-(bromomethyl)acrylate with difluoro-, or trifluoroethanimine **333c–d** generates the respective γ -difluoromethyl- α -methylene- γ -butyrolactams **334c–d** (Eq. 98).³³⁵



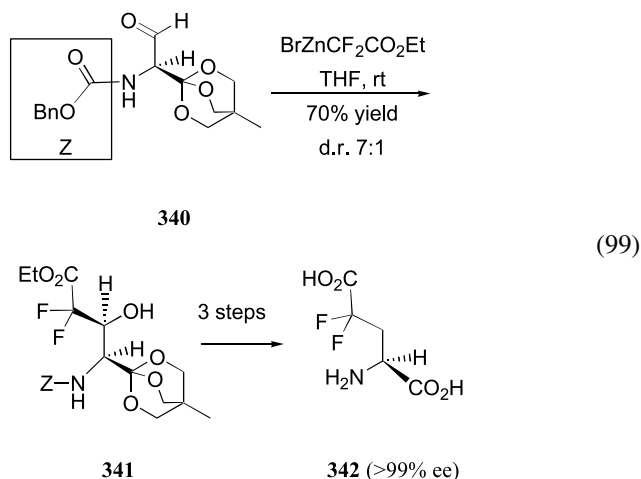
- | | |
|---------------------|---------------------|
| 333a, R = H, X = O | 334a, R = H, X = O |
| 333b, R = F, X = O | 334b, R = F, X = O |
| 333c, R = H, X = NH | 334c, R = H, X = NH |
| 333d, R = F, X = NH | 334d, R = F, X = NH |

Domino Reformatsky-Wittig reactions of 4-bromocrotonates with perfluorinated ketophosphonium salts **335** afford either perfluoroalkylated 1,4-alkadienes **337** resulting from α -attack, or 5-(perfluoroalkyl)-substituted 2,5-alkadienes **339** corresponding to γ -attack (Scheme 13).^{337,338} Regioselectivities of up to 100% in favor of the γ -product **339** are reported.

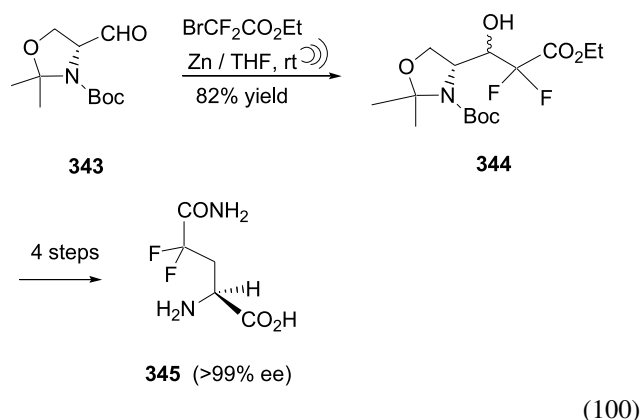
5.1. Advances in stereochemical aspects

Stereochemical control on the Reformatsky reaction of aldehydes and bromodifluoroacetate was briefly mentioned in a short review on asymmetric fluoroalkylations.³³⁹ Some stereoselective developments with fluorinated substrates make use of enzymatic resolution techniques of racemic products using lipase MY from *Candida cylindracea*³⁴⁰ or lipase PS or Novozym 435.³⁴¹ Asymmetric induction of catalytic Cp_2TiCl_2 in Reformatsky reactions of chiral electrophiles, particularly derived from carbohydrates, with bromo- and iododifluoroacetate with diastereoselectivities of up to 95:5 is also reported.^{342–344} Significant progress has been made on diastereo- and enantioselective Reformatsky reactions of fluorinated precursors, taking advantage of some of the methodologies described in Section 4.4 involving chiral precursors or electrophiles, or chiral auxiliaries as additives. For instance, chiral N-protected aldehyde **340**, chemically deriving from L-serine, was used as the key synthetic intermediate for the enantioselective preparation of L-4,4-difluoroglutamic acid **342**, with the crucial step being the 7:1 diastereoselective Reformatsky reaction of $\text{BrCF}_2\text{CO}_2\text{Et}$ to afford **341** (Eq. 99), and the

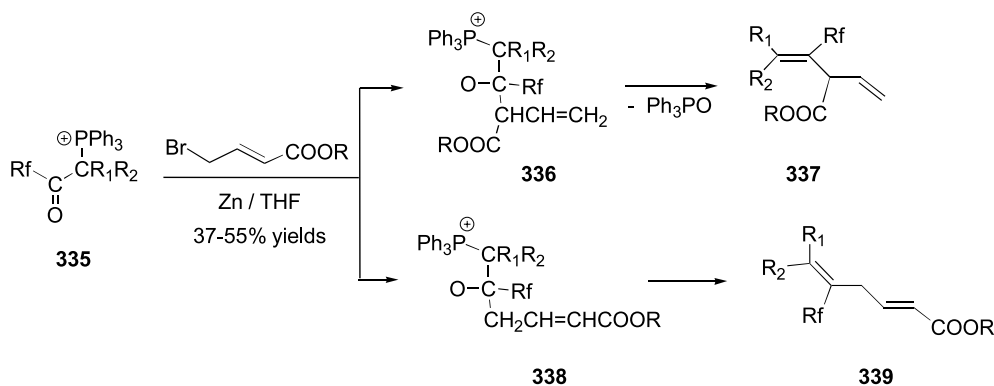
whole sequence was amenable to a large-scale preparation of the enantiopure difluorinated amino acid **342**.²⁶⁷



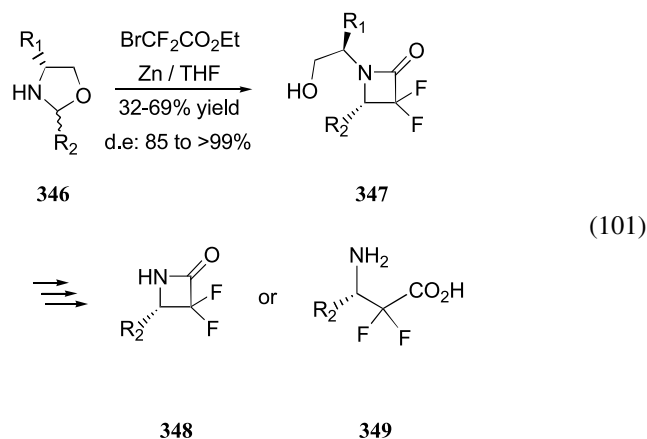
A similar synthetic strategy was applied to the preparation of the structurally related L-4,4-difluoroglutamine **345**, starting in this case from (*R*)-Garner's aldehyde **343**, which bears a chiral oxazolidine moiety (Eq. 100).³⁴⁵



Chiral oxazolidines (e.g., **346**) function themselves as electrophiles with the bromodifluoro Reformatsky reagent, and this reaction is used to stereoselectively generate *N*-alkyl functionalized α,α -difluoro- β -amino acids **347**, which may be converted either to β -lactams **348** or to the respective α,α -difluoro- β -amino acids **349** (Eq. 101).³²⁰

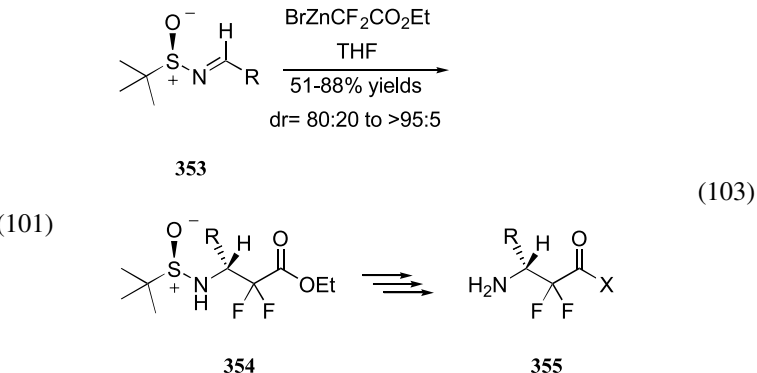
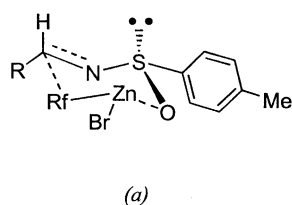
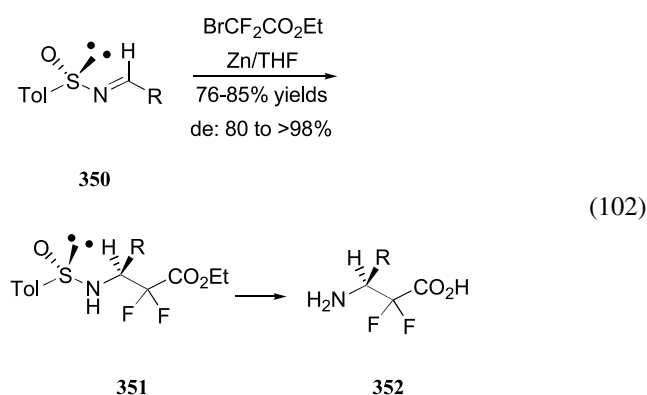


Scheme 13.



Chiral α,α -difluoro- β -amino acids, obtained in this manner, were used recently for the synthesis of difluorinated pseudo-peptides.³⁴⁶

Enantiopure sulfinylimines (e.g., **350** and **353**) have become very promising imine equivalents^{269,270} for use in carrying out stereocontrolled Reformatsky reactions from fluorinated precursors, affording the respective β -[*N*-(alkylsulfinyl)-amino]- α,α -difluoro-esters (**351** and **354**). The highest diastereomeric ratios are obtained with bulky aryl or cycloalkyl R groups,²⁷⁰ and stereodefined products can be clearly anticipated. (*Ss*)-sulfinimines **350** generate (*Ss,3S*) diastereomers **351** (Eq. 102),²⁶⁹ while the respective (*R**s*,3*R*) diastereomers **354** result from the (*R**s*) starting sulfinimines **353** (Eq. 103).²⁷⁰



Diastereoselectivity exhibited by the bromodifluoro- Reformatsky reactions with chiral sulfinimines^{269,270} has been explained in terms of the Ellman and Davis transition state models depicted in Figure 2,³⁴⁷ which account for the predictable addition to the *Re* face of the (*Ss*)-sulfinimine (Fig. 2a) or to the *Si* face of the (*R**s*)-sulfinimine (Fig. 2b and c).

Further transformations of these Reformatsky products give rise to the respective α,α -difluoro- β -amino acids **352** and **355**, or β -branched α,α -difluoro- β -amino esters useful for solid-phase peptide synthesis, and enantiomeric excesses even greater than 99% are reported.^{269,270}

Chiral carbinols or amino alcohols such as ephedrine-¹³⁹ or *N*-methylephedrine derivatives³⁴⁸ of the form **292**, **293** and **294** are similarly effective chiral auxiliaries for enantioselective Reformatsky reactions of the bromodifluoro-Reformatsky reaction, leading to enantiomeric excesses similar to those which have been observed with the non-fluorinated version (as described in Table 7).

5.2. Recent applications in synthesis

Several examples of syntheses of diverse fluorinated targets have been already mentioned in the previous sections, and in addition, some other examples deserve mention. For example, the classical route of carbonyl compounds to olefins was modified to generate 1,1-difluoroolefins **359** via the bromodifluoro- Reformatsky adducts **356** and their respective acids **357** (Scheme 14).^{349,350} By this procedure α,α -difluoro- β -lactones **358** were systematically formed and isolated, and they were then thermally decarboxylated to afford 1,1-difluoroalkenes **359**.

A related methodology has made possible the generation of α -fluoro- β -hydroxyacids **361** and **365** via the non-aqueous alkaline hydrolysis of the respective diastereomeric α -fluoro- β -hydroxyesters **360** and **364** obtained from the

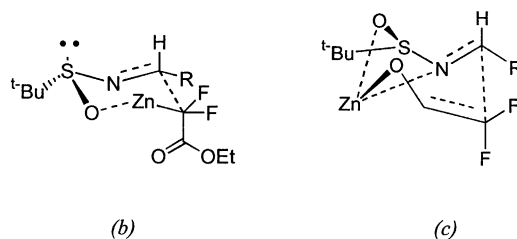
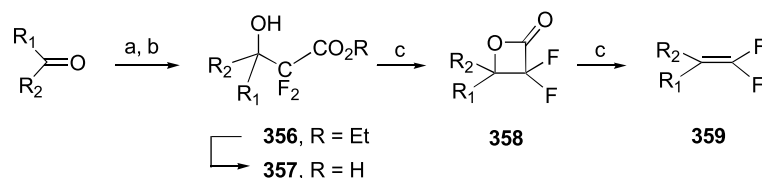


Figure 2.



Scheme 14. (a) $\text{BrCF}_2\text{CO}_2\text{Et}$, Zn/CeCl_3 , THF, rt; (b) NaOH, rt, 12 h, then HCl; (c) $\text{PhSO}_2\text{Cl/Pyr/CHCl}_3$, 0 °C; (d) Heat.

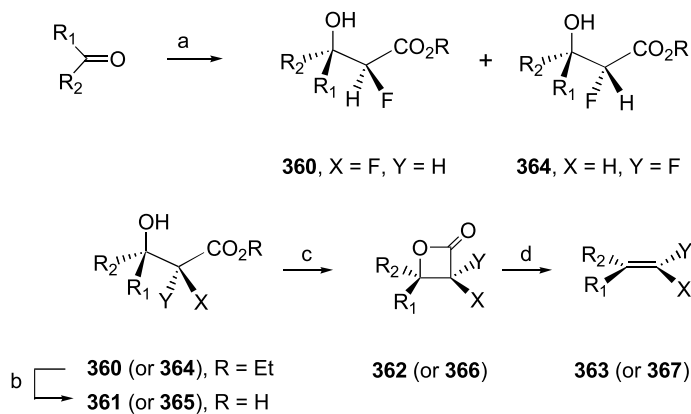
CeCl_3 -catalyzed Reformatsky reaction.⁴⁷ Some of these α -fluoro- β -hydroxyacids could be converted to the novel α -fluoro- β -lactones **362** and **366**,³⁵¹ which are promising synthetic intermediates for the stereospecific generation of (*E*)- or (*Z*)-1-fluoroolefins **363** and **367** (Scheme 15).

Other significant synthetic applications involving fluorinated systems include the synthesis of (+)-10,10-difluorothromboxane A_2 ,¹⁷ and other diverse synthons for medicinal chemistry targets,²⁷⁰ the new quinoline antibacterial agent DQ-113 from ethyl bromofluoroacetate,³⁵² 3,3-difluoro-L-homocystein,³¹⁶ optically pure α,α -difluoro- β -amino acids,³²⁰ a potential inhibitor of human thrombin,³⁵³ a key intermediate of potent proteinase inhibitors,³²² chlorambucil derivatives,³³¹ (*S*)- β,β -difluoromalic acid,³⁴⁰ nucleoside derivatives for antiviral assays,^{342–344} alkyl fluorides,³⁵⁴ cyclopentane derivatives containing a $\text{CF}_2\text{CO}_2\text{Et}$ group,³⁵⁵ (*Z*)-2-fluoroacrylate thioesters and other desired fluoroacrylates,³⁵⁶ antitumoral compounds such as γ -difluoromethyl- and γ -difluoromethyl- α -methylene- γ -butyrolactones,^{335,357} 14,14-difluoro-4-demethoxydaunorubicin,³⁵⁸ enzyme inhibitors (for instance interleukin-1 β converting enzyme (ICE)),³⁵⁹ a leucinal-derived hapten synthesized to elicit catalytic antibodies,³⁶⁰ renin,^{361,362} human leukocyte elastase³⁶³ or aspartate transcarbamoylase,²¹⁹ β -gem-difluoromethylene C-glycosides,^{334,364} vitamin D3 analogues,^{365,366} or antagonists,³⁶⁷ N-substituted dibenzoxazepines (analgesic PGE2 antagonists),³⁶⁸ and, in general, biologically active compounds with CF_2 units as isopolar and isosteric replacement sites for ether oxygen.^{318,369}

6. Concluding remarks

As seen throughout this review, much progress has been made on the asymmetric version of Reformatsky reactions, involving both fluorinated and non-fluorinated Reformatsky reagents. This significant progress has rescued Reformatsky methodology so that it is now even more useful than before for modern organic synthesis. When condensations are needed in planning organic synthesis, there are several factors which make the Reformatsky reaction a choice worth considering. Among those special factors which deserve emphasis are: (i) easy predictable C–C addition without C–O bond formations (with very few exceptions), (ii) demanding of neutral conditions (in contrast to the strong alkaline medium required for aldol condensations), (iii) smoothness of the Reformatsky reagents (in comparison to Grignard reagents), and (iii) the possibility of good levels of stereoselectivity, as described in the latest developments. We have seen that syntheses of a number of natural products very often involve key steps consisting of classical Reformatsky reactions or Reformatsky-type procedures, including cases where diastereomeric or enantiomeric control is desired. Much progress has also been made on the use of diastereoselective intramolecular Reformatsky reactions in assembling rings of varied sizes.

However, there has only been limited progress in developing alternative procedures with the goal of replacing organic solvents. Although there are some few reports on aqueous medium or non-solvent Reformatsky reactions, the quest for



For **360** through **363**, X = F, Y = H. For **364** through **367**, X = H, Y = F

Scheme 15. (a) $\text{BrCHF}_2\text{CO}_2\text{Et}$, Zn/CeCl_3 , THF, rt; (b) NaOH/abs. EtOH, 0 °C, 2 h, then HCl; (c) $\text{MePhSO}_2\text{Cl/DMAP/CHCl}_3$, –50 °C; (d) Heat.

green chemistry is expected to lead to safer and more efficient procedures in the future, including those that produce less waste.

References and notes

- Reformatsky, S. *Ber. Dtsch. Chem. Ges.* **1887**, 20, 1210.
- Orsini, F.; Sello, G. *Curr. Org. Synth.* **2004**, 1, 111–135.
- Fürstner, A. *Synthesis* **1989**, 571–590.
- (a) Fürstner, A. In *Encyclopedia of Reagents for Organic Synthesis*. Paquette, L. A., Ed.; Wiley: New York, 1995; pp 2402–2408. (b) Fürstner, A. In *Organozinc Reagents*. Knochel, P., Jones, P., Eds.; University Press: Oxford, 1999; pp 287–305. (c) Rathke, M. W.; Weipert, P. In *Comprehensive Organic Synthesis*. Trost, B. M., Fleming, I., Eds.; Pergamon Press: Oxford, 1991; Vol. 2, pp 277–299. (d) Rathke, M. W. In *Organic Reactions*. John Wiley and Sons: New York, 1975; Vol. 22, Chapter 2, pp 30–59.
- Zinc-induced Reformatsky reactions should be considered only a small subdivision of organozinc chemistry. (a). For documentation on organozinc chemistry, see: *Organozinc Reagents*. Knochel, P., Jones, P., Eds.; University Press: Oxford, 1999; 354 p. (b) For a review on transition metal catalyzed reactions of organozinc reagents, see: Erdik, E. *Tetrahedron* **1992**, 48, 9577–9648.
- Areias, M. C. C.; Bieber, L. W.; Navarro, M.; Diniz, F. B. *J. Electroanal. Chem.* **2003**, 558, 125–130.
- (a) Horner, W.; Bertagnolli, H. *J. Organomet. Chem.* **2002**, 649, 128–135. (b) Ertel, T. S.; Horner, W.; Abraham, I.; Bertagnolli, H. *J. Phys. IV* **1997**, 7, 947–948.
- For older reviews, see: (a) Shriner, R. L. *Org. React.* **1942**, 1, 1. (b) Rathke, M. W. *Org. React.* **1975**, 22, 423–460. (c) Gaudemar, M. *Organomet. Chem. Rev., Sect. A*, **1972**, 8, 183–233. (d) Nützel, K. *Houben-Weyl*, Vol. XIII/2a, Georg Thieme: Stuttgart, 1973, p 805. (e) Diaper, D. G. M.; Kuksis, A. *Chem. Rev.* **1959**, 59, 89. (f) Brossi, M.; Kaenel, H. R. *Schweiz. Lab.-Z.* **1984**, 41, 55–57. (g) Vul'fson, N. S.; Vinograd, L. K. *Reactions and Research Methods for Organic Compounds Reformatsky Reaction*, Vol. 17; Khimiya: Moscow, 1967.
- Zhao, M.-X.; Wang, M.-X.; Yu, C.-Y.; Huang, Z.-T.; Fleet, G. W. *J. Org. Chem.* **2004**, 69, 997–1000.
- Bowman, M. P.; Senet, J. P. G.; Malfroot, T.; Olofson, R. A. *J. Org. Chem.* **1990**, 55, 5982–5986.
- Shchepin, V. V.; Russkikh, N. Y.; Desyatkov, D. A. *Zh. Org. Khim.* **1993**, 29, 2316–2317.
- (a) Shchepin, V. V.; Chuprikova, T. Y.; Petukhova, I. Y. *Zh. Org. Khim.* **1991**, 27, 1796–1797. (b) Shchepin, V. V.; Efremov, D. I.; Desyatkov, D. A. *Zh. Org. Khim.* **1993**, 29, 410–411.
- For a review on some aspects on the stereochemistry of the Reformatsky reactions, see: Guette, J. P. In *Organometallics Fonct. Ambidents, Recl. Commun., Colloq. Fr.-Bulg.*; Blagoev, B., Gaudemar-Bardone, F., Mladenova, M., Eds., 1980. Acad. Bulg. Sci., Inst. Chim. Org.: Sofia, pp 123–129.
- Akita, H.; Chen, S.-Y.; Nagumo, S. *Tetrahedron: Asymmetry* **1994**, 5, 1207–1210.
- DeGraw, J. I.; Christie, P. H.; Kisliuk, R. L.; Gaumont, Y.; Sirotnak, F. M. *J. Med. Chem.* **1990**, 33, 212–215.
- Khan, R. H.; Rastogi, R. C. *Indian J. Chem. Sect. B* **1992**, 31B, 276.
- Witkowski, S.; Rao, Y. K.; Premchandran, R. H.; Halushka, P. V.; Fried, J. *J. Am. Chem. Soc.* **1992**, 114, 8464–8472.
- Gomes Constantino, M.; Gonzaga de Oliveira Matias, L.; da Silva, G. V. J.; Gomes Heleno, V. C.; Gambardella, M. T. D. P. *Synth. Commun.* **1997**, 27, 4285–4295.
- Izquierdo, I.; Plaza, M. T.; Robles, R.; Mota, A. J.; Franco, F. *Tetrahedron: Asymmetry* **2001**, 12, 2749–2754.
- (a) Gawronski, J. K. *Tetrahedron Lett.* **1984**, 25, 2605. (b) Picotin, G.; Miginiac, P. *J. Org. Chem.* **1987**, 52, 4796. (c) Stephen, C. J. *Tetrahedron Lett.* **1992**, 33, 6193–6196.
- Guertler, S.; Ruf, S.; Johnner, M.; Otto, H. H. *Pharmazie* **1996**, 51, 811–815.
- Gaudemar, M. *Tetrahedron Lett.* **1983**, 24, 2749.
- (a) Kuwajima, I.; Nakamura, E.; Hashimoto, K. *Org. Synth.* **1983**, 61, 122. (b) Stocker, G. E.; Hoffmann, W. F.; Alberts, A. W.; Cragoe, E. J.; Peana, A. A.; Gilfillan, J. L.; Huff, J. W.; Novello, F. C.; Prugh, J. D.; Smith, R. L.; Willard, A. K. *J. Med. Chem.* **1985**, 28, 347. (c) Maruoka, K.; Hashimoto, S.; Kitagawa, Y.; Yamamoto, H.; Nozaki, H. *Bull. Chem. Soc. Jpn* **1980**, 53, 3301. (d) Maruoka, K.; Hashimoto, S.; Kitagawa, Y.; Yamamoto, H.; Nozaki, H. *J. Am. Chem. Soc.* **1977**, 99, 7705. (e) Kuroboshi, M.; Ishihara, T. *Tetrahedron Lett.* **1987**, 28, 6481. (f) Ishihara, T.; Kuroboshi, M. *Chem. Lett.* **1987**, 1145.
- (a) Bellasoued, M.; Arous-Chtara, R.; Gaudemar, M. *J. Organomet. Chem.* **1982**, 231, 185. (b) Lucas, M.; Guette, J. P. *J. Chem. Res. (S)* **1978**, 214. (c) Bellasoued, M.; Gaudemar, M. *J. Organomet. Chem.* **1975**, 93, 9. (d) Gaudemar-Bardone, F.; Gaudemar, M.; Mladenova, M. *Synthesis* **1987**, 1130. (e) Bogavac, M.; Arsenijevic, L.; Arsenijevic, V. *Bull. Soc. Chim. Fr.* **1980**, II, 145.
- Santaniello, E.; Manzocchi, A. *Synthesis* **1977**, 698.
- Slougui, N.; Rousseau, G. *Synth. Commun.* **1987**, 17, 1.
- (a) Csuk, R.; Fürstner, A.; Weidmann, H. *J. Chem. Soc., Chem. Commun.* **1986**, 775. (b) Csuk, R.; Glaenger, B. I. *J. Carbohydr. Chem.* **1990**, 9, 797–807.
- Rieke, R. D.; Li, P. T. J.; Burns, T. P.; Uhm, S. T. *J. Org. Chem.* **1981**, 46, 4323.
- Boudjouk, P.; Thompson, D. P.; Ohrbom, W. H.; Han, B. H. *Organometallics* **1986**, 5, 1257.
- Rieke, R. D.; Uhm, S. J. *Synthesis* **1975**, 452.
- Arnold, R. T.; Kulenovic, S. T. *Synth. Commun.* **1977**, 7, 223.
- Makosza, M.; Grela, K.; Fabianowski, W. *Tetrahedron* **1996**, 52, 9575–9580.
- Malfait, S.; Péliniski, L.; Brocard, J. *Tetrahedron: Asymmetry* **1998**, 9, 2595–2610.
- Ukaji, Y.; Yoshida, Y.; Inomata, K. *Tetrahedron: Asymmetry* **2000**, 11, 733–736.
- Aoyagi, Y.; Tanaka, W.; Ohta, A. *J. Chem. Soc., Chem. Commun.* **1994**, 1225–1226.
- Kataoka, Y.; Makihiro, I.; Yamagata, T.; Tani, K. *Organometallics* **1997**, 16, 4788–4795.
- Curran, T. T. *J. Org. Chem.* **1993**, 58, 6360–6363.
- Zhao, Z.; Ding, Y.; Zhao, G. *Synth. Commun.* **2001**, 31, 2089–2096.
- Ding, Y.; Zhao, Z.; Zhou, C. *Tetrahedron* **1997**, 53, 2899–2906.
- Fylyakova, V. I.; Saitkulova, F. G.; Pashkevich, K. I. *Izv. Akad. Nauk. Ser. Khim.* **1992**, 1228–1229.
- Horiuchi, Y.; Taniguchi, M.; Oshima, K.; Utimoto, K. *Tetrahedron Lett.* **1995**, 36, 5353–5356.

42. Basile, T.; Tagliavini, E.; Trombini, C.; Umani-Ronchi, A. *Synthesis* **1990**, 305–311.
43. Chattopadhyay, A.; Salaskar, A. *Synthesis* **2000**, 561–564.
44. Nozaki, K.; Oshima, K.; Utimoto, K. *Bull. Chem. Soc. Jpn* **1991**, 64, 403–409.
45. Shen, Y.; Qi, M. *J. Fluorine Chem.* **1994**, 67, 229–232.
46. Shen, Y.; Qi, M. *J. Chem. Res. (S)* **1993**, 222–223.
47. Ocampo, R.; Dolbier, W. R.; Abboud, K. A.; Zuluaga, F. *J. Org. Chem.* **2002**, 67, 72–78.
48. (a) Kagayama, A.; Igarashi, K.; Shiina, I.; Mukaiyama, T. *Bull. Chem. Soc. Jpn* **2000**, 73, 2579–2585. (b) Mukaiyama, T.; Kagayama, A.; Igarashi, K.; Shiina, I. *Chem. Lett.* **1999**, 1157–1158.
49. Hayashi, M.; Sugiyama, M.; Toba, T.; Oguni, N. *J. Chem. Soc., Chem. Commun.* **1990**, 767–768.
50. Hashimoto, Y.; Kikuchi, S. *Chem. Lett.* **2002**, 126–127.
51. Shimizu, M.; Kobayashi, F.; Hayakawa, R. *Tetrahedron* **2001**, 57, 9591–9595.
52. (a) Parrish, J. D.; Shelton, D. R.; Little, R. D. *Org. Lett.* **2003**, 5, 3615–3617. (b) Parrish, J. D.; Shelton, D. R.; Park, Y. S.; Little, R. D. *Proc. Electrochem. Soc.* **2003**, 2003–12, 161–164.
53. Sánchez, M.; Bermejo, F. *Tetrahedron Lett.* **1997**, 38, 5057–5060.
54. Kim, S.; Han, W. S.; Lee, J. M. *Bull. Kor. Chem. Soc.* **1992**, 13, 466–467.
55. Katritzky, A. R.; Nichols, D. A.; Qi, M. *Tetrahedron Lett.* **1998**, 39, 7063–7066.
56. (a) Vidal, A.; Nefzi, A.; Houghten, R. A. *J. Org. Chem.* **2001**, 66, 8268–8272. Vidal, A.; Nefzi, A.; Houghten, R. A. Abstr. Pap., 223rd ACS National Meeting, 2002, ORGN-234 and FLUO-008.
57. Katritzky, A. R.; Hong, Q.; Yang, Z. *J. Org. Chem.* **1995**, 60, 3405–3408. (b) Strah, S.; Katritzky, A. R.; Belyakov, S. A. *Abstr. Pap.-Am. Chem. Soc.* 1998, 215th ORGN-191.
58. For a recent review on chromium (II) and chromium (III)-mediated organic synthesis, included the chromium-Reformatsky reaction, see: (a) Fürstner, A. *Chem. Rev.* **1999**, 99, 991–1045. (b) Wessjohann, L. A.; Scheid, G. *Synthesis* **1999**, 1, 1–36.
59. For reviews on the application of SmI_2 in organic synthesis, see: (a) Krief, A.; Laval, A.-M. *Chem. Rev.* **1999**, 99, 745. (b) Molander, G. A.; Harris, C. R. *Chem. Rev.* **1996**, 96, 307. (c) Molander, G. A.; Harris, C. R. *Tetrahedron* **1998**, 54, 3321. (d) Skrydstrup, T. *Angew. Chem., Int. Ed. Engl.* **1997**, 36, 345. (e) Molander, G. A. In *Organic Reactions*; Paquette, L. A., Ed. John Wiley: New York, 1994, Vol. 46, p 211. (f) Imamoto, T. *Lanthanides in Organic Synthesis*; Academic Press: London, 1994. (g) Molander, G. A. *Chem. Rev.* **1992**, 92, 29. (h) Curran, D. P.; Fevig, T. L.; Jaspersen, C. P.; Tottleben, M. L. *Synlett* **1992**, 943. (i) Inanaga, J. *Trends Org. Chem.* **1990**, 1, 23–30.
60. For reviews on indium in organic synthesis, see: (a) Podlech, J.; Maier, T. C. *Synthesis* **2003**, 633. (b) Cintas, P. *Synlett* **1995**, 1087.
61. (a) Wessjohann, L. A.; Wild, H. *Synthesis* **1997**, 512–514. (b) Wessjohann, L. A.; Wild, H. *Synlett* **1997**, 6, 731–733.
62. Buse, C. T.; Heathcock, C. H. *Tetrahedron Lett.* **1978**, 19, 1685.
63. Mainz, J.; Arrieta, A.; Lopez, X.; Ugalde, J. M.; Cossio, F. P.; Lecea, B. *Tetrahedron Lett.* **1993**, 34, 6111–6114.
64. Wessjohann, L. A.; Wild, H. *Synlett* **1997**, 731.
65. Gabriel, T.; Wessjohann, L. A. *Tetrahedron Lett.* **1997**, 38, 1363–1366.
66. Gabriel, T.; Wessjohann, L. A. *Tetrahedron Lett.* **1997**, 38, 4387–4388.
67. Wessjohann, L. A.; Gabriel, T. *J. Org. Chem.* **1997**, 62, 3772–3774.
68. Wittenberg, R.; Beier, C.; Drager, G.; Jas, G.; Jasper, C.; Monenschein, H.; Kirschning, A. *Tetrahedron Lett.* **2004**, 45, 4457–4460.
69. Park, H. S.; Lee, I. S.; Kim, Y. H. *Tetrahedron Lett.* **1995**, 36, 1673–1674.
70. Fukuzawa, S.; Mutoh, K.; Tsuchimoto, T.; Hiyama, T. *J. Org. Chem.* **1996**, 61, 5400–5405.
71. Xu, X. L.; Lu, P.; Zhang, Y.-M. *Chin. Chem. Lett.* **1999**, 10, 729–730.
72. Zhang, J.-M.; Zhang, Y.-M. *Chin. J. Chem.* **2002**, 20, 111–113.
73. Hirano, K.; Fukuzawa, S. *Kidorui* **1998**, 32, 312–313.
74. Fukuzawa, S.; Hirai, K. *J. Chem. Soc., Perkin Trans. 1* **1993**, 1963–1965.
75. Lannou, M. I.; Helion, F.; Namy, J.-L. *Tetrahedron* **2003**, 59, 10551–10565.
76. Chauhan, K. K.; Frost, C. G.; Waite, D. C. Abstr. Pap.-220th ACS Meeting, 2000, ORGN-114.
77. Wang, Z.; Hammond, G. B. *J. Org. Chem.* **2000**, 65, 6547.
78. Johar, P. S.; Araki, S.; Butsugan, Y. *J. Chem. Soc., Perkin Trans. 1* **1992**, 711–713.
79. Bang, K.; Lee, K.; Park, Y. K.; Lee, P. H. *Bull. Kor. Chem. Soc.* **2002**, 23, 1272–1276.
80. Lee, P. H.; Bang, K.; Lee, K.; Sung, S.-Y.; Chang, S. *Synth. Commun.* **2001**, 31, 3781–3789.
81. Chung, W. J.; Higashiya, S.; Welch, J. T. *J. Fluorine Chem.* **2001**, 112, 343–347.
82. Banik, B. K.; Ghatak, A.; Becker, F. F. Abstr. Pap., 223rd ACS National Meeting, 2002, ORGN-308.
83. Izquierdo, I.; Plaza, M. T.; Tamayo, J. A.; Mota, A. J. *Synthesis* **2004**, 1083–1087.
84. Araki, S.; Katsumura, N.; Kawasaki, K.; Butsugan, Y. *J. Chem. Soc., Perkin Trans. 1* **1991**, 499–500.
85. (a) Schick, H.; Ludwig, R.; Schwarz, K.-H.; Kleiner, K.; Kunath, A. *J. Org. Chem.* **1994**, 59, 3161–3164. (b) Schick, H.; Ludwig, R.; Kleiner, K.; Kunath, A. *Tetrahedron* **1995**, 51, 2939–2946.
86. (a) Schwarz, K.-H.; Kleiner, K.; Ludwig, R.; Schick, H. *J. Org. Chem.* **1992**, 57, 4013–4015. (b) Schwarz, K.-H.; Kleiner, K.; Ludwig, R.; Schick, H. *Chem. Ber.* **1993**, 126, 1247–1249.
87. Durandetti, M.; Meignein, C.; Perichon, J. *Org. Lett.* **2003**, 5, 317–320.
88. Tussa, L.; Lebreton, C.; Mosset, P. *Chem.-A Eur. J.* **1997**, 3, 1064–1070.
89. Hirashita, T.; Kinoshita, K.; Yamamura, H.; Kawai, M.; Araki, S. *J. Chem. Soc., Perkin Trans. 1* **2000**, 825–828.
90. Conan, A.; Sibille, S.; Perichon, J. *J. Org. Chem.* **1991**, 56, 2018–2024.
91. Mcharek, S.; Sibille, S.; Nedelec, J. Y.; Perichon, J. *J. Organomet. Chem.* **1991**, 401, 211–215.
92. Shen, Z.; Zhang, J.; Zou, H.; Yang, M. *Tetrahedron Lett.* **1997**, 38, 2733–2736.
93. Kagoshima, H.; Hashimoto, Y.; Saigo, K. *Tetrahedron Lett.* **1998**, 39, 8465–8466.
94. Han, Y.; Chi, Z.-F.; Huang, Y. Z. *Chin. Chem. Lett.* **1996**, 7, 713–716.

95. (a) Kagoshima, H.; Hashimoto, Y.; Oguro, D.; Saigo, K. *J. Org. Chem.* **1998**, *63*, 691–697. (b) Kagoshima, H.; Hashimoto, Y.; Oguro, D.; Kutsuna, T.; Saigo, K. *Tetrahedron Lett.* **1998**, *39*, 1203–1206.
96. (a) Orsini, F.; Pelizzoni, T.; Pulici, M.; Vallarino, L. M. *J. Org. Chem.* **1994**, *59*, 1–3. (b) Orsini, F. *J. Org. Chem.* **1997**, *62*, 1159. (c) Orsini, F.; Pulici, M. In *Trends in Organometallic Chemistry*, Research Trends Ed., Trivandrum: India, 1994; Vol. 1, pp 625–667. (d) Orsini, F.; Pulici, M.; Vallarino, L. M. *J. Organomet. Chem.* **1995**, *495*, C1–C5.
97. Orsini, F. *Tetrahedron Lett.* **1998**, *39*, 1425–1428.
98. Orsini, F.; Lecciolli, C. *Tetrahedron: Asymmetry* **1997**, *8*, 4011–4019.
99. Pettit, G. R.; Grealish, M. P. *J. Org. Chem.* **2001**, *66*, 8640–8642.
100. Duffield, J. J.; Pettit, G. R. *J. Nat. Prod.* **2001**, *64*, 472–479.
101. (a) Kanai, K.; Wakabayashi, H.; Honda, T. *Org. Lett.* **2000**, *2*, 2549–2551. (b) Marshall, J. A. *Chemtracts* **2000**, *13*, 705–707.
102. Honda, T.; Wakabayashi, H.; Kanai, K. *Chem. Pharm. Bull.* **2002**, *50*, 307–308.
103. Review: Csuk, R.; Glänzer, B. I.; Fürstner, A. *Adv. Organomet. Chem.* **1988**, *28*, 85–137.
104. Csuk, R.; Schröder, C.; Krieger, C. *Tetrahedron* **1997**, *53*, 12947–12960.
105. Hanessian, S.; Girard, C. *Synlett* **1994**, 865–867.
106. Yokoyama, Y.; Tsuchikawa, T.; Hiramatsu, M.; Mochida, K. *Main Group Metal Chem.* **2000**, *23*, 691–693.
107. Takai, K.; Ueda, T.; Hayashi, T.; Moriwake, T. *Tetrahedron Lett.* **1996**, *37*, 7049–7052.
108. Suh, Y.-S.; Rieke, R. D. *Tetrahedron Lett.* **2004**, *45*, 1807–1809.
109. Takai, K.; Kakiuchi, T.; Utimoto, K. *J. Org. Chem.* **1994**, *59*, 2671–2673.
110. (a) Fürstner, A.; Hupperts, A. *J. Am. Chem. Soc.* **1995**, *117*, 4468–4475. (b) Knochel, P.; Yeh, M. C. P.; Berk, S. C.; Talbert, J. *J. Org. Chem.* **1988**, *53*, 2390–2392. (c) Gaudemar, M. *Bull. Soc. Chim. Fr.* **1962**, 974–987.
111. Kakiya, H.; Nishimae, S.; Shinokubo, H.; Oshima, K. *Tetrahedron* **2001**, *57*, 8807–8815.
112. Demir, A. S.; Tanyeli, C.; Mahasneh, A. S.; Aksoy, H. *Synthesis* **1994**, 155–157.
113. Davies, S. G.; Edwards, A. J.; Evans, G. B.; Mortlock, A. A. *Tetrahedron* **1994**, *50*, 6621–6642.
114. Shibata, I.; Yamaguchi, T.; Baba, A.; Matsuda, H. *Chem. Lett.* **1993**, 97–100.
115. Shibata, I.; Suwa, T.; Sakakibara, H.; Baba, A. *Org. Lett.* **2002**, *4*, 301–303.
116. (a) Kuznetsov, N. Y.; Khrustalev, V. N.; Terentiev, A. B.; Belokon, Y. N. *Russ. Chem. Bull.* **2001**, *50*, 548–550. (b) Terent'ev, A. B.; Vasil'eva, T. T.; Kuz'mina, N. A.; Chakhovskaya, O. V.; Brodsky, E. S.; Belokon', Y. N. *Russ. Chem. Bull.* **2000**, *49*, 722–723. (c) Terent'ev, A. B.; Vasil'eva, T. T.; Kuz'mina, N. A.; Mysov, E. I.; Ikonnikov, N. S.; Kuznetsov, N. Y.; Belokon', Y. N. *Russ. Chem. Bull.* **1999**, *48*, 1121–1127.
117. Terentiev, A. B.; Vasilieva, T. T.; Kuzmina, N. A.; Mysov, E. I.; Kuznetsov, N. Y.; Belokon, Y. N. *J. Chem. Res. (S)* **1998**, 306 [*J. Chem. Res. (M)* **1998**, 1281–9].
118. (a) Roberts, C. D.; Schutz, R.; Leumann, C. J. *Synlett* **1999**, 6, 819–821. (b) Cantin, M.; Schutz, R.; Leumann, C. J. *Tetrahedron Lett.* **1997**, *38*, 4211–4214.
119. De Vos, E.; Esmans, E. L.; Alderweireldt, F. C.; Balzarini, J.; De Clercq, E. *J. Heterocycl. Chem.* **1993**, *30*, 1245–1252.
120. Yanagisawa, A.; Takahashi, H.; Arai, T. *Chem. Commun.* **2004**, 580–581.
121. For a review on non-conventional reaction conditions (ultrasound, high pressure and microwave heating) in organic synthesis, see: Giguere, R. *J. Org. Synth.: Theory Appl.* **1989**, *1*, 103–172.
122. For a review on solvent-free reactions, see: Tanaka, K.; Toda, F. *Chem. Rev.* **2000**, *100*, 1025–1074.
123. For a review on water medium, see: Li, C. J. *Chem. Rev.* **1993**, *93*, 2023.
124. (a) Gholap, A. R.; Chavan, A. P. *J. Chem. Res. (S)* **2003**, 374–376. (b) Tanaka, K.; Kishigami, S.; Toda, F. *J. Org. Chem.* **1991**, *56*, 4333–4334.
125. (a) Latouche, R.; Texier-Boullet, F.; Hamelin, J. *Bull. Soc. Chim. Fr.* **1993**, *130*, 535–546. (b) Latouche, R.; Texier-Boullet, F.; Hamelin, J. *Tetrahedron Lett.* **1991**, *32*, 1179–1182.
126. Wang, G.-W.; Murata, Y.; Komatsu, K.; Wan, T. S. M. *Chem. Commun.* **1996**, 2059–2060.
127. Wang, Z.; Meier, M. S. Abstr. Pap.-223rd ACS Meeting 2002, ORGN-134.
128. Dellinger, D. J.; Sheehan, D. M.; Christensen, N. K.; Lindberg, J. G.; Caruthers, M. H. *J. Am. Chem. Soc.* **2003**, *125*, 940–950.
129. Bieber, L. W.; Malvestiti, I.; Storch, C. *J. Org. Chem.* **1997**, *62*, 9061–9064.
130. Rollin, Y.; Gebehenne, C.; Derien, S.; Dunach, E.; Perichon, J. *J. Organomet. Chem.* **1993**, *461*, 9–13.
131. Adrian, J. C.; Snapper, M. L. *J. Org. Chem.* **2003**, *68*, 2143–2150.
132. Hilgers, G.; Hess, J. Ger. Offen. German Patent: DE 90-4016076 9900518, 1991, 6 pp.
133. Lim, C. L.; Han, B. H. *J. Kor. Chem. Soc.* **1991**, *35*, 762–764.
134. (a) Ross, N. A.; Bartsch, R. A. *J. Org. Chem.* **2003**, *68*, 360–366. (b) Ross, N. A.; Bartsch, R. A.; Marchand, A. P. *ARKIVOC* **2003**, 27–30.
135. Ross, N. A.; MacGregor, R. R.; Bartsch, R. A. *Tetrahedron* **2004**, *60*, 2035–2041.
136. Kitazume, T. *Ultrasonics* **1990**, *28*, 322–325.
137. Baldoli, C.; Del Buttero, P.; Licandro, E.; Papagni, A. *Tetrahedron* **1996**, *52*, 4849–4856.
138. Coe, P. L.; Löhr, M.; Rochin, C. *J. Chem. Soc., Perkin Trans. I* **1998**, 2803–2812.
139. Andrés, J. M.; Martinez, M. A.; Pedrosa, R.; Perez-Encabo, A. *Synthesis* **1996**, 1070–1072.
140. Altenburger, J. M.; Schirlin, D. *Tetrahedron Lett.* **1991**, *32*, 7255–7258.
141. (a) Ross, N. A.; Bartsch, R. A. *J. Heterocycl. Chem.* **2001**, *38*, 1255–1258. (b) Ross, N. A.; Bartsch, R. A. Abstr. Pap.-Am. Chem. Soc. 2002, 224th ORGN-888.
142. Durant, A.; Delplancke, J. L.; Libert, V.; Reisse, J. *Eur. J. Org. Chem.* **1999**, 2845–2851.
143. Makosza, M.; Nieczypor, P.; Grela, K. *Tetrahedron* **1998**, *54*, 10827–10836.
144. Mylari, B. L.; Zembrowki, W. J. *J. Org. Chem.* **1991**, *56*, 2587–2589.
145. Andrés, C.; González, A.; Pedrosa, R.; Perez-Encabo, A. *Tetrahedron Lett.* **1992**, *33*, 2895–2898.
146. Bott, K. *Tetrahedron Lett.* **1994**, *35*, 555–556.
147. Syed, J.; Forster, S.; Effenberger, F. *Tetrahedron: Asymmetry* **1998**, *9*, 805–815.

148. Adrian, J. C.; Barkin, J. L.; Hassib, L. *Tetrahedron Lett.* **1999**, 34, 2457–2460.
149. Evans, D. A.; Hoveyda, A. H. *J. Am. Chem. Soc.* **1990**, 112, 6447–6449.
150. Bitterwolf, T. E.; Dai, X. *J. Organomet. Chem.* **1992**, 440, 103–112.
151. Ferraz, H. M. C.; Silva, L. F. *Tetrahedron* **2001**, 57, 9939–9949.
152. Olughade, T. A.; Waigh, R. D.; Mackay, S. P. *J. Chem. Soc., Perkin Trans. 1* **1990**, 2657–2660.
153. Krasuts'kii, P. O.; Semenova, I. G.; Voitsits'ka, O. D.; Simurova, N. V. *Ukr. Khim. Zh.* **1997**, 63, 107–110.
154. Olah, G. A.; Reddy, V. P.; Casanova, J.; Prakash, G. K. *J. Org. Chem.* **1992**, 57, 6431–6434.
155. Andrés, J. M.; Pedrosa, R.; Pérez, A.; Pérez-Encabo, A. *Tetrahedron* **2001**, 57, 8521–8530.
156. Yates, P.; Kaldas, M. *Can. J. Chem.* **1992**, 70, 1492–1505.
157. Izquierdo, I.; Plaza, M. T.; Robles, R.; Mota, A. *Tetrahedron: Asymmetry* **1997**, 8, 2597–2606.
158. Dener, J. M.; Zhang, L. H.; Rapoport, H. *J. Org. Chem.* **1993**, 58, 1159–1166.
159. (a) Michael, J. P.; de Koning, C. B.; Stanbury, T. V. *Tetrahedron Lett.* **1996**, 37, 9403–9406. (b) Michael, J. P.; Hosken, G. D.; Howard, A. S. *Tetrahedron* **1988**, 44, 3025–3026. (c) Michael, J. P.; de Koning, C. B.; Hosken, G. D.; Stanbury, T. V. *Tetrahedron* **2001**, 57, 9635–9648.
160. Shchepin, V. V.; Gladkova, G. E. *Zh. Org. Khim.* **1995**, 31, 1094.
161. Shchepin, V. V.; Fatukhova, Y. K.; Kirillov, N. T.; Russkikh, N. Y.; Litvinov, D. N. *Russ. J. Org. Chem.* **2000**, 36, 1120–1123.
162. Ito, Y.; Terashima, S. *Tetrahedron* **1991**, 47, 2821–2834.
163. Ito, Y.; Sasaki, A.; Tamoto, K.; Sunagawa, M.; Terashima, S. *Tetrahedron* **1991**, 47, 2801–2820.
164. Kobayashi, Y.; Ito, Y.; Terashima, S. *Tetrahedron* **1992**, 48, 55–66.
165. Mori, M.; Oida, S. *Bioorg. Med. Chem. Lett.* **1993**, 3, 2389–2392.
166. (a) Fukuzawa, S.; Matsuzawa, H.; Yoshimitsu, S. *J. Org. Chem.* **2000**, 65, 1702–1706. (b) Fukuzawa, S.; Tatsuzawa, M.; Hirano, K. *Tetrahedron Lett.* **1998**, 39, 6899–6900.
167. Seki, M.; Kondo, K.; Iwasaki, T. *J. Chem. Soc., Perkin Trans. 1* **1996**, 2851–2856.
168. (a) Kondo, K.; Seki, M.; Kuroda, T.; Yamanaka, T.; Iwasaki, T. *J. Org. Chem.* **1995**, 60, 1096–1097. (b) Kondo, K.; Seki, M.; Kuroda, T.; Yamanaka, T.; Iwasaki, T. *J. Org. Chem.* **1997**, 62, 2877–2884.
169. (a) Pyun, D. K.; Jeong, W. J.; Jung, H. J.; Kim, J. H.; Lee, J. S.; Lee, C. H.; Kim, B. *J. Synlett* **2001**, 1950–1952. (b) Jeong, W. J.; Pyun, D. K.; Jung, H. J.; Kim, J. H.; Lee, C. H.; Kim, B. *J. Abstr. Pap.-Am. Chem. Soc.* 2001, 221st ORGN-555.
170. Lu, X.; Xu, Z.; Yang, G.; Fan, R. *Org. Proc. Res. Dev.* **2001**, 5, 186–188.
171. Andrés, J. M.; Pedrosa, R.; Perez-Encabo, A. *Tetrahedron* **2000**, 56, 1217–1223.
172. Shchepin, V. V.; Gladkova, G. E.; Russkikh, N. Y. *Zh. Org. Khim.* **1992**, 28, 1156–1162.
173. Shchepin, V. V.; Tryatsin, A. A.; Shchepin, R. V.; Kalyuzhnyi, M. M.; Lewis, S. B. *Russ. J. Org. Chem.* **2001**, 37, 1596–1599.
174. Shchepin, V. V.; Gladkova, G. E. *Zh. Org. Khim.* **1993**, 29, 474–479.
175. For reviews, see: (a) Werstiuk, N. H. *Tetrahedron* **1982**, 39, 205. (b) Hoppe, D. *Angew. Chem., Int. Ed. Engl.* **1984**, 23, 932.
176. Fukuzawa, S.; Sumimoto, N.; Fujinami, T.; Sakai, S. *J. Org. Chem.* **1990**, 55, 1628–1631.
177. (a) Oler, E.; Reininger, K.; Schmidt, U. *Angew. Chem., Int. Ed. Engl.* **1970**, 9, 457. (b) Löffler, A.; Pratt, D.; Pucknat, J.; Gelbard, G.; Dreiding, A. S. *Chimia* **1969**, 23, 413.
178. Ikramuddeen, T. M.; Chandrasekara, N.; Ramarajan, K.; Garrison, G. L.; Berlin, K. *Indian J. Chem. Sect. B* **1994**, 33B, 362–364.
179. For reviews on the synthesis of α -methylene- γ -butyrolactones: (a) Hoffmann, H. M. R.; Rabe, J. *Angew. Chem., Int. Ed. Engl.* **1985**, 24, 94. (b) Grieco, P. A. *Synthesis* **1975**, 67. (c) Petragnani, N.; Feraz, H. M. C.; Silva, G. V. *J. Synthesis* **1986**, 157.
180. Richter, F.; Bauer, M.; Perez, C.; Maichle-Möessmer, C.; Maier, M. E. *J. Org. Chem.* **2002**, 67, 2474–2480.
181. Maier, M. E.; Pérez, C. *Synlett* **1998**, 159–160.
182. Rauter, A. P.; Ferreira, M. J.; Font, J.; Virgili, A.; Figueredo, M.; Figueiredo, J. A.; Ismael, M. I.; Canda, T. L. *J. Carbohydr. Chem.* **1995**, 14, 929–948.
183. Rauter, A. P.; Figueiredo, J.; Ismael, M.; Canda, T.; Font, J.; Figueredo, M. *Tetrahedron: Asymmetry* **2001**, 12, 1131–1146.
184. (a) Kim, J. C.; Kim, J.-A.; Kim, Si.-H.; Park, J. I.; Kim, Se.-H.; Choi, S.-K.; Park, W.-W. *Arch. Pharm. Res.* **1996**, 19, 235–239. (b) Kim, J. C.; Kim, J.-A.; Park, J. I.; Kim, S.-H.; Kim, S.-H.; Choi, S.-K.; Park, W.-W. *Arch. Pharm. Res.* **1997**, 20, 253–258. (c) Huang, B.-R.; Lee, K.-H.; Tseng, T.-L.; Wang, H.-M.; Chen, C.-F.; Tzeng, C.-C. *Gaox. Yix. Kex. Zaz.* **1993**, 9, 707–711. (d) Huang, B.-R.; Lee, K.-H.; Hwang, L. C.; Han, C. H.; Chen, Y. L.; Tzeng, C. C.; Chen, C. F. *Zhong. Yaox. Zaz.* **1991**, 43, 447–455.
185. Kim, J. C.; Kim, S.-H.; Kim, J.-A.; Choi, S.-K.; Park, W.-W. *Arch. Pharm. Res.* **1998**, 21, 458–464.
186. (a) Tzeng, C.-C.; Lee, K.-H.; Wang, T.-C.; Han, C.-H.; Chen, Y.-L. *Pharm. Res.* **2000**, 17, 715–719. (b) Tzeng, C.-C.; Chen, I.-L.; Chen, Y.-L.; Wang, T.-C.; Chang, Y.-L.; Teng, C.-M. *Helv. Chim. Acta* **2000**, 83, 349–358. (c) Hsu, S.-L.; Chen, Y.-L.; Tzeng, C.-C.; Fang, K.-C.; Sheu, J.-Y. *Helv. Chim. Acta* **2001**, 84, 874–879. (d) Chen, Y.-L.; Chen, I.-L.; Tzeng, C.-C.; Wang, T.-C. *Helv. Chim. Acta* **2000**, 83, 989–994. (e) Wang, T.-C.; Chen, Y.-L.; Tzeng, C.-C.; Liou, S.-S.; Tzeng, W.-F.; Chang, Y.-L.; Teng, C.-M. *Helv. Chim. Acta* **1998**, 81, 1038–1047. (f) Tzeng, C.-C.; Zhao, Y.-L.; Chen, Y.-L.; Liou, S.-S.; Wang, T.-C.; Chang, Y.-L.; Teng, C.-M. *Helv. Chim. Acta* **1997**, 80, 2337–2344. (g) Tzeng, C.-C.; Chen, Y.-L.; Wang, C.-J.; Wang, T.-C.; Chang, Y.-L.; Teng, C.-M. *Helv. Chim. Acta* **1997**, 80, 1161–1168. (h) Wang, T.-C.; Chen, Y.-L.; Tzeng, C.-C.; Liou, S.-S.; Chang, Y.-L.; Teng, C.-M. *Helv. Chim. Acta* **1996**, 79, 1620–1626. (i) Chen, Y.-L.; Wang, T.-C.; Lee, H.-H.; Tzeng, C.-C.; Chang, Y.-L.; Teng, C.-M. *Helv. Chim. Acta* **1996**, 79, 651–657. (j) Chen, Y.-L.; Wang, T.-C.; Fang, K.-C.; Chang, N.-C.; Tzeng, C.-C. *Heterocycles* **1999**, 50, 453–462.
187. Wang, T.-C.; Zhao, Y.-L.; Kuo, D.-H. *Eur. J. Med. Chem.* **2001**, 36, 909–914.
188. Sawant, M. S.; Katoch, R.; Trivedi, G. K.; Desai, U. R. *J. Chem. Soc., Perkin Trans. 1* **1998**, 843–846.
189. Dembélé, Y. A.; Belaud, C.; Hitchcock, P.; Villiéras, J. *Tetrahedron: Asymmetry* **1992**, 3, 351–354.

190. Yamamoto, H.; Oritani, T.; Yamashita, K. *Agric. Biol. Chem.* **1990**, *54*, 1923–1929.
191. Mastalerz, H.; Menard, M. *J. Org. Chem.* **1994**, *59*, 3223–3226.
192. Sakai, K.; Takahashi, K.; Nukano, T. *Tetrahedron* **1992**, *48*, 8229–8238.
193. Mecozzi, T.; Petrini, M. *Tetrahedron Lett.* **2000**, *41*, 2709–2712.
194. Jain, S.; Jain, R.; Singh, J.; Anand, N. *Tetrahedron Lett.* **1994**, *35*, 2951–2954.
195. Shchepin, V. V.; Rodygin, A. S.; Balandina, M. I. *Zh. Org. Khim.* **1991**, *27*, 2035–2039.
196. Csuk, R.; Glaenger, B. I. *J. Carbohydr. Chem.* **1990**, *9*, 809–822.
197. Fürstner, A.; Kollegger, G.; Weidmann, H. *J. Organomet. Chem.* **1991**, *414*, 295–305.
198. Kuroda, C.; Ito, K. *Bull. Chem. Soc. Jpn* **1996**, *69*, 2297–2303.
199. (a) Akai, S.; Tsuzuki, Y.; Matsuda, S.; Kitagaki, S.; Kita, Y. *J. Chem. Soc., Perkin Trans. 1* **1992**, 2813–2820. (b) Akai, S.; Tsuzuki, Y.; Matsuda, S.; Kitagaki, S.; Kita, Y. *Synlett* **1991**, 911–912.
200. (a) Palomo, C.; Aizpurua, J. M.; Concepcion Lopez, M.; Aurrekoetxea, N. *Tetrahedron Lett.* **1990**, *31*, 2205–2208. (b) Palomo, C.; Aizpurua, J. M.; Aurrekoetxea, N. *Tetrahedron Lett.* **1990**, *31*, 2209–2210.
201. Jacobsen, M. F.; Turks, M.; Hazell, R.; Skrydstrup, T. *J. Org. Chem.* **2002**, *67*, 2411–2417.
202. Ricci, M.; Blakskjr, P.; Skrydstrup, T. *J. Am. Chem. Soc.* **2000**, *122*, 12413–12421.
203. Dutta, L. N.; Bhattacharyya, M.; Sarkar, A. K. *Can. J. Chem.* **1995**, *73*, 1556–1562.
204. Shchepin, V. V.; Korzun, A. E.; Sazhneva, Y. K.; Nedugov, A. N. *Chem. Heterocycl. Compd.* **2001**, *37*, 374–375.
205. (a) Shchepin, V. V.; Fotin, D. V.; Nedugov, A. N.; Fotin, V. V. *Russ. J. Org. Chem.* **2002**, *38*, 256–258. (b) Shchepin, V. V.; Fotin, D. V.; Nedugov, A. N.; Fotin, V. V.; Vakhrin, M. I. *Chem. Heterocycl. Compd.* **2001**, *37*, 650.
206. Shchepin, V. V.; Kirillov, N. F. *Chem. Heterocycl. Compd.* **2000**, *36*, 1110–1111.
207. Kirillov, N. F.; Shchepin, V. V. *Russ. J. Org. Chem.* **2001**, *37*, 811–813.
208. Kirillov, N. F.; Shchepin, V. V. *Russ. J. Org. Chem.* **2001**, *37*, 1223–1224.
209. Kise, N.; Yamazaki, H.; Mabuchi, T.; Shono, T. *Tetrahedron Lett.* **1994**, *35*, 1561–1564.
210. Nishiyama, T.; Kishi, H.; Kitano, K.; Yamada, F. *Bull. Chem. Soc. Jpn* **1994**, *67*, 1765–1768.
211. Alberola, A.; Álvarez, M. A.; Andrés, C.; González, A.; Pedrosa, R. *Synthesis* **1990**, 1057–1058.
212. Datta, A.; Ila, H.; Junjappa, H. *J. Org. Chem.* **1990**, *55*, 5589–5594.
213. Ukaji, Y.; Takenaka, S.; Horita, Y.; Inomata, K. *Chem. Lett.* **2001**, 254–255.
214. Shankar, B. B.; Kirkup, M. P.; McCombie, S. W.; Clader, J. W.; Ganguly, A. K. *Tetrahedron Lett.* **1996**, *37*, 4095–4098.
215. Robinson, A. J.; Wyatt, P. B. *Tetrahedron* **1993**, *49*, 11329–11340.
216. Lee, J.-H.; Lee, Y.-Y.; Goo, Y.-M.; Kim, K. *Bull. Kor. Chem. Soc.* **1993**, *14*, 172–174.
217. Hart, D. J.; HA, D.-C. *Chem. Rev.* **1989**, *89*, 1447.
218. Fotin, V. V.; Shchepin, V. V.; Fotin, D. V.; Vakhrin, M. I. *Russ. J. Org. Chem.* **1999**, *35*, 1278–1281.
219. Lindell, S. D.; Turner, R. M. *Tetrahedron Lett.* **1990**, *31*, 5381–5384.
220. Schick, H.; Ludwig, R. *Synthesis* **1992**, 369–370.
221. Wang, M.-X.; Liu, Y.; Gao, H.-Y.; Zhang, Y.; Yu, C.-Y.; Huang, Z.-T.; Fleet, G. W. J. *J. Org. Chem.* **2003**, *68*, 3281–3286.
222. Brandeenge, S.; Holmgren, E.; Leijonmarck, H.; Rodríguez, B. *Acta Chem. Scand.* **1995**, *49*, 922–928.
223. Yassin, F. A.; El-Faragy, A. F.; El-Mobayed, M. M.; El-Kady, M. Y.; Abd El-Maksoud, M. R. *J. Chem. Soc. Pakist.* **1991**, *13*, 267–272.
224. El-Safty, M. A.; El-Bahaei, S. A.; Yassin, F. A. *Afinidad* **1992**, *49*, 131–134.
225. (a) Kashima, C.; Huang, X. C.; Harada, Y.; Hosomi, A. *J. Org. Chem.* **1993**, *58*, 793–794. (b) Kashima, C.; Kita, I.; Takahashi, K.; Hosomi, A. *J. Heterocycl. Chem.* **1995**, *32*, 723–725.
226. (a) Kielbasinski, P.; Mikolajczyk, M. *Synthesis* **1995**, 144–146. (b) Fagadar-Cosma, E.; Fagadar-Cosma, G. *Rev. Roum. Chim.* **2003**, *48*, 211–217.
227. Lee, H. K.; Kim, J.; Pak, C. S. *Tetrahedron Lett.* **1999**, *40*, 2173–2174.
228. Michael, J. P.; de Koning, C. B.; Petersen, R. L.; Stanbury, T. V. *Tetrahedron Lett.* **2001**, *42*, 7513–7516.
229. Chandrasekharan, M.; Bhat, L.; Ila, H.; Junjappa, H. *Tetrahedron Lett.* **1993**, *34*, 6439–6442.
230. Hannick, S. M.; Kishi, Y. J. *J. Org. Chem.* **1983**, *48*, 3833–3835.
231. Mohri, K.; Yoshida, Y.; Mataga, S.; Hisatsune, Y.; Isobe, K.; Tsuda, Y. *Heterocycles* **1997**, *45*, 2405–2412.
232. Dartiguelongue, C.; Payan, S.; Duval, O.; Gomès, L. M.; Waigh, R. *Bull. Soc. Chim. Fr.* **1997**, *134*, 769–772.
233. Duffield, J. J.; Regan, A. C. *Tetrahedron: Asymmetry* **1996**, *7*, 663–666.
234. Poehlauer, P.; Hartmann, M.; Mayrhofer, H. Abstr. Pap.-Am. Chem. Soc. 2001, 221st ORGN-507.
235. Bernardi, L.; Bonini, B. F.; Capito, E.; Dessole, G.; Fochi, M.; Comes-Franchini, M.; Ricci, A. *Synlett* **2003**, 1778–1782.
236. Sheppard, A.; Miller, M. J. *J. Chem. Soc., Perkin Trans. 1* **1990**, 2519–2525.
237. Akhmetvaleev, R. R.; Imaeva, L. R.; Belogaeva, T. A.; Baikova, I. P.; Miftakhov, M. S. *Russ. J. Org. Chem.* **1999**, *35*, 242–245.
238. Lavalley, J. F.; Spino, C.; Ruel, R.; Hogan, K. T.; Deslongchamps, P. *Can. J. Chem.* **1992**, *70*, 1406–1426.
239. (a) Ruel, R.; Deslongchamps, P. *Tetrahedron Lett.* **1990**, *31*, 3961–3964. (b) Ruel, R.; Deslongchamps, P. *Can. J. Chem.* **1992**, *70*, 1939–1949.
240. Zegelman, L.; Hassner, A.; Mendel, Z.; Dunkelblum, E. *Tetrahedron Lett.* **1993**, *34*, 5641–5644.
241. Dyer, J.; Keeling, S.; Moloney, M. G. *Tetrahedron Lett.* **1996**, *37*, 4573–4576.
242. Batra, S.; De, D.; Seth, M.; Bhaduri, A. P. *J. Chem. Res. (S)* **1993**, 202 [J. Chem. Res. (M) **1993**, 1228].
243. (a) Shchepin, V. V.; Fotin, D. V. *Chem. Heterocycl. Compd.* **2002**, *38*, 1430–1431. (b) Shchepin, V. V.; Fotin, D. V.; Shurov, S. N. *Russ. J. Org. Chem.* **2003**, *39*, 1231–1233.
244. Menicagli, R.; Samaritani, S. *Tetrahedron* **1996**, *52*, 1425–1432.

245. Mazumdar, S. N.; Mahajan, M. P. *Tetrahedron Lett.* **1990**, 31, 4215–4218.
246. Vedejs, E.; Duncan, S. M. *J. Org. Chem.* **2000**, 65, 6073–6081.
247. Heathcock, C. H.; Ruggeri, R. B.; McClure, K. F. *J. Org. Chem.* **1992**, 57, 2585–2594.
248. Inanaga, J.; Yokoyama, Y.; Handa, Y.; Yamaguchi, M. *Tetrahedron Lett.* **1991**, 32, 6371–6374.
249. Molander, G. A.; Etter, J. B.; Harring, L. S.; Thorel, P. J. *J. Am. Chem. Soc.* **1991**, 113, 8036–8045.
250. Fujita, K.; Mori, K. *Eur. J. Org. Chem.* **2001**, 493–502.
251. Shiina, I.; Uoto, K.; Mori, N.; Kosugi, T.; Mukaiyama, T. *Chem. Lett.* **1995**, 181–182.
252. Takemura, T.; Nishii, Y.; Takahashi, S.; Kobayashi, J.; Nakata, T. *Tetrahedron* **2002**, 58, 6359–6365.
253. Inoue, S.; Iwabuchi, Y.; Irie, H.; Hatakeyama, S. *Synlett* **1998**, 735–736.
254. Reddy, P. P.; Yen, K.-F.; Uang, B.-J. *J. Org. Chem.* **2002**, 67, 1034–1035.
255. (a) Inoue, M.; Sasaki, M.; Tachibana, K. *J. Org. Chem.* **1999**, 64, 9416–9429. (b) Inoue, M.; Sasaki, M.; Tachibana, K. *Tetrahedron* **1999**, 55, 10949–10970. (c) Inoue, M.; Sasaki, M.; Tachibana, K. *Tetrahedron Lett.* **1997**, 38, 1611–1614.
256. Ichikawa, S.; Minakawa, N.; Shuto, S.; Tanaka, M.; Sasaki, T.; Matsuda, A. *Nucl. Acids Symp. Ser.* 1996, 35 (33th Symposium on Nucleic Acids Chemistry, 1996), 13–14. (b) Ichikawa, S.; Shuto, S.; Minakawa, N.; Matsuda, A. *J. Org. Chem.* **1997**, 62, 1368–1375.
257. Yamada, S.; Tanaka, A.; Oritani, T. *Biosci. Biotech. Biochem.* **1995**, 59, 1657–1660.
258. Nagamitsu, T.; Tacano, D.; Fukuda, T.; Otoguro, K.; Kuwajima, I.; Harigaya, Y.; Omura, S. *Org. Lett.* **2004**, 6, 1865–1867.
259. Molander, G. A.; Brown, G. A.; Storch de Gracia, I. *J. Org. Chem.* **2002**, 67, 3459–3463.
260. Kim, S.-J.; Kang, H.-Y.; Sherman, D. H. *Synthesis* **2001**, 1790–1793.
261. Fukuzawa, S.; Miura, M.; Matsuzawa, H. *Tetrahedron Lett.* **2001**, 42, 4167–4169.
262. Nerz-Stormes, M.; Thornton, E. R. *J. Org. Chem.* **1991**, 56, 2489.
263. Obringer, M.; Colobert, F.; Neugnot, B.; Solladié, G. *Org. Lett.* **2003**, 5, 629–632.
264. Basavaiah, D.; Bharathi, T. K. *Tetrahedron Lett.* **1991**, 32, 3417–3420.
265. Kende, A. S.; Kawamura, K.; DeVita, R. J. *J. Am. Chem. Soc.* **1990**, 112, 4070–4072.
266. Blaskovich, M. A.; Lajoie, G. A. *J. Am. Chem. Soc.* **1993**, 115, 5021–5030.
267. Ding, Y.; Wang, J.; Abboud, K. A.; Xu, Y.; Dolbier, W. R.; Richards, N. G. J. *J. Org. Chem.* **2001**, 66, 6381–6388.
268. Brocard, J.; Mahmoudi, M.; Pelinski, L.; Maciejewski, L. *Tetrahedron* **1990**, 46, 6995–7002.
269. (a) Sorochinsky, A.; Voloshin, N.; Markovsky, A.; Belik, M.; Yasuda, N.; Uekusa, H.; Ono, T.; Berbasov, D. O.; Soloshonok, V. A. *J. Org. Chem.* **2003**, 68, 7448–7454. (b) Soloshonok, V. A.; Ohkura, H.; Sorochinsky, A.; Voloshin, N.; Markovsky, A.; Belik, M.; Yamazaki, T. *Tetrahedron Lett.* **2002**, 43, 5445–5448. (c) Sorochinsky, A.; Berbasov, D. O.; Soloshonok, V. A. *Abstr. Pap.-Am. Chem. Soc.* 2003, 226th. ORGN-530.
270. Staas, D. D.; Savage, K. L.; Homnick, C. F.; Tsou, N. N.; Ball, R. G. *J. Org. Chem.* **2002**, 67, 8276–8279.
271. Ribeiro, C. M. R.; Santos, E. S.; Jardim, A. H. O.; Maia, M. P.; da Silva, F. C.; Moreira, A. P. D.; Ferreira, V. F. *Tetrahedron: Asymmetry* **2002**, 13, 1703–1706.
272. (a) Mi, A.; Wang, Z.; Zhang, J.; Jiang, Y. *Synth. Commun.* **1997**, 27, 1469–1473. (b) Mi, A.; Wang, Z.; Chen, Z.; Jiang, Y. *Tetrahedron: Asymmetry* **1995**, 6, 2641–2642.
273. Andrés, J. M.; Martin, Y.; Pedrosa, R.; Perez-Encabo, A. *Tetrahedron* **1997**, 53, 3787–3794.
274. Soai, K.; Kawase, Y. *Tetrahedron: Asymmetry* **1991**, 2, 781–784.
275. Pini, D.; Uccello-Barretta, G.; Mastantuono, A.; Salvadori, P. *Tetrahedron* **1997**, 53, 6065–6072.
276. (a) Pini, D.; Mastantuono, A.; Salvadori, P. *Tetrahedron: Asymmetry* **1994**, 5, 1875–1876. (b) Mastantuono, A.; Pini, D.; Rolfini, C.; Salvadori, P. *Chirality* **1995**, 7, 499–504.
277. Soai, K.; Oshio, A.; Saito, T. *J. Chem. Soc., Chem. Commun.* **1993**, 811–812.
278. Seki, K.; Tatzawa, M.; Fukuzawa, S. *Kidorui* **1997**, 30, 282–283.
279. (a) Ojida, A.; Yamano, T.; Taya, N.; Tasaka, A. *Org. Lett.* **2002**, 4, 3051–3054. (b) Park, D. H.; Choi, H. J.; Lee, S.-G. *J. Kor. Chem. Soc.* **2003**, 47, 597–600.
280. Zhang, Y.; Wu, W. *Tetrahedron: Asymmetry* **1997**, 8, 3575–3578.
281. (a) Wang, Z. Y.; Shen, J.; Jiang, C. S.; You, T. P. *Chin. Chem. Lett.* **2000**, 11, 659–662. (b) Jiang, C.; Li, D.; Cheng, Q.; Ke, Y.; You, T. *Huaxue Tongbao* **2001**, 637–640.
282. Yadav, J. S.; Praveen Kumar, T. K.; Maniyan, P. P. *Tetrahedron Lett.* **1993**, 34, 2969–2972.
283. Greiner, E.; Folk, J. E.; Jacobson, A. E.; Rice, K. C. *Bioorg. Med. Chem.* **2004**, 12, 233–238.
284. Kigoshi, H.; Kita, M.; Ogawa, S.; Itoh, M.; Uemura, D. *Org. Lett.* **2003**, 5, 957–960.
285. VanSickle, A. P.; Rapoport, H. *J. Org. Chem.* **1990**, 55, 895–901.
286. Orsini, F.; Pelizzoni, F. *Carbohydr. Res.* **1993**, 243, 183–189.
287. Fkyerat, A.; Burki, N.; Tabacchi, R. *Tetrahedron: Asymmetry* **1996**, 7, 2023–2028.
288. Zubaidha, P. K.; Chavan, S. P.; Racherla, U. S.; Ayyangar, N. R. *Tetrahedron* **1991**, 47, 5759–5768.
289. Kapha, T. K.; Ghosal, M.; Mukherjee, D. *Indian J. Chem. Sect. B.* **1990**, 301–303.
290. Lee, S. H.; Lee, S.-G.; Song, C. E.; Kim, I. O.; Chung, B. Y. *Kor. J. Med. Chem.* **1997**, 7, 34–40.
291. Johnson, T. R.; Silverman, R. B. *Bioorg. Med. Chem.* **1999**, 7, 1625–1636.
292. Kalyanakumar, R.; Chadha, M. S.; Chattopadhyay, A.; Mamdapur, V. R. *Indian J. Chem. Sect. B.* **1996**, 35B, 356–358.
293. Grassberger, V.; Berger, A.; Dax, K.; Fechter, M.; Gradnig, G.; Stuetz, A. E. *Liebigs Ann. Chem.* **1993**, 379–380.
294. McKinney, A. R.; Ridley, D. D. *Aust. J. Chem.* **2001**, 54, 757–761.
295. Iwashima, M.; Nagaoka, H.; Kobayashi, K.; Yamada, Y. *Tetrahedron Lett.* **1992**, 33, 81–82.
296. Vaezi, M. F.; Alam, M.; Sani, B. P.; Rogers, T. S.; Simpson-Herren, L.; Wille, J. J.; Hill, D. L.; Doran, T. I.; Brouillette, W. J.; Muccio, D. D. *J. Med. Chem.* **1994**, 37, 4499–4507.
297. Cheng, C. Y.; Wu, S. C.; Hsin, L. W.; Tam, S. W. *J. Med. Chem.* **1992**, 35, 2243–2247.
298. Lu, Q.; Duan, W. H.; Cai, J. C. *Chin. Chem. Lett.* **2002**, 13, 113–114.

299. Kurosawa, T.; Fujiwara, M.; Nakano, H.; Sato, M.; Yoshimura, T.; Murai, T. *Steroids* **2001**, 66, 499–504.
300. Yuan, W.; Munoz, B.; Wong, C.-H.; Haeggstroem, J. Z.; Wetterholm, A.; Samuelsson, B. *J. Med. Chem.* **1993**, 36, 211–220.
301. McPherson, D. W.; DeHaven-Hudkins, D. L.; Callahan, A. P.; Knapp, F. F. *J. Med. Chem.* **1993**, 36, 848–854.
302. Adamczyk, M.; Johnson, D. D.; Reddy, R. E. *Bioconjugate Chem.* **1998**, 9, 403–408.
303. Zhang, M.; Zhu, L.; Ma, X. *Tetrahedron: Asymmetry* **2003**, 14, 3447–3453.
304. (a) Presser, A.; Potschger, I.; Haslinger, E.; Hufner, A. *Monatsh. Chem.* **2002**, 133, 231–239. (b) Presser, A.; Haslinger, E.; Weis, R.; Huefner, A. *Monatsh. Chem.* **1998**, 129, 921–930.
305. Cambie, R. C.; Moratti, S. C.; Rutledge, P. S.; Weston, R. J.; Woodgate, P. D. *Aust. J. Chem.* **1990**, 43, 1151–1162.
306. Chavan, S. P.; Zubaidha, P. K.; Ayyangar, N. R. *Tetrahedron Lett.* **1992**, 33, 4605–4608.
307. Vostrikov, N. S.; Abutkov, A. V.; Spirikhin, L. V.; Fatykhov, A. A.; Miftahov, M. S. *Russ. Chem. Bull.* **2001**, 50, 654–658.
308. Goto, K.; Kubo, T.; Yamamoto, K.; Nakasuji, K.; Sato, K.; Shiomi, D.; Takui, T.; Kubota, M.; Kobayashi, T.; Yakusi, K.; Ouyang, J. *J. Am. Chem. Soc.* **1999**, 121, 1619–1620.
309. Ceccherelli, P.; Curini, M.; Marcotullio, M. C.; Rosati, O. *J. Org. Chem.* **1990**, 55, 311–315.
310. Atkins, R. J.; Breen, G. F.; Crawford, L. P.; Grinter, T. J.; Harris, M. A.; Hayes, J. F.; Moores, C. J.; Saunders, R. N.; Share, A. C.; Walsgrove, T. C.; Wicks, C. *Org. Proc. Res. Dev.* **1997**, 1, 185–197.
311. Rauter, A. P.; Oliveira, O.; Canda, T.; Leroy, E.; Ferreira, H.; Ferreira, M. J.; Ascenso, J. A. *J. Carbohydr. Chem.* **2002**, 21, 257–273.
312. Balakumar, A.; Janardhanam, S.; Rajagopalan, K. *Indian J. Chem. Sect. B* **1993**, 32B, 313–317.
313. Hisatome, M.; Watanabe, J.; Kawajiri, Y.; Yamakawa, K.; Iitaka, Y. *Organometallics* **1990**, 9, 497–503.
314. Obara, R.; Wawrzenczyk, C. *J. Cosmet. Sci.* **1998**, 49, 299–308.
315. For a review on fluorinated organozinc reagents in general, see: Davis, C. R.; Burton, D. J. In *Organozinc Reagents*; Knochel, P., Jones, P., Eds.; University Press: Oxford, 1999, p 57–76.
316. Baldwin, J. E.; Lynch, G. P.; Schofield, C. J. *Tetrahedron* **1992**, 48, 9085–9100.
317. Angelastro, M. R.; Bey, P.; Mehdi, S.; Peet, N. P. *Bioorg. Med. Chem. Lett.* **1992**, 2, 1235–1238.
318. Otake, A.; Watanabe, J.; Yukimasa, A.; Sasaki, Y.; Watanabe, H.; Kinoshita, T.; Oishi, S.; Tamamura, H.; Fujii, N. *J. Org. Chem.* **2004**, 69, 1634–1645.
319. (a) Tsukamoto, T.; Kitazume, T. *J. Chem. Soc., Chem. Commun.* **1992**, 540–541. (b) Tsukamoto, T.; Kitazume, T. *J. Chem. Soc., Perkin Trans. 1* **1993**, 1177–1181.
320. Marcotte, S.; Pannecoucke, X.; Feasson, C.; Quirion, J.-C. *J. Org. Chem.* **1999**, 64, 8461–8464.
321. Lacroix, S.; Cheguillaume, A.; Gerard, S.; Marchand-brynaert, J. *Synthesis* **2003**, 2483–2486.
322. (a) Sham, H. L.; Betebenner, D. A.; Wideburg, N. E.; Kempf, D. J.; Plattner, J. J.; Norbeck, D. W. *J. Fluorine Chem.* **1995**, 73, 221–224. (b) Sham, H. L.; Betebenner, D. A. *J. Chem. Soc., Chem. Commun.* **1991**, 1134–1135.
323. Wang, Y.; Zhu, S. *Tetrahedron Lett.* **2001**, 42, 5741–5744.
324. Wang, Y.; Zhu, S. *Synthesis* **2002**, 1813–1818.
325. Tsukamoto, T.; Kitazume, T. *Synlett* **1992**, 977–979.
326. Hu, Q.-S.; Hu, C.-M. *J. Fluorine Chem.* **1997**, 83, 87–88.
327. Kumadaki, I. Abstr. Pap.-226th ACS Meeting 2003, FLUO-010.
328. Kuroboshi, M.; Ishihara, T. *Bull. Chem. Soc. Jpn* **1990**, 63, 428–437.
329. Yoshida, M.; Suzuki, D.; Iyoda, M. *Synth. Commun.* **1996**, 26, 2523–2529.
330. Yoshida, M.; Suzuki, D.; Iyoda, M. *J. Chem. Soc., Perkin Trans. 1* **1997**, 643–648.
331. (a) Coe, P. L.; Markou, M.; Tatlow, J. C. *J. Fluorine Chem.* **1997**, 84, 113–118. (b) Coe, P. L.; Markou, M.; Tatlow, J. C. *J. Fluorine Chem.* **1998**, 89, 183–188.
332. Allmendinger, T.; Lang, R. W. *Tetrahedron Lett.* **1991**, 32, 339–340.
333. Kaneko, S.; Yamazaki, T.; Kitazume, T. *J. Org. Chem.* **1993**, 58, 2302–2312.
334. Gong, Y.; Kato, K. *J. Fluorine Chem.* **2001**, 111, 77–80.
335. Murata, K.; Kaneko, S.; Kitazume, T. *Bioorg. Med. Chem. Lett.* **1993**, 3, 2685–2688.
336. Watanabe, S.; Sakai, Y.; Kitazume, T.; Yamazaki, T. *J. Fluorine Chem.* **1994**, 68, 59–61.
337. Shen, Y.; Xiang, Y. *J. Fluorine Chem.* **1991**, 51, 349–355.
338. Shen, Y.; Gao, S.; Xiang, Y. *J. Fluorine Chem.* **1993**, 63, 151–155.
339. Iseki, K. In *Enantiocontrolled Synthesis of Fluoro-Organic Compounds: Stereochemical Challenges and Biomedical Targets*; Soloshonok, V. A., Ed.; John Wiley & Sons, Ltd., 1999, chapter 2, pages 33–62.
340. Tsukamoto, T.; Yoshiyama, T.; Kitazume, T. *Tetrahedron: Asymmetry* **1991**, 2, 759–762.
341. Watanabe, S.; Fujita, T.; Sakamoto, M.; Takeda, H.; Kitazume, T.; Yamazaki, T. *J. Fluorine Chem.* **1997**, 82, 1–7.
342. Matsumura, Y.; Fujii, H.; Nakayama, T.; Morizawa, Y.; Yasuda, A. *J. Fluorine Chem.* **1992**, 57, 203–207.
343. Kotra, L. P.; Xiang, Y.; Newton, M. G.; Schinazi, R. F.; Cheng, Y.-C.; Chu, C. K. *J. Med. Chem.* **1997**, 40, 3635–3644.
344. Kotra, L. P.; Newton, M. G.; Chu, C. K. *Carbohydr. Res.* **1998**, 306, 69–80.
345. Meffre, P.; Dave, R. H.; Leroy, J.; Badet, B. *Tetrahedron Lett.* **2001**, 42, 8625–8627.
346. Gouge, V.; Jubault, P.; Quirino, J.-C. *Tetrahedron Lett.* **2004**, 45, 773–776.
347. (a) Cogan, D. A.; Liu, G.; Ellman, J. A. *Tetrahedron* **1999**, 55, 8883–8904. (b) Davis, F. A.; Reddy, R. T.; Reddy, R. E. *J. Org. Chem.* **1992**, 57, 6387–6389.
348. Braun, M.; Vonderhagen, A.; Waldmueller, D. *Liebigs Ann.* **1995**, 1447–1450.
349. (a) Dolbier, W. R.; Ocampo, R.; Paredes, R. *J. Org. Chem.* **1995**, 60, 5372–5373. (b) Ocampo, R.; Dolbier, W. R.; Paredes, R. *J. Fluorine Chem.* **1998**, 88, 41–50.
350. Ocampo, R.; Dolbier, W. R.; Bartberger, M. D.; Paredes, R. *J. Org. Chem.* **1997**, 62, 109–114.
351. Ocampo, R.; Dolbier, W. R.; Zuluaga, F. *Coll. Czech. Chem. Commun.* **2002**, 67, 1325–1334.
352. Inagaki, H.; Sugita, K.; Miyauchi, R. N.; Miyauchi, S.; Takeda, T.; Itoh, M.; Takahashi, H.; Takemura, M. *ARKIVOC* **2003**, 112–117.
353. Altenburger, J. M.; Schirlin, D. *Tetrahedron Lett.* **1991**, 32, 7255–7258.
354. Takeuchi, Y.; Kanada, A.; Kawahara, S.; Koizumi, T. *J. Org. Chem.* **1993**, 58, 3483–3485.

355. Morikawa, T.; Uejima, M.; Kobayashi, Y.; Taguchi, T. *J. Fluorine Chem.* **1993**, *65*, 79–89.
356. Pirrung, M. C.; Rowley, E. G.; Holmes, C. P. *J. Org. Chem.* **1993**, *58*, 5683–5689.
357. Fukuda, H.; Kitazume, T. *Heterocycles* **1997**, *46*, 275–285.
358. Matsuda, F.; Matsumoto, T.; Ohsaki, M.; Terashima, S. *Bull. Chem. Soc. Jpn* **1991**, *64*, 2983–2989.
359. Robinson, R. P.; Donahue, K. M. *J. Org. Chem.* **1992**, *57*, 7309–7314.
360. Shen, R.; Priebe, C.; Patel, C.; Rubo, L.; Su, T.; Kahn, M.; Sugawara, R. *Tetrahedron Lett.* **1992**, *33*, 3417–3420.
361. Doherty, A. M.; Kaltenbronn, J. S.; Hudspeth, J. P.; Repine, J. T.; Roark, W. H.; Sircar, I.; Tinney, F. J.; Connolly, C. J.; Hodges, J. C.; et al *J. Med. Chem.* **1991**, *34*, 1258–1271.
362. Doherty, A. M.; Sircar, I.; Kornberg, B. E.; Quin, J.; Winters, R. T.; Kaltenbronn, J. S.; Taylor, M. D.; Batley, B. L.; Rapundalo, S. R.; et al *J. Med. Chem.* **1992**, *35*, 2–14.
363. Skiles, J. W.; Miao, C.; Sorcek, R.; Jacober, S.; Mui, P. W.; Chow, G.; Weldon, S. M.; Possanza, G.; Skoog, M.; et al *J. Med. Chem.* **1992**, *35*, 4795–4808.
364. Marcotte, S.; D’Hooge, F.; Ramadas, S.; Feasson, C.; Pannecoucke, X.; Quirion, J.-C. *Tetrahedron Lett.* **2001**, *42*, 5879–5882.
365. Konno, K.; Ojima, K.; Hayashi, T.; Takayama, H. *Chem. Pharm. Bull.* **1992**, *40*, 1120–1124.
366. Gill, H. S.; Londowski, J. M.; Corradino, R. A.; Zinsmeister, A. R.; Kumar, R. *J. Med. Chem.* **1990**, *33*, 480–490.
367. Saito, N.; Matsunaga, T.; Fujishima, T.; Anzai, M.; Saito, H.; Takenouchi, K.; Miura, D.; Ishizuka, S.; Takayama, H.; Kittaka, A. *Org. Biomol. Chem.* **2003**, *1*, 4396–4402.
368. Hallinan, E. A.; Hagen, T. J.; Husa, R. K.; Tsybalov, S.; Rao, S. N.; vanHoeck, J. P.; Rafferty, M. F.; Stapelfeld, A.; Savage, M. A.; Reichman, M. *J. Med. Chem.* **1993**, *36*, 3293–3299.
369. Fukuda, H.; Kitazume, T. *J. Fluorine Chem.* **1995**, *74*, 171–176.

Biographical sketch



Rogelio Ocampo was born on July 9, 1962 in Marquetalia, Caldas (Colombia). He performed undergraduate studies of Education in Biology and Chemistry at Universidad de Caldas in Manizales (Colombia) where he obtained his degree in 1984. His graduated studies were performed at Universidad del Valle in Cali (Colombia) from where he obtained the MSc degree (1990) and PhD degree (1997). He received his doctoral training as a visitor scholar (partially granted by COLCIENCIAS) at University of Florida (1994–1996) from Professors William R. Dolbier, Jr. and Rodrigo Paredes (Universidad del Valle). His doctoral dissertation was graded *Cum Laude*, and he was awarded with the Colombian IV Prize of Chemistry (second place) in 2001. After his PhD degree he was back to Universidad de Caldas, where he is at present Professor of Organic Chemistry. His current interest is focused on synthesis and physical organic chemistry of fluorinated β -lactones and related compounds, working in collaborative projects of Universidad de Caldas-Universidad del Valle-University of Florida. At present, he is member of the research group *SIMERQO* granted by COLCIENCIAS (Colombia).



William R. Dolbier, Jr. Bill Dolbier is currently the Col. Allen R. and Margaret G. Crow Professor of Chemistry at the University of Florida. He received his BS in Chemistry from Stetson University in 1961 and obtained his PhD in organic chemistry from Cornell University in 1965, working with Mel Goldstein. After one and a half years of postdoctoral work with Bill Doering at Yale, he joined the faculty at UF in 1966, where he has been ever since, serving as Chairman from 1983 to 1988. Bill's research interests continue to be physical organic in nature, and he maintains long-term interests in thermal homolytic reactions, pericyclic reactions, and free radical reactivity. Since 1975, his efforts have mainly focused on the study of molecules containing fluorine. In recent years, his efforts have increasingly been devoted to development of new synthetic methods in organofluorine chemistry. Bill received the ACS award for Creative Work in Fluorine Chemistry in 2000, and is currently a member of the Executive Committee of the Fluorine Division of the ACS. When not immersed in such activities, Bill's main interests continue to be his wife, Jing, son, Stephen, three grandchildren, and a little handball.